

Foundations: 0D through 4D

Typed-Delta Structures: A Dimensional Skeleton, 0D–4D

Axioms, Classification, and Verification from 0D to 4D

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Abstract

This paper develops an axiomatic framework for typed-differentiation structures, built up from a zero-dimensional base state through four dimensions, each layer self-contained and forcing a specific finite automorphism group: the full symmetric group at one dimension, the dihedral group of order 8 at two dimensions, and, for every n simultaneously rather than dimension by dimension, the hyperoctahedral group B_n once a local-saturation axiom is imposed. Without that axiom, two systematic families of proper subgroups, E_n and P_n , also satisfy the weaker typing conditions — and at $n=5$ we report a confirmed counterexample to the natural conjecture that these two families are exhaustive: an explicit order-640 structure isomorphic to neither B_n , E_n , nor P_n . The most natural way of extending this counterexample to $n=7$ is tested directly and shown to fail, for a fully derived reason. Every numerical claim is independently verified computationally, with twenty-two checks reported in full. Left open are the general-cardinality case of two dimensions, a point-set-level characterization at each n , and whether further exceptions to E_n/P_n exhaustiveness exist beyond $n=5$.

Keywords: hyperoctahedral group · Hamming scheme · Bose–Mesner algebra · Schur ring · permutation group · association scheme

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1. Introduction

This paper studies (S, Δ) : a set together with a typed differentiation relation, axiomatized layer by layer from Section 2 onward. From three dimensions on, S is taken to be $\{0,1\}^n$ under Hamming distance — a purely combinatorial choice, fixed by the axioms themselves (Sections 14–15), not by any geometric picture. Where this set is, by a separately cited theorem rather than by assumption, isomorphic to the vertex set of what the literature calls the hypercube graph Q_n [1], that is a fact about which abstract group acts on it — supplying a citable name, B_n , and nothing more. No proof anywhere in this paper draws on a drawing, an embedding in Euclidean space, or any reasoning that depends on picturing $\{0,1\}^n$ as a geometric solid; Sections 14–15 state this discipline explicitly and enforce it without exception (Section 1.4 discusses one place this distinction was tested directly), and Appendix O collects the notation used throughout, including this point about Q_n specifically.

1.1 Motivation

Any structural theory requires a starting point that does not itself presuppose prior structure. We argue that this starting point — the zero-dimensional state, $0D$ — must be treated as a foundational layer in its own right, with its own internal axioms, rather than as a limiting case described in the vocabulary of higher-dimensional structures. Built on this foundation, the same typed-differentiation discipline turns out to force a single specific finite symmetry group — the hyperoctahedral group B_n — at every dimension $n \geq 2$, with one documented exception at $n=5$ that is, in a precise sense made below, the central surprise of this paper.

1.2 Problem Statement

$0D$ cannot be described using higher-dimensional constructs without begging the question of where those constructs come from. It requires a self-contained formalism: one that can refer to $0D$ without presupposing that $0D$ already contains the structure used to refer to it. The same discipline, suitably adapted, must extend to $1D$ once $1D$ is characterized as an object with genuine internal structure of its own. From three dimensions onward, the underlying question is a classical one in finite permutation group theory: which finite automorphism groups satisfy a natural typed-differentiation axiom system, and are the obvious candidates — the hyperoctahedral group B_n and its two natural index-2 subgroups — the only ones that do?

1.3 Contribution

- A direct counting argument proving that a local-saturation axiom (Axiom 12') forces the hyperoctahedral group B_n — the group of signed permutations of n coordinates — as the unique automorphism group satisfying it, at every $n \geq 2$ simultaneously rather than dimension by dimension (Theorem 3.15); together with an identification of two systematic families of proper subgroups, E_n and P_n , that satisfy the weaker typing axioms (Axioms 10–11) without satisfying this stronger one (Section 12.5).

- A confirmed counterexample, at $n=5$, to the natural conjecture that E_n and P_n are the only proper subgroups satisfying Axioms 10–11 without Axiom 12': an explicit order-640 structure, found by exhaustive search of B_5 's subgroup lattice and identified via a Frobenius-group construction, isomorphic to neither B_5 , E_5 , nor P_5 (Appendix M). We further show that the most natural way of extending this construction to the next prime, $n=7$, fails for a fully derived reason — it satisfies the required rank condition if and only if $n \leq 5$ — narrowing, though not resolving, the classification question at higher n (Appendix N).
- A complete axiom system for oD (Section 3).
- A rigorous derivation of oD 's internal properties (Sections 5–8).
- A formal proof of the minimal distinction relating oD and $1D$, characterized via a distinction operator rather than any notion of displacement (Section 7, Theorem 3.3).
- An explicit meta-language/object-language separation that inoculates the axiom system against self-reference objections (Appendix A).
- A self-contained axiomatization of the internal anatomy of $1D$, and a proof — independent of, and compatible with, the meta-linguistic argument of Theorem 3.3 — that no one-dimensional structure admits a nontrivial coordinatizing partition (Section 9).
- A threshold theorem identifying the unique minimal relaxation under which a structure admitting such a coordinatizing partition first becomes possible, together with an interface-level definition of $2D$ (Section 9.4).
- A self-contained axiomatization of the internal anatomy of $2D$ at the smallest non-trivial finite cardinality, forcing a unique automorphism group and ruling out a third independent difference type via the associated Schur ring (Section 10).
- An explicit identification of $2D$ as the $n=2$ instance of the general- n framework developed independently in Sections 14–15, together with a structural explanation — via Lagrange's theorem at the smallest cases — of why $2D$ never required the saturation axiom needed at every $n \geq 3$ (Section 11).
- A general- n extension of the typed-differentiation axioms, together with a self-contained axiomatization of the internal anatomy of $3D$ at the smallest non-trivial finite cardinality, requiring a local-saturation axiom beyond typing alone and identifying the two specific proper subgroups that axiom is needed to exclude (Section 12).
- A self-contained axiomatization of the internal anatomy of $4D$ at the smallest non-trivial finite cardinality, instantiating the general- n results directly and ruling out a fifth independent difference type by explicit computation (Section 13).
- A standalone computational verification package, independently re-implementing and checking every numerical claim across all four dimensional layers (twenty-two checks in total, Appendix I).

1.4 Related Work

The classification problem this paper reaches from three dimensions onward — which finite groups can be the full automorphism group of a structure satisfying a given local typing condition — sits within an established area of finite permutation group theory and algebraic

combinatorics. This section places the present results within that literature; the axiomatic route taken to reach them (Sections 2–15) is otherwise independent of it.

The hyperoctahedral group B_n , forced as the unique automorphism group by Theorem 3.15, is the group of signed permutations of n coordinates; equivalently, it is the full stabilizer of the natural Cartesian decomposition of $\{0,1\}^n$ into n binary coordinates, in the sense developed systematically by Praeger and Schneider [2]. The dimension-counting arguments used throughout Sections 12 and 14–15 to exclude additional independent difference types (Theorems 3.8, 3.11, 3.14, 3.17) rest on the classical theory of Schur rings [3] and, in their general- n form, on the theory of association schemes founded by Delsarte [4] and developed systematically by Bannai and Ito [5]: the relevant object throughout is the Hamming scheme $H(n,2)$, whose Bose–Mesner algebra has dimension exactly $n+1$ (Lemma 2.13; corroborated by [6]) and whose automorphism group is B_n itself [1]. One point is worth flagging here, ahead of Sections 14–15 where it becomes load-bearing: citing this literature to name a group — “hyperoctahedral group,” “hypercube graph Q_n ” — is a citation of established nomenclature, not a license to reason from a geometric picture. Throughout this paper, (S, Δ) is nothing more than a set together with the typed relation axiomatized at each layer; no proof anywhere in this paper argues from a drawing, an embedding in Euclidean space, or a word like “rotate” or “face” in place of Δ itself. Sections 14–15 state this discipline explicitly and maintain it without exception, including one recorded instance (Section 12.3) where an argument leaning on exactly such a picture was tried and shown to fail for that reason.

The specific classification question this paper leaves open — whether E_n and P_n exhaust the proper subgroups of B_n that are transitive with rank $n+1$ and satisfy the local condition of Axiom 11 — is a question about rank- $(n+1)$ permutation groups, a topic with a literature of its own: Liebeck and Saxl’s classification of primitive permutation groups by minimal degree [7] and Kivva’s confirmation that rank-4 primitive coherent configurations other than the Johnson and Hamming schemes have linear minimal degree [8] both bear on closely related questions, though neither has been checked against the specific counterexample reported in Appendix M (an order-640 subgroup of B_5 , built from the Frobenius group $AGL(1,5)$). A search of the readily available literature did not locate a prior discussion of this specific structure under any name (Appendix M); this is reported as the result of a literature search, not as a claim that no such discussion exists.

A related parallel, distinct from the classification question above but structurally suggestive, is worth flagging explicitly rather than left implicit. Kantor’s classification of k -homogeneous, non- k -transitive permutation groups is complete for $k=2,3,4$ [25, 26], resolved case by case with no single formula covering all three values of k . This paper’s own small- n results — the forced D_4 at $n=2$ (Theorem 3.7), the two extra transitive rank-4 subgroups of B_3 found by exhaustive search at $n=3$ (Section 12.3), and the confirmation that E_4 and P_4 are exactly the rank-5 candidates at $n=4$ (Section 13.4) — were likewise obtained case by case, one dimension at a time, before Theorem 3.15’s general- n counting argument closed the sufficiency question uniformly. The two classification problems are not the same — Kantor’s concerns k -homogeneity of an action on an arbitrary set under k -subsets, this paper’s concerns transitive rank- $(n+1)$ subgroups of B_n acting specifically on $\{0,1\}^n$ — and no claim is made that either result follows from or subsumes the

other. Appendix M.5 records this parallel in full, including why the identification stops short of a shared theorem.

Every exhaustive search reported in this paper — the $|S|=6,8$ enumeration of Appendix L, the $\{0,1\}^3$ witness-model search of Section 12.3, and the full subgroup-lattice search of B_5 in Appendix M — was carried out computationally, using standard methods of computational group theory [9]; all such claims are independently re-verified in a standalone package reported in full in Appendix I.

The relation between this paper and the literature just cited runs in one direction only. B_n is forced here as a consequence of an axiom system motivated by typed differentiation (Axioms 10–12'), not assumed or studied directly; this paper does not draw on the cited classification results to prove anything, and none of its theorems depend on them. The connection runs the other way: once B_n is forced, the further question of which proper subgroups can satisfy the weaker axioms alone is exactly a rank- $(n+1)$ classification question of the kind this literature addresses, and it is in that direction — as an input to, not an output of, that literature — that the order-640 counterexample at $n=5$ (Appendix M) and its failure to generalize to $n=7$ (Appendix N) are offered.

1.5 Statement of Main Results

The results below are restated here without proof, gathered in one place for reference; each is proved in full at the section cited. This section may be skipped on a first reading in favor of Section 2.

Theorem 3.3 (the minimal distinction relating 0D and 1D is unique; proved in Section 7).

Let $\delta \neq id$ mark a distinction involving 0D. The structure thereby distinguished is, up to meta-linguistic equivalence, exactly the one-dimensional structure 1D: δ marks neither fewer nor more distinguishable relations than this.

Theorem 3.5 (no coordinatization of 1D; proved in Section 9).

Let (S, Δ) satisfy Axioms 6–7. Then (S, Δ) admits no coordinatizing partition compatible with $Aut(S, \Delta)$; equivalently, no nontrivial product decomposition $S \cong S_1 \times S_2$ compatible with $Aut(S, \Delta)$ exists.

Theorem 3.6 (threshold theorem; proved in Section 9.4).

A structure (S, Δ) can admit a coordinatizing partition compatible with $Aut(S, \Delta)$ only if Axiom 6 (Binary Differentiation) does not hold of it. This relaxation is necessary, and non-vacuous: Appendix H exhibits a structure satisfying the relaxed condition that genuinely admits such a partition.

Theorem 3.7 (forced dihedral symmetry at 2D; proved in Section 10).

Let (S, Δ) satisfy Axioms 8–9 with $|S|=4$. Then $Aut(S, \Delta) \cong D_4$, with rank exactly 3.

Theorem 3.8 (no third independent type at 2D; proved in Section 10).

Under the hypotheses of Theorem 3.7, Σ cannot be extended by a third type σ_3 , distinguishable from σ_1 and σ_2 , compatible with $\text{Aut}(S, \Delta)$.

Theorem 3.15 (general-n sufficiency: Axiom 12' forces B_n ; the central general result of this paper, proved in Section 12.5.1).

Let $n \geq 2$ and let (S, Δ) satisfy Axioms 10–11 with $S = \{0,1\}^n$ under Hamming distance. If $\text{Aut}(S, \Delta)$ additionally satisfies Axiom 12', then $\text{Aut}(S, \Delta) \cong B_n$. The proof is a single direct counting argument holding at every n simultaneously, not an exhaustive search repeated dimension by dimension.

Theorem 3.11 and Theorem 3.14 (no fourth or fifth independent type at $n=3,4$; proved in Sections 14–15, generalized to every n by Theorem 3.17).

Under the hypotheses of Theorem 3.15 at $n=3$ (respectively $n=4$), Σ cannot be extended by a fourth type σ_4 (respectively a fifth type σ_5), distinguishable from $\sigma_1, \dots, \sigma_n$, compatible with $\text{Aut}(S, \Delta)$.

A counterexample at $n=5$ (the central surprise of this paper; established in Appendix M, with the most natural attempt to generalize it tested and shown to fail in Appendix N).

Theorem 3.15 forces B_n under Axiom 12', but under the strictly weaker Axioms 10–11 alone, E_n and P_n are not the only proper subgroups of B_n that are transitive with rank $n+1$ and satisfy Axiom 11's local condition. At $n=5$, an exhaustive search of B_5 's entire subgroup lattice (3840 elements, 953 conjugacy classes of subgroups) finds a fourth such structure, of order 640, isomorphic to neither B_5 , E_5 , nor P_5 . This structure is built from the Frobenius group $\text{AGL}(1,5)$, order 20, acting on the 5 coordinates — a construction available specifically because 5 is prime. The most natural way of extending it to the next prime, $n=7$, fails: the same construction satisfies the required rank condition if and only if $n \leq 5$, a fact fully derived rather than merely observed (Appendix N). Whether some different construction produces a further exception at $n=7$ or beyond is the principal question this paper leaves open.

1.6 Structure of the Paper

This section maps the paper's structure for reference; readers may skip to Section 2 directly.

Section 2 introduces the minimal apparatus needed to discuss oD. It also specifies what the paper deliberately omits (Section 2.4). Sections 3–4 give the axioms and definitions. Sections 5–8 derive propositions, lemmas, theorems, and corollaries.

Section 14 discusses the self-referential character of the oD layer specifically, the paper's methodological position within a larger programme, and — once oD through 4D have each been established — the pattern common to every layer's forcing argument and the role exhaustive

computation played at $n=5$. Section 15 considers implications across several fields, including the rank- $(n+1)$ classification literature this paper's general- n results connect to.

Section 9 characterizes the internal anatomy of 1D and establishes the threshold condition for 2D. Section 10 characterizes the internal anatomy of 2D at the smallest non-trivial finite cardinality. Section 11 clarifies how 2D relates to the general- n framework developed in Sections 14–15.

Section 12 extends the typed-differentiation axioms to general n and characterizes the internal anatomy of 3D at the smallest non-trivial finite cardinality, including the general- n sufficiency theorem (Theorem 3.15) that forces B_n at every dimension simultaneously. Section 13 characterizes the internal anatomy of 4D at the smallest non-trivial finite cardinality, instantiating these general results directly. Section 16 concludes.

The appendices carry the full formal apparatus, listed here for reference:

- Appendix A — the meta-language disclaimer.
- Appendix B — the defense against the infinite-regress objection.
- Appendix C — the structural algebra of oD, including its role as the null element of the operations that preserve it.
- Appendix D — the external-framework compatibility note.
- Appendix E — the explicit construction of the minimal oD/1D distinction.
- Appendix F — the argument for the inapplicability of physical formalism at oD, introduced in Section 1.8.
- Appendix G — worked finite examples illustrating Section 9.
- Appendix H — an explicit 2D structure witnessing the non-vacuity of Theorem 3.6.
- Appendix I — full computational verification results for Sections 14–15 (twenty-two checks).
- Appendix J — genuine applications of the verified machinery from Sections 14–15, tested directly and reported with both accepted and rejected proposals.
- Appendix K — minimal standalone scripts verifying the key claims of Section 12, each independently runnable without the full verification package.
- Appendix L — the exhaustive computational verification behind §11.4's conjecture that Axioms 8–9 admit exactly two non-isomorphic structures at $|S|=6$ and at $|S|=8$, including the methodological missteps actually made during this verification.
- Appendix M — a confirmed counterexample, at $n=5$, to the classification question (§13.9) of whether E_n and P_n exhaust all rank- $(n+1)$ subgroups of B_n satisfying Axioms 10–11; an explicit fourth structure is exhibited, with full identification and a statement of what this does and does not affect.
- Appendix N — a tested negative result at $n=7$: the construction underlying Appendix M's order-640 structure does not generalize to the next prime, with a derived criterion ($n \leq 5$) explaining exactly why, narrowing rather than resolving the general classification question.

- Appendix O — a one-place reference summary of the notation used throughout, including the precise sense in which Q_n names a combinatorial graph and not a geometric solid.

Appendices A–M are kept, together with this paper’s reference list, in a single master appendices file (shared with no other paper), so that appendix and reference content does not diverge across separately-edited working copies of this same document.

1.7 Position Within a Larger Programme

This paper is the first in a planned sequence building the dimensional architecture from oD upward, and it covers the entire foundational tier of that sequence: oD in full, the minimal distinction relating oD and 1D, the internal anatomy of 1D, the internal anatomy of 2D at the smallest non-trivial finite cardinality (Section 10), and the internal anatomy of 3D and 4D at their respective smallest non-trivial finite cardinalities (Sections 14–15), together with the general- n sufficiency theorem (Theorem 3.15) that establishes the central forcing result at every dimension $n \geq 2$ simultaneously. The general (arbitrary cardinality, possibly infinite) case at each layer is left to subsequent work, as is the question — stated precisely as open where it arises — of whether the rank- $(n+1)$ subgroups identified here exhaust every alternative to B_n at every n .

The broader programme adopts a two-tier methodology, stated here for transparency and revisited in Section 14.5. The foundational tier — oD through 4D — is treated with the same axiomatic rigor as the present paper: formal definitions, explicit operators, and proofs. Beyond this foundational tier, a separate, heuristic, exploratory layer proposes structural analogies that are not yet established as formal correspondences; this exploratory layer is outside the scope of the present paper and is not discussed further here.

This division is a deliberate methodological choice, not an indication of differing confidence in the underlying architecture. The foundational layer (oD–4D) must withstand independent scrutiny before any claim built on it can be entertained. Work beyond this foundational layer is addressed only in separate material, once it exists in a form that can itself withstand the same scrutiny.

Theorem 3.3 establishes a minimal structural relation between oD and 1D. The threshold theorem of Section 9.4, relating 1D to 2D, is likewise a statement of logical conditionality: it identifies what must be relaxed for a structure to admit a nontrivial coordinatizing partition.

(A note on what these results do not claim: Theorem 3.3 characterizes a relation, not a direction of priority, origination, or process between oD and 1D. The threshold theorem identifies a logical condition, not a claim about which structure is prior to, causes, or gives rise to the other. No claim about priority, origination, or process is made or required anywhere in this paper.)

1.8 On the Inapplicability of Physical Formalism at oD

This paper deliberately employs no physical vocabulary. This is not a provisional omission but a categorical constraint, defended fully in Appendix F. The reason is structural: the mathematical apparatus of physical theory presupposes the existence of at least one degree of freedom on which such structures can act.

By Axioms 0–5, oD possesses none. Any physical notation applied directly to oD would therefore describe the meta-language used to discuss oD, not oD itself — a violation of the discipline established in Appendix A.o. The absence of physical formalism is thus a structural necessity of the dimensional origin, not a gap to be filled. (Section 2.4 specifies precisely which structures this entails omitting.)

This absence is itself the paper's claim about physics: a theory of the preconditions for physical description must be stated in a vocabulary that does not already assume those preconditions hold.

PART I — THE ZERO-DIMENSIONAL STATE (oD)

Sections 2–10. Foundations, axioms, definitions, propositions, lemmas, theorems, corollaries, discussion, and implications for oD.

2. Foundations

This section introduces the minimal vocabulary this paper needs before any axiom can be stated: what counts as a meta-language describing oD, and what a "distinction" is as the smallest possible meta-linguistic act. Nothing here presupposes dimension, structure, or position — only what is needed to talk about a structural entity prior to any of those.

2.1 Preliminaries

We introduce only the minimal mathematical vocabulary needed to talk about oD without presupposing dimension: the notion of a meta-language in which oD is described, and the notion of a distinction as the smallest possible meta-linguistic act (formalized in Section 7 and Appendix E). No metric, order, or coordinate structure is assumed at the outset.

2.2 Internal Citation Framework

Results are organized in the standard order Definitions → Propositions → Lemmas → Theorems → Corollaries, with each later result citing the earlier ones on which it depends. This is a self-referential citation system internal to the paper, not a structural feature of oD itself.

2.3 The Role of oD in Structural Language

oD is positioned as the base layer of structure as such: the unique entity exhibiting no structural distinction, related by the minimal such distinction to 1D (Section 7).

2.4 What This Paper Does Not Attempt

This paper does not construct a Hilbert space, define an operator algebra, specify a symmetry group, introduce observables, or develop a measurement theory. Such structures presuppose the existence of degrees of freedom and belong to layers of appearance and physical description that are absent at oD (Theorem 3.3).

The scope of this work is strictly limited to the internal axioms of oD and the minimal distinction relating oD and 1D; all physically formalizable structures lie

beyond the domain of the present paper and are addressed only in later stages of the programme (Section 1.7). The precise dimension at which such structures first become applicable is not asserted here; see Appendix F.7.

3. Axioms of the Zero-Dimensional State

Each axiom below is referenced throughout the remainder of the paper. The axioms describe the internal constitution of oD; they say nothing about the meta-language used to state them (see Appendix A).

Axiom 0 (Existence).

The zero-dimensional state exists as the base entity of the structural sequence developed in this paper. "Base" here names its position in that sequence, not an internal structural property; Axioms 1–5 specify what oD does and does not contain.

Axiom 1 (Indivisibility).

oD cannot be decomposed into parts.

Axiom 2 (Absence of Boundary).

oD contains no internal or external boundary.

Axiom 3 (Zero Tension).

oD has no internal tension or curvature.

Axiom 4 (Non-Orientation).

oD cannot encode direction, order, or orientation.

Axiom 5 (Minimal Information).

oD contains zero structural information.

4. Definitions

This section collects the formal definitions used throughout the paper — for oD itself, and for the group-theoretic objects (such as E_n and P_n) introduced once the argument reaches general n . Each is referenced by number wherever it is used, not restated in full.

Definition 0.1 (Zero-Dimensional State).

oD is the unique structural entity satisfying Axioms 0–5. Its designation here is a meta-linguistic act of reference (Appendix A) and does not assert that oD internally contains the apparatus used to refer to it.

Definition 0.2 (Structural Nullity).

oD functions as the null element of the structural algebra developed in Appendix C.

Definition 0.3 (Zero Boundary Condition).

A formalization, internal to the axiom system, of the absence of boundary asserted in Axiom 2.

Definition 0.4 (Zero Tension Field).

A formalization, internal to the axiom system, of the absence of tension asserted in Axiom 3.

Definition 0.5 (Distinction Operator (meta-language device)).

*δ is a meta-language device that marks structural difference relative to *oD*. An application of δ is trivial if it marks no difference ($\delta = \text{id}$, identified with *oD* itself). A non-trivial application ($\delta \neq \text{id}$) marks some difference relative to *oD*. This definition does not specify, presuppose, or stipulate how many distinguishable relations a non-trivial application marks; that question is settled by Theorem 3.3, not by this definition.*

*δ is not part of *oD*'s internal content; it is the minimal apparatus by which the meta-language can mark a difference involving *oD*. The existence of δ as a meta-language device does not contradict Axiom 5, since δ is external to *oD*, not an internal feature of it — precisely as the embedding of the empty set into a larger set is not an internal feature of the empty set.*

δ is deliberately not called an “offset”: that term carries connotations of displacement and metric magnitude that this paper does not assume and explicitly disclaims. δ names a distinction, not a movement, and the relation it marks carries no implication of priority or process between the structures it relates (Section 1.7).

5. Propositions (Direct Consequences of Axioms)

*The propositions below follow from the axioms of Section 3 in one or two steps, with no intermediate construction required. They establish *oD*'s most basic properties: no internal difference, no orientation, no smaller structural unit.***Proposition**

1.1 (No Internal Difference).

**oD* cannot contain distinguishable subcomponents. (From Axioms 1, 5.)*

Proposition 1.2 (No Orientation).

**oD* cannot encode left/right, before/after, or any directional relation. (From Axiom 4.)*

Proposition 1.3 (Minimal Structural Unit).

**oD* is the unique minimal element in the structural poset. (From Axioms 0, 1.)*

Proposition 1.4 (Zero Curvature).

*Curvature requires dimension; *oD*, having none, has zero curvature. (From Axiom 3.)*

6. Lemmas (Intermediate Results)

The lemmas here go one step beyond Section 5's direct consequences, each requiring a short argument rather than an immediate read-off from a single axiom. They establish the structural facts the theorems of Section 7 depend on.

Lemma 2.1 (Zero Boundary $\Rightarrow$ Zero Gradient).

Without boundary (Axiom 2), no gradient can be defined on oD .

Lemma 2.2 (Zero Curvature $\Rightarrow$ Zero Tension).

Curvature is a precondition for tension; by Proposition 1.4, oD has neither.

Lemma 2.3' (Atomicity of the Distinction Operator).

Let $\delta \neq id$ be a non-trivial distinction operator marking a difference relative to oD . δ is operationally indecomposable: there do not exist two operators δ_1, δ_2 , each distinct from id , such that δ is equivalent to their composite (or to any specification of their joint or sequential application).

Proof.

Assume, in the meta-language, that

$$\delta \equiv \delta_2 \circ \delta_1$$

for some $\delta_1, \delta_2 \neq id$. For the expression $\delta_2 \circ \delta_1$ to denote a *single* operator, there must exist an object—call it T —that serves simultaneously as:

- the referent relative to which δ_1 marks a difference from oD , and
- the referent relative to which δ_2 marks its own difference.

This is a **co-reference requirement** for the expression to be well-defined. It does **not** introduce any temporal or procedural ordering: nothing in this assumption presupposes a process, a passage, or a sequence of events. It concerns only the semantic condition under which the composite can be treated as a single operator.

Two consequences follow, neither requiring any notion of process.

First.

Fixing the role of T requires an ordering that Axiom 4 forbids. By the syntax of composition,

$$\delta_2 \circ \delta_1 \neq \delta_1 \circ \delta_2,$$

*because the two expressions assign **different roles** to the co-referent T : in the first, T is the target of δ_1 ; in the second, it is the source for δ_2 . To determine which of these two roles T occupies is already to impose an **ordering relation** between δ_1 and δ_2 . But Axiom 4 (Non-Orientation) states that oD cannot license any such ordering. Thus, the role-fixing required for the composite to be well-defined is structurally unavailable.*

Second.

*T requires a structural resource that Axiom 5 withholds. For T to be co-referenced by both δ_1 and δ_2 without collapsing either into oD or into the output of δ , T must possess some structural feature that keeps it distinct from both. However, Axiom 5 (Minimal Information) asserts that oD provides **no** such resource. Therefore, the co-reference condition cannot be satisfied without violating the minimality of oD .*

Without that resource, T collapses, contradicting the assumption.

Without such a resource, T has no distinguishing predicate separating it from oD or from δ 's output. By Lemma 2.5 (Meta-Isomorphism), anything lacking a distinguishing predicate is meta-linguistically identified with what it cannot be distinguished from; T therefore collapses into oD (forcing $\delta_1 = \text{id}$) or into the output (forcing $\delta_2 = \text{id}$). Either case contradicts the assumption that the decomposition is non-trivial.

Therefore δ admits no decomposition, and is atomic — not by stipulation, but because the bare syntax of stating a decomposition already requires a co-referenced object T , and supplying T with the structural resources needed to keep it distinct requires exactly what Axioms 4 and 5 withhold at oD . No step of this argument depends on oD itself undergoing, hosting, or licensing any process; the argument concerns only what the meta-language must presuppose to state a decomposition, which is then shown to be unavailable.

□

Lemma 2.4 (oD Is a Fixed Point of Folding).

No folding operator can reduce oD further than oD itself; oD is the absolute fixed point of any folding operation (cf. Corollary 4.4).

Lemma 2.5 (Meta-Isomorphism Lemma).

Let X and Y be two structures each satisfying Axioms 0–5. Then X and Y are indistinguishable at the level of internal structure: no predicate expressible in the structural vocabulary of this paper can differentiate X from Y .

Proof.

Any distinguishing predicate must appeal to some structural feature — a boundary, an orientation, or an internal relation — that could differentiate $[X](ca://s? q = \text{Explain}_{\text{role}_X})$ from $[Y](ca://s? q = \text{Explain}_{\text{role}_Y})$. But by Axioms 1–5, neither X nor Y possesses any such internal feature.

Consequently, any apparent distinction between X and Y can arise only from the **meta-language's own act of labeling**, not from anything in the objects themselves.

□

We emphasize the scope of this result. Lemma 2.5 concerns **indistinguishability with respect to internal structure**, not absolute identity. Two structures satisfying Axioms 0–5 may receive different labels in the meta-language — they may be called “ X ” and “ Y ,” appear in different parts of an argument, or be introduced at different stages of an exposition — yet none of this labeling constitutes a structural difference between them.

This is the same sense in which two models of a theory may be non-identical as labeled objects but still **elementarily equivalent**: they cannot be distinguished by any sentence expressible in the theory's own vocabulary.

Accordingly, the uniqueness asserted in [Theorem 3.2](ca://s?q=Explain_Theorem_3_2) is uniqueness **up to structural equivalence**, not a claim that meta-linguistic labels could never differ.

7. Theorems (Core Results)

This section states the core results following from oD's axioms — including the result the rest of the paper depends on most directly: that the minimal distinction relating oD to a one-dimensional structure is unique (Theorem 3.3), which Part II builds on.

Theorem 3.1 (Zero Structural Energy).

From Axioms 2–3 and Lemmas 2.1–2.2, oD has total structural energy zero. This result is derived purely internally; its relation to the Well-formed Energy Mapping Protocol (WEMP), an external framework introduced and addressed separately in Appendix D, has no bearing on this proof.

Theorem 3.2 (Uniqueness of oD).

There exists exactly one zero-dimensional state up to meta-linguistic equivalence, as established by Lemma 2.5.

We emphasize that “isomorphism” here denotes meta-language indistinguishability, not an internal morphism of oD; oD itself admits no morphism structure, internal or otherwise.

Theorem 3.3 — The Minimal Distinction Relating oD and 1D

This is the central theorem of the paper.

Let $\delta \neq \text{id}$ mark a distinction involving oD (Definition 0.5). We claim the structure thereby distinguished is, up to meta-linguistic equivalence (Lemma 2.5), exactly the one-dimensional structure 1D: that δ marks neither fewer nor more distinguishable relations than this.

(i) Not less than 1D.

By Definition 0.5, any non-trivial application $\delta \neq \text{id}$ marks a difference relative to oD; marking no difference is, by definition, the identity case and is already excluded. To mark *some* difference is to mark at least one **distinguishable relation** — a “this versus not-this,” the minimal structural seed of orientation. Section 7 identifies precisely this minimal structural content as **one-dimensional**.

Therefore, the distinction marked by $[\delta](ca://s?q = \text{Explain}_{t h e r o l e o f d e l t a})$ cannot be of dimension less than $[1D](ca://s?q = \text{Explain}_{w h y 1 D i s m i n i m a l})$.

(ii) Not more than 1D.

Suppose, for contradiction, that a single non-trivial application of $[\delta](ca://s?q = \textit{Explain}_t \textit{he}_r \textit{ole}_o \textit{f}_d \textit{elta})$ marks more than one distinguishable relation. For this to amount to more than one — rather than a single relation under two names — the meta-language must be able to tell one relation apart from another. Doing so requires some structural resource that expresses a contrast between them: an orientation, an ordering, or a relational predicate fixing which is which.

We now ask whether such a resource is available at the stage where δ is applied.

If such a resource **is** available, then δ has marked not only the two relations but also the **arrangement** that keeps them apart. This arrangement requires an ordering that Axiom 4 (Non-Orientation) forbids oD from licensing, and a structural encoding that Axiom 5 (Minimal Information) denies oD the capacity to supply. Asserting that δ marks this arrangement in addition to the relations themselves would therefore assign oD more structural information than Axiom 5 permits.

If, instead, no such resource is available, then the two candidate relations lack any distinguishing predicate. By **Lemma 2.5** (Meta-Isomorphism) anything without a distinguishing predicate is meta-linguistically identified with what it cannot be distinguished from, and the two candidate relations collapse into one.

(iii) Uniqueness.

By (i) and (ii), any $\delta \neq \text{id}$ marks exactly the structure characterized as 1D. By Lemma 2.5, any two such results are meta-linguistically indistinguishable, hence identified as the same 1D.

Therefore the minimal non-trivial distinction relating oD and 1D is unique, and no smaller non-trivial distinction is available within this formalism. An explicit, concrete construction of this distinction is given in Appendix E. This theorem characterizes a relation between oD and 1D; it asserts nothing about priority, origination, or process between them (Section 1.7).

□

Theorem 3.4 (The Iterated Distinction Structure).

All nD structures stand in an iterated distinction relation to oD, each stage's distinguishing power grounded in the distinction-content already established at the prior stage (cf. Appendix B). This theorem characterizes a relational hierarchy among the nD structures; it asserts nothing about which stage is prior to, causes, or is more fundamental than another.

8. Corollaries (Immediate Consequences)

The corollaries below follow directly from Section 7's theorems, with no further argument required; they are recorded separately for ease of reference, not because each needs its own proof.

Corollary 4.1 (oD Cannot Contain Substructure).

Directly from indivisibility (Axiom 1).

Corollary 4.2 (oD Is the Only Zero-Tension State).

All higher dimensions, once distinguished, admit tension; oD alone does not (Theorem 3.1).

Corollary 4.3 (oD and the Existence of Structure).

Every structure stands in the iterated distinction relation to oD established in Theorem 3.4; none lacks this relation.

Corollary 4.4 (oD Is the Absolute Folding Fixed Point).

No folding operator can reduce oD (Lemma 2.4).

- Summary

oD is definable without circularity, axiomatizable in a small mutually consistent set of axioms (Axioms 0–5), and uniquely determined (Theorem 3.2). The minimal distinction relating oD to a one-dimensional structure is likewise unique (Theorem 3.3), established by two independent arguments — one meta-linguistic, one object-linguistic — that converge on the same conclusion. Part II takes up that one-dimensional structure’s own internal anatomy.

PART II — THE ONE-DIMENSIONAL STRUCTURE (1D)

Section 9. Internal anatomy of 1D and the threshold condition for 2D.

9. The One-Dimensional State and the Threshold to Two Dimensions

Theorem 3.3 and Appendix E characterize:

$$[1D](ca://s? q = Explain_1 D_{def}inition)$$

meta-linguistically, as the unique structure marked by a single distinction relative to:

$$[0D](ca://s? q = Explain_0 D_{def}inition)$$

This section complements that treatment in three steps. First, it characterizes the internal anatomy of 1D directly. Second, it establishes — through an independent route — a conclusion compatible with Theorem 3.3(ii): that no one-dimensional structure admits a non-trivial coordinatizing partition, the structural prerequisite for hosting more than one degree of freedom. Third, it identifies the unique minimal condition under which that limitation is lifted, giving an interface-level characterization of:

$$[2D](ca://s? q = Explain_2 D_{def}inition)$$

0D's axioms (0–5) govern an entity with no internal content of its own, requiring the meta-language/object-language separation articulated in Appendix A.0. The structure characterized in this section is different: it has genuine internal content — a state space and a relation on it. This provides an object-language realization of the “exactly one degree of freedom” already established abstractly in Definition E.4. No meta-language/object-language tension arises here, since the state space and relation introduced below are not devices for describing 1D from outside; they constitute what 1D internally is.

9.1 The Internal Structure of 1D

Axiom 6 (Binary Differentiation).

A one-dimensional structure consists of a state space S together with a differentiation relation Δ on S taking exactly two values: absence of difference, and a single, undifferentiated mark of difference. No finer gradation, type, or magnitude of difference is admitted: $\Delta(x, y)$ either does not hold, or holds without further qualification.

Axiom 7 (Universal Non-Triviality).

For all $x, y \in S$ with $x \neq y$, $\Delta(x, y)$ holds. No two distinct elements of S are exempt from the differentiation relation.

Axioms 6–7 jointly fix Δ completely: $\Delta(x, y)$ holds exactly when $x \neq y$. Nothing about S itself is fixed by these axioms — not its cardinality, nor any further structure upon it. S may be any set with at least two elements. This is intentional: the axioms characterize the differentiation relation, not the particular set it differentiates. (See Appendix G for several worked examples at different finite cardinalities, all instantiating the same structure in the sense made precise below.)

9.2 Forced Maximal Symmetry

Proposition 1.5 (Forced Symmetry).

Let (S, Δ) satisfy Axioms 6–7, with S any set such that $|S| \geq 2$. Then $\text{Aut}(S, \Delta)$, the group of bijections of S preserving Δ , is the full symmetric group on S .

Proof.

By Axioms 6–7, $\Delta(x, y)$ holds exactly when $x \neq y$. Any bijection $\sigma: S \rightarrow S$ preserves the relation $x \neq y$ by definition of being a bijection ($\sigma(x) = \sigma(y)$ if and only if $x = y$). Hence every bijection of S preserves Δ , and $\text{Aut}(S, \Delta)$ is the full symmetric group on S . □

This holds regardless of the cardinality of S . Different choices of S satisfying Axioms 6–7 are not competing characterizations of 1D; by an extension of Lemma 2.5’s meta-isomorphism to this setting, any two such structures are indistinguishable by any predicate available in this paper’s vocabulary, and are identified as the same 1D.

9.3 No Nontrivial Coordinatization of 1D

Definition 0.7 (Coordinatizing Partition).

*Let (S, Δ) be a structure. A **coordinatizing partition** of S compatible with*

$\text{Aut}(S, \Delta)$ ($\text{Aut}(S, \Delta)$)

is a partition of S into blocks — neither the discrete partition into singletons nor the indiscrete partition $\{S\}$ — such that every automorphism in $\text{Aut}(S, \Delta)$ maps each block onto some block of the same partition. In other words, the partition is $\text{Aut}(S, \Delta)$ -invariant.

*A structure (S, Δ) admits a **non-trivial product decomposition***

$$S \cong S_1 \times S_2, |S_1|, |S_2| \geq 2$$

*Compatible with $\text{Aut}(S, \Delta)$ exactly when the fibers of one of the coordinate projections form such a coordinatizing partition. This is the standard notion of a **system of blocks** in permutation group theory, where the blocks correspond to the orbits of the action induced by fixing one coordinate.*

Lemma 2.6 (Universal Primitivity of 2-Transitive Actions).

Let G act 2-transitively on a set S with $|S| \geq 2$ (S of any cardinality, finite or infinite). Then the action admits no coordinatizing partition (Definition 0.7): G is primitive on S .

Proof.

For finite S this is classical [11, Ch. 7]; the following argument is cardinality-independent and covers the infinite case as well [cf. 12, Ch. 1–2].

Suppose toward contradiction that a coordinatizing partition exists. Let B be a block containing at least two elements x, y with $x \neq y$, and let $z \notin B$ (such a z exists because B is not all of S). Since the action is 2-transitive, there exists $g \in G$ with

$$g(x) = x, g(y) = z$$

Because $x \in B$ and $g(x) = x$, we have $x \in g(B)$, so $B \cap g(B) \neq \emptyset$. By the defining property of a block system, any two blocks with non-empty intersection must coincide; hence $g(B) = B$. But $y \in B$, so $g(y) \in g(B) = B$. Since $g(y) = z$, this implies $z \in B$, contradicting $z \notin B$.

□

Theorem 3.5 (No Nontrivial Coordinatization of 1D).

Let (S, Δ) satisfy Axioms 6–7. Then (S, Δ) admits no coordinatizing partition compatible with $\text{Aut}(S, \Delta)$ (Definition 0.7); equivalently, no nontrivial product decomposition $S \cong S_1 \times S_2$ compatible with $\text{Aut}(S, \Delta)$ exists.

Proof.

By Proposition 1.5, $\text{Aut}(S, \Delta)$ is the full symmetric group on S , which acts 2-transitively on S for any $|S| \geq 2$ [11]. By Lemma 2.6, this action is primitive: it admits no coordinatizing partition. Hence no nontrivial product decomposition compatible with $\text{Aut}(S, \Delta)$ exists, by the equivalence given in Definition 0.7.

□

This conclusion is reached entirely within the object-language structure of (S, Δ) , independently of the meta-linguistic argument given for Theorem 3.3(ii). The two arguments use different formal apparatus — one meta-linguistic, one grounded in permutation group theory — and arrive at compatible conclusions about the same structure: a structure satisfying the minimal non-triviality required of 1D admits no internal coordinate structure of the kind that would be needed to host more than one degree of freedom. Neither argument depends on the other; their agreement is a consistency check, not a restatement.

(A note on an earlier, replaced formulation: an earlier version of this result was stated in terms of "independent distinguishing operators." On closer examination, this was not well-posed as stated. Disjoint transpositions in a symmetric group, for instance, satisfy a naive independence test without producing any invariant partition. It has been replaced by the coordinatization formulation given here, which is not vulnerable to that issue and ties directly to the standard notion of primitivity.)

9.4 The Threshold to 2D

Axioms 6–7 are not the only way to satisfy non-triviality (some difference exists) while excluding the structureless case (oD). Axiom 6 in particular — that Δ takes only two values, with no further type or gradation of difference — is a substantive restriction, and it is this restriction that forces $\text{Aut}(S, \Delta)$ to be maximal in Proposition 1.5.

Theorem 3.6 (Threshold Theorem).

A structure (S, Δ) can admit a coordinatizing partition compatible with $\text{Aut}(S, \Delta)$ (Definition 0.7) only if Axiom 6 (Binary Differentiation) does not hold of it — that is, only if Δ admits at least two non-equivalent types of marked difference, rather than the single undifferentiated mark Axiom 6 requires. Relaxing Axiom 6 in this way is necessary. It is also non-vacuous: Appendix H exhibits an explicit structure satisfying this relaxed condition that genuinely admits such a coordinatizing partition. This establishes that the relaxation identified here is not merely permitted but realized by at least one structure; it does not establish that every structure satisfying the relaxed condition admits such a partition, a stronger claim this paper does not make.

Proof.

Step 1 — Identify the only candidates for relaxation.

Axioms 6–7 jointly and completely fix Δ (§9.1): there is no third axiom at this layer governing the differentiation relation, and S itself is already left maximally general — any set of at least two elements — so no constraint on S is available to relax. The only candidates for relaxation internal to the 1D layer are therefore Axiom 6 and Axiom 7 themselves, not because other axioms have been considered and rejected one by one, but because no other axiom in the system governs this layer's structure to begin with.

Axioms 0–5 are not candidates at all, for a different reason: they govern oD, a layer with no internal content of its own (§9, opening remarks), whereas the structure under discussion here — a state space together with a relation on it — has genuine internal content by hypothesis.

Relaxing an axiom that governs an entity with no internal content does not engage this layer's structure at all, so Axioms 0–5 are simply not among the axioms defining the class of structures in question.

Step 2 — Theorem 3.5 forces some relaxation, of Axiom 6, Axiom 7, or both.

By Theorem 3.5, any (S, Δ) satisfying Axioms 6–7 unmodified admits no coordinatizing partition. Admitting one therefore requires relaxing Axiom 6, Axiom 7, or both — by Step 1, there is nowhere else to look.

Step 3 — Rule out relaxing Axiom 7 as a route on its own.

Appendix G works through, for a finite illustrative case, both ways one might try to relax Axiom 7 instead of Axiom 6 while still aiming for a coordinatizing partition. Allowing some distinct pairs to satisfy $\Delta(x, y) = 0$ collapses those pairs into one state entirely (by Lemma 2.5), reducing the structure rather than enriching it — no partition to admit remains. Retaining $\Delta(x, y) = 1$ for all distinct pairs but quietly assigning different sub-types to different

pairs, without declaring this typing as a feature of Δ itself, turns out to just be Axiom 6's relaxation under a different name (Appendix G, §G.3) — not a way of avoiding it. Neither route yields a coordinatizing partition while leaving Axiom 6 intact.

Step 4 — Conclude: relaxing Axiom 6 is the only remaining route, hence necessary.

With Step 3 closing off Axiom 7's relaxation as a genuine alternative, and Step 1 limiting the search to Axioms 6–7 in the first place, the only way to block the proof of Proposition 1.5 — and hence the route to Theorem 3.5 — is for some bijection of S to fail to preserve Δ . With Axiom 7 (universal non-triviality) necessarily retained, this can only happen if Δ distinguishes among differences themselves: if Δ takes more than the two values Axiom 6 permits. This is exactly Axiom 6 failing, establishing necessity.

Appendix H establishes the non-vacuity claim by explicit construction, separately from the necessity argument above.

□

Definition 0.6 (2D (Interface Characterization Only)).

A two-dimensional structure is a structure (S, Δ) in which Δ admits at least two non-equivalent types of marked difference, and which admits a coordinatizing partition compatible with $\text{Aut}(S, \Delta)$ (Definition 0.7) — equivalently, a non-trivial product decomposition

$$S \cong S_1 \times S_2$$

compatible with $\text{Aut}(S, \Delta)$. This definition identifies the interface at which 2D becomes a candidate structure. Appendix H exhibits at least one structure satisfying it, and Section 12 characterizes this structure fully at the smallest non-trivial finite cardinality ($|S| = 4$), including a 2D analogue of Proposition 1.5 (Theorem 3.7) and a 2D analogue of Theorem 3.5 (Theorem 3.8).

This paper does not characterize the internal anatomy of 2D at arbitrary (larger or infinite) cardinality, and does not address the threshold from 2D to 3D. These topics are deferred to subsequent work under the same rigor established here (Section 1.7).

This theorem and definition are statements of logical conditionality. They identify what must be relaxed for a structure to admit a coordinatizing partition — the structural prerequisite for hosting more than one degree of freedom. They make no claim about temporal, causal, or ontological priority between 1D and 2D (Section 1.7).

This relation is symmetric in its logical content, not merely in its presentation. "A coordinatizing partition requires relaxing Axiom 6" and its contrapositive, "retaining Axiom 6 precludes a coordinatizing partition" (the content of Theorem 3.5), are the same relation viewed from either side; neither is prior to the other.

The sequence in which this paper presents 0D, 1D, and 2D reflects only the order in which each result’s proof draws on results already established. (Theorem 3.6’s proof, for instance, depends on Proposition 1.5 and Lemma 2.5 from the 1D development.) It does not assert that the relation between 1D and 2D itself has a preferred direction.

The table below summarizes the technical handoff between Section 9 and Section 10, for reference. Every entry restates a fact already established in the surrounding text; the table adds no new claim.

Layer	Forcing axioms	$\text{rank}(\text{Aut}(\mathbf{S}, \Delta))$	Forced group	Tool that excludes alternatives
1D	Axioms 6–7	2	Sym(S) (Prop. 1.5)	2-transitivity / primitivity (Lemma 2.6, Thm 3.5)
2D, $ \mathbf{S} =4$	Axioms 8–9	3	D ₄ (Thm 3.7)	Direct exclusion of C ₄ , V ₄ (Lemma 2.7); Schur ring dimension count (Thm 3.8)

The pattern in the last column is not a stylistic choice repeated by coincidence. As §10.4 argues in detail below, the rank forced by each layer’s axioms determines which branch of permutation group theory applies to exclude alternatives — 2-transitivity at rank 2, Schur rings at rank 3. This is restated as a general principle, not merely illustrated by these two rows, in the discussion following Theorem 3.8.

- Summary

A state space with a single binary differentiation relation (Axioms 6–7) is forced to maximal symmetry (Proposition 1.5), confirmed independently by a meta-linguistic and an object-linguistic argument (Theorem 3.6), and admits no nontrivial coordinatizing partition. Relaxing Axiom 6 to permit a second type of difference is the unique minimal change under which this limitation no longer applies (Theorem 3.6) — the threshold Part III characterizes.

PART III — THE TWO-DIMENSIONAL STRUCTURE (2D)

Section 10. Internal anatomy of 2D at the smallest non-trivial finite cardinality.

10. The Internal Structure of 2D

Theorem 3.6 and Appendix H establish that a coordinatizing partition is non-vacuously realizable once Axiom 6 is relaxed, but characterize only a single witness structure. This section gives a self-contained axiomatization of 2D's own internal anatomy, continuing directly from Axioms 6–7 in the same spirit in which Section 9 characterized 1D. It shows that, at the smallest non-trivial finite cardinality, the witness structure of Appendix H is not merely an example but the unique structure satisfying the axioms.

The meta-language/object-language question already settled in Section 9.1 for 1D applies unchanged here and is not revisited as a fresh issue. Axioms 8–9 govern the same kind of object as Axioms 6–7: a state space and a relation on it, constituting genuine internal content rather than a device for describing the structure from outside. They differ only in how many types the relation Δ is permitted to take, and in what symmetry properties $\text{Aut}(S, \Delta)$ is required to have. No new instance of the Appendix A.o discipline is required at this layer.

10.1 Symmetric Typed Differentiation

Axiom 8 (Typing of Difference (Symmetric Form)).

*A structure (S, Δ) satisfies **Typing of Difference** if $\Delta: S \times S \rightarrow \Sigma \cup \{0\}$ for some set Σ of types, and the following conditions hold:*

1. **Multiplicity of types.** *The set Σ contains at least two distinct elements σ_1, σ_2 .*
2. **Non-degeneracy at each point.** *For every $x \in S$, both*

$$\Gamma_1(x) := \{y \in S : \Delta(x, y) = \sigma_1\}, \Gamma_2(x) := \{y \in S : \Delta(x, y) = \sigma_2\}$$

are nonempty. This ensures that each point witnesses at least two different types of differences.

3. **Symmetry of typing.** *For all $x, y \in S$,*

$$\Delta(x, y) = \Delta(y, x)$$

*This symmetry is a requirement on the **assigned type**, not an additional structure on S itself. It states that the type of the difference between x and y does not depend on which element is taken as the reference.*

Clause (iii) is essential and is not a stylistic preference. Without it, a structure could smuggle in an orientation through the back door of type assignment, rather than through any explicit directional primitive — for instance, marking σ_1 for "clockwise neighbor" and σ_2 for "counterclockwise neighbor" relative to some

labeling. Such an assignment can be made consistent with a proper subgroup of a larger automorphism group, but only by treating x and y asymmetrically. This is exactly the kind of directional structure Axioms 0–7 were built, over many rounds of revision, to exclude. Clause (iii) closes this off: the type of a difference is a property of the unordered pair $\{x,y\}$, not of an ordered reference.

Axiom 9 (Homogeneous Anisotropy).

A structure (S, Δ) satisfying Axiom 8 satisfies Homogeneous Anisotropy if $\text{Aut}(S, \Delta)$ is transitive on S but not 2-transitive on S , and for every $x \in S$, the point stabilizer $G_x := \{g \in \text{Aut}(S, \Delta) : g(x) = x\}$ acts transitively on $\Gamma_1(x)$ and, independently, transitively on $\Gamma_2(x)$.

Axiom 9 is what distinguishes a genuine two-type structure from a relabeled one-type structure. Transitivity of G_x on each $\Gamma_i(x)$ separately ensures that within a fixed type, no point is structurally privileged over another. The failure of 2-transitivity on S as a whole ensures something different: that the two types are not merged into a single orbit — the situation Proposition 1.5 and Theorem 3.5 already show 1D is forced into.

10.2 Exclusion of the Cyclic and Klein Four-Group Cases at $|S|=4$

At the smallest cardinality admitting two nonempty, disjoint, non-trivial sets $\Gamma_1(x), \Gamma_2(x)$ for every x — namely $|S|=4$ — three candidate transitive permutation groups merit examination as the automorphism group of some (S, Δ) satisfying Axioms 8–9: the cyclic group C_4 , the Klein four-group V_4 , and the dihedral group D_4 (order 8). The witness structure of Appendix H realizes D_4 . The following lemma shows that C_4 and V_4 cannot be realized as the full automorphism group of any structure satisfying Axiom 8's symmetry clause.

Lemma 2.7 (Exclusion of C_4 and V_4 at $|S|=4$).

Let $|S|=4$. There is no (S, Δ) satisfying Axioms 8–9 with $\text{Aut}(S, \Delta) = C_4$ or $\text{Aut}(S, \Delta) = V_4$.

Proof.

Write $S = \{1,2,3,4\}$. The non-diagonal pairs of S split into three orbits under each of

$$C_4 = \langle (1234) \rangle \text{ and } V_4 = \{e, (12)(34), (13)(24), (14)(23)\}$$

Under Axiom 8(i)–(ii), these three orbits must be distributed across exactly two types σ_1, σ_2 , so two of the three orbits must receive the same type.

Case C_4 .

The three orbits under $\langle (1234) \rangle$ are

$$O_1 = \{(1,2), (2,3), (3,4), (4,1)\}, O_2 = \{(1,3), (2,4), (3,1), (4,2)\}, O_3 = \{(1,4), (2,1), (3,2), (4,3)\}.$$

By Axiom 8(iii), $\Delta(1,2) = \Delta(2,1)$. Since $(1,2) \in O_1$ and $(2,1) \in O_3$, this forces O_1 and O_3 to carry the same type — the symmetry clause leaves no freedom here, because O_1 and O_3 are exactly the orbit and its reverse. Assign σ_1 to $O_1 \cup O_3$ and σ_2 to O_2 ; up to relabeling $\sigma_1 \leftrightarrow \sigma_2$, this is the only assignment satisfying (i)–(iii).

Step — Verify (24) preserves Δ , on all twelve ordered pairs, not a sample of them.

The transposition (24) fixes 1 and 3 and swaps 2 and 4. Applied to all eight pairs of $O_1 \cup O_3$:

$$(1,2) \mapsto (1,4), (2,3) \mapsto (4,3), (3,4) \mapsto (3,2), (4,1) \mapsto (2,1), \\ (2,1) \mapsto (4,1), (3,2) \mapsto (3,4), (4,3) \mapsto (2,3), (1,4) \mapsto (1,2),$$

and every image again lies in $O_1 \cup O_3$. Thus (24) maps this 8-element type- σ_1 set to itself.

Applied to all four pairs of O_2 :

$$(1,3) \mapsto (1,3), (3,1) \mapsto (3,1)$$

(both fixed, since 1 and 3 are fixed by (24)), while

$$(2,4) \mapsto (4,2), (4,2) \mapsto (2,4)$$

(swapped within the set). Every image again lies in O_2 , so (24) maps this 4-element type- σ_2 set to itself. Together these eight-plus-four images account for all twelve ordered pairs, so (24) preserves Δ entirely, not merely on the handful of pairs checked above as illustration.

But (24) $\notin C_4$: the non-identity elements of C_4 are the 4-cycle (1234), the double-transposition (13)(24), and the inverse of (1234), namely (1432); none of these is the single transposition (24). Hence $\text{Aut}(S, \Delta)$ properly contains C_4 , so C_4 is not the full automorphism group.

Case V4.

$$O_4 = \{(1,2), (2,1), (3,4), (4,3)\}, O_5 = \{(1,3), (3,1), (2,4), (4,2)\}, O_6 = \{(1,4), (4,1), (2,3), (3,2)\}.$$

Each orbit is already symmetric under reversal, so Axiom 8(iii) imposes no forced merger here — unlike the C_4 case. By the pigeonhole principle, two of the three orbits must still share a type under (i)–(ii). Assign, without loss of generality, σ_1 to $O_4 \cup O_6$ and σ_2 to O_5 . Then

$$\Gamma_1(1) = \{2,4\}, \Gamma_2(1) = \{3\}.$$

Step — Again verify (24) preserves Δ on all twelve ordered pairs.

Applied to all eight pairs of $O_4 \cup O_6$:

$$(1,2) \mapsto (1,4), (2,1) \mapsto (4,1), (3,4) \mapsto (3,2), (4,3) \mapsto (2,3), \\ (1,4) \mapsto (1,2), (4,1) \mapsto (2,1), (3,2) \mapsto (3,4), (2,3) \mapsto (4,3),$$

and every image again lies in $O_4 \cup O_6$. Thus (24) maps this 8-element type- σ_1 set to itself.

Applied to all four pairs of O_5 :

$$(1,3) \mapsto (1,3), (3,1) \mapsto (3,1)$$

(both fixed), while

$$(2,4) \mapsto (4,2), (4,2) \mapsto (2,4)$$

(swapped within the set). Every image again lies in O_5 , so (24) maps this 4-element type- σ_2 set to itself. As before, all twelve ordered pairs are accounted for, so (24) preserves Δ entirely.

But (24) $\notin V_4$: every non-identity element of V_4 is a product of two disjoint transpositions, whereas (24) is a single transposition. Hence $\text{Aut}(S, \Delta)$ properly contains V_4 .

In both cases, any type-assignment satisfying Axiom 8 on the C_4 - or V_4 -orbit structure forces an additional automorphism not belonging to the candidate group, so neither C_4 nor V_4 can be the full (maximal) automorphism group of any (S, Δ) at $|S| = 4$.

□

10.3 The Forced Structure at $|S|=4$

Theorem 3.7 (Forced Dihedral Symmetry).

Let (S, Δ) satisfy Axioms 8–9 with $|S|=4$. Then $\text{Aut}(S, \Delta) \cong D_4$, and $\text{rank}(\text{Aut}(S, \Delta)) = 3$ (three orbits on $S \times S$: the diagonal, the σ_1 -pairs, and the σ_2 -pairs).

Proof.

By Axiom 9, G_x has a non-trivial orbit on $\Gamma_1(x)$ and a non-trivial orbit on $\Gamma_2(x)$, so $\text{Aut}(S, \Delta)$ has at least three orbits on $S \times S$: the diagonal, together with at least one orbit refining Γ_1 and at least one refining Γ_2 . By Axiom 8(i)–(ii), there are exactly two non-diagonal types, so there are exactly three orbits on $S \times S$ in total; hence

$$\text{rank}(\text{Aut}(S, \Delta)) = 3$$

By Axiom 9, $\text{Aut}(S, \Delta)$ is transitive but not 2-transitive on S . By Lemma 2.7, it is neither C_4 nor V_4 . Among the transitive permutation groups on four points —

$$C_4, V_4, D_4, A_4, S_4,$$

the latter two are 2-transitive (S_4 trivially; A_4 is 2-transitive on four points by direct inspection of its order-12 action). Both are excluded by the non-2-transitivity clause of Axiom 9. The only remaining candidate is D_4 . Appendix H exhibits a structure realizing it, so D_4 is both necessary and achieved.

□

$\text{rank}(\text{Aut}(S, \Delta)) = 3$ is therefore not an additional hypothesis. It is a direct consequence of Axioms 8–9, in exactly the sense that forced maximal symmetry (Proposition 1.5) was a direct consequence of Axioms 6–7. The same discipline — deriving structural invariants from the axioms, rather than assuming them — is maintained at this layer.

10.4 No Third Independent Difference Type

Theorem 3.8 (No Third Type at $|S|=4$).

Let (S, Δ) satisfy Axioms 8–9 with $|S|=4$, so that $\text{Aut}(S, \Delta) \cong D_4$ with rank 3 (Theorem 3.7). Then Σ cannot be extended by a third type σ_3 , distinguishable from σ_1 and σ_2 , compatible with $\text{Aut}(S, \Delta)$.

Proof.

Step 1 — The Schur ring has dimension exactly 3.

The orbits of $\text{Aut}(S, \Delta)$ on $S \times S$ correspond to the basis elements of the Schur ring (the adjacency algebra spanned by the orbital matrices) generated by D_4 's action [3].

A structure of rank r has a Schur ring of dimension r : the identity matrix (the diagonal orbit) together with $r-1$ further linearly independent adjacency matrices, one per non-diagonal orbit.

By Theorem 3.7, $\text{rank}(\text{Aut}(S, \Delta)) = 3$, so this algebra has dimension exactly 3, spanned by the identity matrix I and the two adjacency matrices M_1, M_2 corresponding to σ_1, σ_2 .

Step 2 — A third type would need a fourth independent matrix, and there is no room for one.

A candidate third type σ_3 , distinguishable from σ_1 and σ_2 and compatible with $\text{Aut}(S, \Delta)$, would correspond to a further adjacency matrix M_3 in this algebra, linearly independent from I, M_1, M_2 . But the algebra has dimension exactly 3, so any matrix in it — including any candidate M_3 — is a linear combination of I, M_1, M_2 .

Hence M_3 cannot be independent of M_1 and M_2 : this is a direct consequence of dimension counting in the Schur ring, not an appeal to any result established elsewhere in this paper. Any pair this candidate type would distinguish is already determined, as a matter of which combination of σ_1 - and σ_2 -adjacency it falls under, by the existing two-type structure.

Hence no third independent type can be added to Σ while remaining compatible with $\text{Aut}(S, \Delta) \cong D_4$.

□

This theorem plays the same role at the 2D/3D boundary that Theorem 3.5 plays at the 1D/2D boundary. It identifies an upper bound on the structure a fixed automorphism group can support, using a tool appropriate to the rank already forced — the Schur ring, via [3] — rather than the tool appropriate to 1D's rank-2 primitive case (2-transitivity and Lemma 2.6). As with Theorem 3.5 and Theorem 3.3(ii), this is offered as a relation, not a claim about which dimension is prior to or gives rise to another. (Section 1.7; see also the bidirectionality remark at the end of Section 9.4, which applies equally here.)

The change of tool across these two boundaries is not an ad hoc choice repeated by coincidence: it follows the rank of $\text{Aut}(S, \Delta)$ actually forced at each layer. Rank 2 (1D, Proposition 1.5) is precisely the regime classically governed by 2-transitivity and primitivity ([11, 12]). Rank 3 (2D, Theorem 3.7) is precisely the regime classically governed by Schur rings and rank-3 association schemes ([3]).

The axioms at each layer determine the rank. The rank, by an existing stratification within permutation group theory rather than by any choice made in this paper, determines which

apparatus applies. This gives a principled rather than coincidental account of why the proof technique changes.

It also yields a concrete, checkable expectation for future work, not a license to assume the pattern: whatever rank 3D's automorphism group is forced to have, once 3D's own axioms are constructed, will determine which branch of the theory is relevant there. This is to be verified against 3D's actual forced rank, not assumed in advance. (See the Speculative Items in the accompanying sequence note.)

11. Clarifying 2D's Relation to the General-n Framework

Sections 14–15 develop a general-n axiomatization (Axioms 10–12') covering every $n \geq 2$ at once, including a sufficiency theorem (Theorem 3.15) stated for $n \geq 2$ without restriction. Since Section 10's own 2D axiomatization (Axioms 8–9) was developed independently, before the general-n framework existed, this section makes explicit how the two relate at $n = 2$ — a relation that was implicit throughout Sections 11–12 but never stated as its own result.

11.1 The Exact Sizes of $\Gamma_1(x)$ and $\Gamma_2(x)$ at $|S|=4$

At $|S| = 4$, the two non-diagonal orbits forced by Theorem 3.7 do not have equal size. From the computation in Lemma 2.7,

$$\Gamma_1(1) = \{2, 4\}(\text{size } 2), \Gamma_2(1) = \{3\}(\text{size } 1).$$

By Axiom 9, the action is transitive, so every point has Γ_1 and Γ_2 of these same two sizes.

This is stated explicitly because it is easy to misstate — for example, by treating both as singletons. An earlier version of Section 14.1 did exactly this before the inconsistency with Lemma 2.7's computation was caught and corrected.

The sizes 2 and 1 are not arbitrary: they match the orbit sizes of D_4 's own natural degree-4 action, restricted to the three points other than a fixed one — a split into a size-2 orbit and a size-1 orbit that holds for this action of D_4 on any 4-point set, independent of any particular coordinatization. $\Gamma_1(x)$, under this correspondence, is the size-2 orbit and $\Gamma_2(x)$ is the singleton orbit.

11.2 Why 2D Never Needed a Saturation Axiom

Sections 14–15 introduce Axiom 12' specifically because, at $n \geq 3$, transitivity of the point stabilizer G_x on $\Gamma_1(x)$ (Axiom 9's own requirement) is strictly weaker than G_x inducing the full symmetric group on $\Gamma_1(x)$:

$\text{Sym}(3)$ has a proper nontrivial transitive subgroup, the cyclic group C_3 , so transitivity alone leaves room for a smaller candidate

This is exactly the source of the two order-24 alternatives Lemma 2.8 excludes at $n=3$. Axiom 12' is needed to close this gap at every $n \geq 3$.

At $n = 2$, this gap does not exist, for a reason that is now precise rather than a loose appeal to small size. $\Gamma_1(x)$ has size 2 (§11.1), and $\text{Sym}(2)$ — having prime order 2— has no proper non-trivial subgroup at all, by Lagrange's theorem. Every transitive action on a 2-element set is therefore automatically the full symmetric action on it. Axiom 9's transitivity requirement, with no further axiom, already forces everything Axiom 12' would separately impose at higher n .

This is why Section 12's original axiomatization (Axioms 8–9 alone) was sufficient at $|S| = 4$, and why no analogue of Axiom 12' was ever missing from it — not because 2D is exempt from the underlying requirement, but because the requirement is automatically satisfied there.

11.3 $B_2 \cong D_4$: The General- n Machinery Already Contains the 2D Result

Axioms 10–11, specialized to $n = 2$, read identically to Axioms 8–9 (this identification is already noted in Section 14.3). Theorem 3.15 (general- n sufficiency) is stated for $n \geq 2$ and concludes

$$\text{Aut}(S, \Delta) \cong B_n$$

At $n = 2$,

$$B_2 = (\mathbb{Z}_2)^2 \rtimes S_2$$

has order 8. Direct computation of its element-order profile —

$$\{1:\text{one element}, 2:\text{five elements}, 4:\text{two elements}\}$$

— matches D_4 's profile exactly, confirming

$$B_2 \cong D_4$$

The proof of Theorem 3.15 (§12.5.1) goes through unchanged at $n = 2$. Lemma 2.12 (no internal type-splitting in $\Gamma_1(x)$) holds at every $n \geq 2$ by its own general statement, and Theorem 3.9's derivation of full local symmetry from Axiom 12' does not treat $n = 2$ as a special or excluded case. Theorem 3.15 therefore already establishes — as a special case of a result proved for entirely different reasons in Section 14 — exactly what Theorem 3.7 established directly for 2D:

$$\text{Aut}(S, \Delta) \cong D_4 (\cong B_2) \text{ at } |S| = 4$$

The two proofs are independent — one via exhaustive exclusion of C_4 and V_4 (Lemma 2.7), the other via the general counting argument of Theorem 3.9 — and their agreement is a consistency check of the same kind already noted between Theorem 3.3 and Theorem 3.5 (§9.3), not a restatement of either.

This does not make Section 10 redundant. Theorem 3.8 (no third independent type at $|S|=4$, via the rank-3 Schur ring) has no stated general- n counterpart until Theorem 3.17 in Section 13, and Section 10's direct, elementary treatment at the smallest case remains the most accessible route to the $|S|=4$ result on its own terms.

11.4 Beyond $|S|=4$: A Classification Conjecture, Not a Uniqueness Theorem

Section 12 characterizes 2D only at $|S|=4$. A natural question is whether Axioms 8–9 force a unique automorphism group at every larger cardinality $|S|=2m$, in the same way that Axioms 6–7 do at every cardinality for 1D (Proposition 1.5). They do not. This was checked directly, exhaustively, by computer: at $|S|=6$, two non-isomorphic structures satisfy Axioms 8–9, not one. The same holds at $|S|=8$. Full details, including the verification method and how to reproduce it independently, are given in Appendix L.

The two structures found at each cardinality are not arbitrary; they have clean, identifiable descriptions. One is the hyperoctahedral group B_m acting on the $2m$ signed coordinate directions

$$\{\pm e_1, \dots, \pm e_m\}$$

The same family already used at 3D and 4D (Sections 14–15), but in a smaller, rank-3 representation rather than the rank- $(m+1)$ Hamming-cube representation used there. The other is the automorphism group of the complete bipartite graph $K_{m,m}$. At $|S|=4$ ($m=2$), these two families coincide — $B_2 \cong D_4$, and $K_{2,2}$'s automorphism group is also D_4 . This is exactly why Section 12 found only one structure: the two families are indistinguishable at the smallest case, not because there is genuinely one family rather than two.

Conjecture (stated as a conjecture, not a theorem; verified only at $m=2,3,4$, not proved for general m): for every $m \geq 2$, the structures satisfying Axioms 8–9 at $|S|=2m$ are exactly two, up to isomorphism — the signed-axis representation of B_m and the automorphism group of $K_{m,m}$ — and no third family exists at any m . This is offered as a precisely-stated open question for future work, in the same spirit as the E_n/P_n exhaustiveness question already left open in Section 13; it is not claimed to be established here.

- Summary

At the smallest non-trivial cardinality, Axioms 8–9 uniquely force D_4 , with a third independent type ruled out by a Schur-ring dimension count (Theorems 3.7–3.8). This uniqueness does not generalize: at $|S|=6$ and $|S|=8$ it is replaced by exactly two non-isomorphic structures, a finding this paper did not anticipate and records as a conjecture, not a theorem (§11.4, Appendix L). Part IV develops the general- n machinery within which this same question is resolved differently — by a saturation axiom rather than by cardinality-by-cardinality classification.

PART IV — THE THREE-DIMENSIONAL STRUCTURE (3D)

Section 12. Internal anatomy of 3D, extending Axioms 8–9 to general- n form and forcing the hyperoctahedral group B_n at $n=3$.

12. The Internal Structure of 3D

Theorem 3.10 (below) is confirmed in two independent ways: by an exhaustive computational verification at the $\{0, 1\}^3$ witness model (Section 14.3), and unconditionally by the general- n sufficiency theorem proved later in this section — *Theorem 3.15* (Section 14.5.1) — which does not depend on exhaustive search at all. The technical condition previously stated as an independent “Axiom 12” has been derived (*Theorem 3.9*) from a more primitive saturation principle (*Axiom 12'*), in the same style as $\text{rank}(G) = 3$ was derived rather than assumed at 2D.

A specific alternative shortcut — excluding the even-parity subgroup directly via the already-proven 2D *Lemma 2.7*, without any new 3D axiom — was proposed, checked, and found to rest on a category error; this is recorded below so it is not re-attempted. The 3D analogue of *Theorem 3.8* (no fourth independent type) is established below as *Theorem 3.11*. What remains genuinely open is the analogous statement for arbitrary cardinality (not just $|S| = 8$); this has not been attempted.

12.1 Why the Direct Generalization of Axioms 8–9 Is Not Enough

At 2D, At 2D, $|S| = 4$ and $\Gamma_1(x), \Gamma_2(x)$ have sizes 2 and 1 respectively (computed directly in *Lemma 2.7*: $\Gamma_1(1) = \{2, 4\}, \Gamma_2(1) = \{3\}$). For sets of size 1 or 2, *Axiom 9*’s requirement that the point stabilizer act transitively on each $\Gamma_i(x)$ is therefore automatically maximal: $\text{Sym}(1)$ and $\text{Sym}(2)$ each have no proper non-trivial subgroup (the latter by Lagrange’s theorem, since 2 is prime). A transitive action on a set of size 1 or 2 is automatically the full symmetric action on it — there was no room for a weaker candidate to slip through.

This is the first point at which that coincidence breaks: the natural 3D witness model has $\Gamma_1(x)$ of size 3, and $\text{Sym}(3)$ does have a proper non-trivial transitive subgroup — the cyclic group C_3 — so transitivity on a 3-element set is strictly weaker than the point stabilizer acting as the full symmetric group on that set. A direct hand computation (below) exhibits two distinct candidate automorphism groups of order 24, in addition to the order-48 target, on the same 8-point structure — all transitive, all of *rank 4*, all satisfying the direct analogue of “stabilizer transitive on each $\Gamma_i(x)$.” The 2D axioms, generalized in the most obvious way, do not by themselves distinguish them.

12.2 The Witness Model: $\{0,1\}^3$ Under Hamming Distance

Let $S = \{0,1\}^3$, the set of all length-3 binary strings. Define $\Delta(x,y)$ for $x \neq y$ as the Hamming distance between x and y — the number of coordinate positions in which they differ — taking values in $\{1,2,3\}$, and $\Delta(x,x)=0$. This gives $\Delta: S \times S \rightarrow \{0,1,2,3\}$, automatically symmetric (Hamming distance does not depend on order), with three non-trivial types $\sigma_1, \sigma_2, \sigma_3$ corresponding to distance 1, distance 2, and distance 3.

For any fixed x ,

$$|\Gamma_1(x)| = 3, |\Gamma_2(x)| = 3, |\Gamma_3(x)| = 1$$

(exactly one point of S is at distance 3 from x). This asymmetry in the sizes of the three $\Gamma_i(x)$ is itself worth flagging: unlike 2D, where $\Gamma_1(x)$ and $\Gamma_2(x)$ were the same size by construction, this *witness model's* natural *type structure* is not symmetric across types in this sense. It is not yet established whether this asymmetry should be required by an axiom or is simply a feature of this particular model that need not generalize.

Any bijection of S preserving Δ (equivalently, preserving *Hamming distance* for every pair, not just every distance-1 pair) is exactly an automorphism of the hypercube graph Q_3 . A graph automorphism preserves graph distance between every pair of points by definition, and Hamming distance on $\{0,1\}^3$ coincides with graph distance in Q_3 : any two points at Hamming distance d are joined by a path of exactly d edges (flipping the differing coordinates one at a time), and no shorter path exists since each edge changes Hamming weight from a fixed reference by exactly ± 1 . This argument is independent of n .

The automorphism group of Q_n , for every n , is the hyperoctahedral group

$$B_n = (\mathbb{Z}_2)^n \rtimes S_n$$

of order $2^n \cdot n!$ [1]. This justifies restricting the search for candidate automorphism groups of (S, Δ) to subgroups of B_3 , rather than all of $\text{Sym}(8)$: no permutation outside B_3 can preserve Δ in the first place.

12.2.1 General-n Axioms (Formalizing What 2D Implicitly Generalized)

Axioms 8–9, as stated for 2D, are written specifically for exactly two types; they are 2D-specific. The remainder of this section needs a precise statement of what "the natural generalization of Axioms 8–9" means at $n=3$ and beyond, rather than continuing to gesture at it informally. The following two axioms state this once, for every n , replacing all informal references below (and in any later section building on this one) to "the direct generalization of Axioms 8–9."

Axiom 10 (Typing of Difference (General n -Type Symmetric Form)).

*A structure (S, Δ) satisfies n -Typing of Difference** if*

$$\Delta: S \times S \rightarrow \Sigma \cup \{0\}$$

for some set Σ of types, and:

1. $[n \text{ distinct types}](ca://s? q = \text{Explain}_{distinct_types}) - \Sigma$ contains at least n distinct elements $\sigma_1, \dots, \sigma_n$.

2. $[\text{Nonempty } \Gamma_i(x)](ca://s? q = \text{Explain}_{\Gamma_i}) - \text{for every } x \in S, \text{ the sets}$

$$\Gamma_i(x) := \{y \in S: \Delta(x, y) = \sigma_i\}$$

are nonempty for each $i = 1, \dots, n$.

3. $[\text{Symmetry}](ca://s? q = \text{Explain}_{symmetric_difference_condition}) - \text{for all } x, y \in S,$

$$\Delta(x, y) = \Delta(y, x).$$

Axiom 11 (Homogeneous Anisotropy (General n-Type Form)).

A structure (S, Δ) satisfying Axiom 10 (with n types) satisfies **Homogeneous Anisotropy** if:

- $\text{Aut}(S, \Delta)$ is transitive on S but not 2-transitive on S ; and
- for every $x \in S$, the point stabilizer G_x acts transitively on each

$$\Gamma_i(x) = \{y \in S: \Delta(x, y) = \sigma_i\}, i = 1, \dots, n,$$

independently.

At $n=2$ these are exactly Axiom 8 and Axiom 9. The $\{0,1\}^3$ witness model of Section 12.2 satisfies Axiom 10 with $n=3$ (three nonempty types $\Gamma_1(x)$, $\Gamma_2(x)$, $\Gamma_3(x)$ for every x , as computed above) and Axiom 11 (B_3 is transitive, not 2-transitive — a 2-transitive group on 8 points would need to move any point-pair to any other, but the three distinct $\Gamma_i(x)$ sizes already show the stabilizer's orbits on $S \setminus \{x\}$ are not all of $S \setminus \{x\}$, ruling out 2-transitivity directly). Every occurrence of "Axioms 8–9" below, when applied to this 3D witness model or to general n , means Axiom 10–11 at the relevant n ; the label "Axioms 8–9" is retained only where the discussion is specifically about the 2D case itself.

12.3 Exhaustive Verification: Exactly Three Transitive Rank-4 Subgroups of B_3

All subgroups of B_3 generated by every single element, every pair of elements, and every triple of elements of B_3 were constructed explicitly and checked for transitivity and rank on $S \times S$. This is not a fully general proof that no subgroup requires four or more generators to produce (a theoretical possibility for an arbitrary finite group), but for a group of order 48 it is exhaustive in practice; the search was independently cross-checked by three separate methods with identical results each time (full methodology and element-order profiles in Appendix I), giving high confidence the list below is complete:

- B_3 itself, order 48.
- Two **Two distinct subgroups of order 24**. One is the index-2 subgroup of B_3 consisting of all elements of even combined parity (the combination of the coordinate-permutation's parity and the number of coordinate flips applied; this is the standard determinant-sign decomposition of B_3), isomorphic to S_4 , with element orders 1, 2, 2, 2, 2, 2, 2, 2, 2, 2, 3, 3, 3, 3, 3, 3, 4, 4, 4, 4, 4, 4.

The other is a genuinely different order-24 subgroup (verified non-identical by direct computation, and distinguishable by its *element-order profile*, which includes elements of order 6, absent from the even-parity subgroup).

No other transitive rank-4 subgroup of B_3 was found by this method.

Deriving the Saturation Condition: Axiom 12' and Theorem 3.9

Axiom 12' (Directional Internal Indistinguishability).

every $x \in S$ and every type σ_i , with

$$\Gamma_i(x) := \{y \in S : \text{type}(\Delta(x, y)) = \sigma_i\}$$

if π is a permutation of $\Gamma_i(x)$ such that, for all $y, z \in \Gamma_i(x)$,

$$\text{type}(\Delta(y, z)) = \text{type}(\Delta(\pi(y), \pi(z)))$$

i.e., π preserves all Δ -types among pairs internal to $\Gamma_i(x)$, then there exists a global automorphism $g \in \text{Aut}(S, \Delta)$ with $g(x) = x$ and g restricted to $\Gamma_i(x)$ equal to π .

Axiom 12' is stated entirely in terms of the type structure already given by Axioms 10–11: it does not stipulate a target group directly. The content is a saturation principle in the same family as Axiom 9's requirement that the stabilizer not be artificially restricted below what the type structure permits — if a rearrangement of $\Gamma_1(x)$ introduces no Δ -type distinction that was not already present, there is no structural reason for the meta-language to forbid it from being realized as a genuine symmetry.

Theorem 3.9 (Directional Indistinguishability Forces Full Local Symmetry).

Let (S, Δ) be a finite structure satisfying Axioms 10–11 and Axiom 12', and suppose that for some x and type σ_1 , every pair of distinct points within $\Gamma_1(x)$ carries the same Δ -type (i.e., $\Gamma_1(x)$ has no internal type-splitting). Then the point stabilizer G_x induces the full symmetric group on $\Gamma_1(x)$.

Proof.

Since every pair within $\Gamma_1(x)$ carries the same type, every permutation π of $\Gamma_1(x)$ trivially preserves all Δ -types among its pairs (there is only one type present, so no permutation can turn it into a different one). By Axiom 12', every such π therefore extends to a global automorphism fixing x . Hence

$$\text{Sym}(\Gamma_1(x)) \subseteq G_x \upharpoonright_{\Gamma_1(x)}.$$

The reverse inclusion

$$G_x \upharpoonright_{\Gamma_1(x)} \subseteq \text{Sym}(\Gamma_1(x))$$

holds trivially (any induced action is by definition a subgroup of the full symmetric group on the set acted on). Hence

$$G_x|_{\Gamma_1(x)} = \text{Sym}(\Gamma_1(x)).$$

□

On the $\{0,1\}^3$ witness model, the hypothesis is verified directly: $\Gamma_1(x)$ (the three distance-1 neighbors of x) are pairwise at Hamming distance 2 from each other, so all three pairs carry the same type σ_2 , with no internal splitting. Theorem 3.9 therefore applies, and the technical condition previously stated as an independent axiom (full symmetric action on $\Gamma_1(x)$) is now a derived consequence of Axioms 10–12', in exactly the sense that $\text{rank}(G)=3$ was a derived consequence of Axioms 8–9 at 2D rather than a separately assumed condition. We retain the label "Axiom 12" below only as a name for this now-derived condition, for continuity with Lemma 2.8 and Theorem 3.10's statements; it is a Corollary of Theorem 3.9, not an axiom in its own right.

A Rejected Alternative: Why Restriction to a Coordinate-Fixed Subset via Lemma 2.7 Does Not Work

A tempting shortcut was proposed and checked: fix one coordinate to a single value, giving a 4-point subset of S (there are six such subsets: 3 coordinates $\times$ 2 values). The setwise stabilizer of one such subset, within the even-parity subgroup of Section 12.3, has order $24/6=4$ (the even-parity subgroup is transitive on the six coordinate-fixed subsets, by an orbit-stabilizer count) and induces a cyclic group of order 4 on that subset (verified directly: the induced permutations have element orders 1,2,4,4, the signature of C_4). One might hope this directly contradicts Lemma 2.7 (which excludes C_4 as the full automorphism group of any 2D structure at $|S|=4$) and thereby excludes the even-parity subgroup without needing Axiom 12' at all.

This does not work, and the reason is a category distinction, not a numerical error: the computation that the induced group on this 4-point subset is C_4 is correct. But Lemma 2.7 is a statement about the full (maximal) automorphism group of an independently defined 2D structure — it excludes C_4 from being everything that preserves some 2D Δ . It says nothing about whether C_4 can arise as the image of some proper subgroup of a larger structure's automorphism group, restricted to a subset. Indeed, this 4-point subset inherits a perfectly good 2D structure from the ambient Hamming distance (distance 1 within the subset corresponding to one type, distance 2 within the subset corresponding to the other), and the FULL automorphism group of that 2D structure, considered on its own (independent of any embedding in S), is D_4 (order 8, by Theorem 3.7) — not C_4 . The even-parity subgroup's setwise stabilizer of this subset is the subgroup $C_4 \leq D_4$, which is an entirely unremarkable fact (C_4 is a perfectly good subgroup of D_4) and raises no contradiction with Lemma 2.7 whatsoever. Excluding the even-parity subgroup therefore genuinely requires a 3D-level argument (Axiom 12' and Theorem 3.9 above); it cannot be obtained for free by appeal to the already-proven 2D result.

More generally: no argument in this section may infer a property of (S,Δ) as a whole from the appearance of some coordinate-fixed subset considered in isolation, without independently

justifying why the restriction of the whole structure's automorphism group to that subset must equal the full automorphism group the subset would have if it were considered as a standalone structure. This was the specific gap in the argument just rejected, and the same gap reappears, in the reverse direction, in any attempt to build a general-cardinality version of Theorem 3.10 by an inductive "restrict to a coordinate-fixed subset and require the restriction to be full" strategy; both directions of this strategy have been checked and rejected for the same underlying reason (see Section 12.5).

Excluding Both Order-24 Candidates, and the Resulting Uniqueness Theorem

Lemma 2.8 (Exclusion of Both Order-24 Candidates).

Neither order-24 transitive rank-4 subgroup of B_3 satisfies the full-symmetric-action condition on $\Gamma_1(x)$ established in Theorem 3.9 (retained here under the name "Axiom 12" for continuity with the statement of Theorem 3.10 below).

Proof.

For the even-parity subgroup: the stabilizer of a fixed point x has order $24/8 = 3$ (by orbit-stabilizer, since the group is transitive on the 8 points) and consists of the identity together with the two elements that cyclically permute the three coordinate positions while applying no coordinate flips. This cyclic permutation of positions fixes x (permuting identical coordinate values leaves the point unchanged) and cyclically permutes the three distance-1 neighbors (each has its single differing coordinate in a different position). Thus the induced action on $\Gamma_1(x)$ is only the cyclic subgroup of order 3, not the full symmetric group of order 6.

For the second order-24 subgroup: direct computation gives the same result — a stabilizer of order 3 inducing only a cyclic action of order 3 on $\Gamma_1(x)$.

For B_3 : the stabilizer of x has order $48/8 = 6$ and induces all six permutations of $\Gamma_1(x)$ (verified directly).

Hence among the three transitive rank-4 subgroups of B_3 identified in Section 14.3, only B_3 itself satisfies **Axiom 12**.

□

Theorem 3.10 (Forced Hyperoctahedral Symmetry at $|S|=8$ ($\{0,1\}^3$ witness model)).

Among the transitive rank-4 subgroups of $\text{Sym}(8)$ that preserve a Hamming-distance-type Δ on $\{0,1\}^3$, satisfying Axioms 10–11 at $n=3$ (typed, symmetric Δ with three types; transitive, non-2-transitive automorphism group; stabilizer transitive on each $\Gamma_i(x)$) and Axiom 12 (full symmetric action on $\Gamma_1(x)$, derived from Axiom 12' in Theorem 3.9), $\text{Aut}(S,\Delta) \cong B_3$ uniquely.

Proof.

By the graph-automorphism argument of Section 12.2, any candidate $\text{Aut}(S,\Delta)$ is a subgroup of B_3 . By the exhaustive search of Section 12.3, the only transitive rank-4 subgroups of B_3 are B_3 itself and the two order-24 subgroups. By Lemma 2.8, both order-24 candidates fail Axiom 12. Hence B_3 is the only candidate satisfying all stated axioms.

□

This establishes *Theorem 3.10* at the specific $\{0, 1\}^3$ witness model, with the candidate pool exhaustively (not merely spot-) checked and the exclusion condition (“Axiom 12”) now derived from *Axiom 12’* rather than independently assumed. The 3D analogue of *Theorem 3.8* (ruling out a fourth independent type) is established below as *Theorem 3.11*. What remains open: the analogous statement for arbitrary cardinality (not just $|S| = 8$) has not been attempted.

Theorem 3.10’s necessity argument no longer needs to rest on the exhaustive search alone. The general- n sufficiency result (*Theorem 3.15*, proved directly from Axioms 10–12’ by a counting argument with no dependence on enumerating candidates) gives the $n = 3$ instance of *Theorem 3.10* unconditionally: any (S, Δ) satisfying Axioms 10–12’ with $S = \{0, 1\}^3$ has

$$\text{Aut}(S, \Delta) \cong B_3$$

directly, without first needing to know that E_3 and P_3 are the only other rank-4 candidates.

The exhaustive search of Section 14.3 and the exclusion argument of Lemma 2.8 remain valuable as independent confirmation, and as the source of the necessity-side conclusion (Corollary 2.9.1: rank and transitivity alone — i.e., Axioms 10–11 without Axiom 12’ — never suffice). But *Theorem 3.10*’s own conclusion is now secured by the same general argument used at every other n , not by this section’s exhaustive search in isolation.

12.4 A Rejected Inductive Strategy: “Sectional Forcing”

Before the general- n route of Section 12.5 was found, a natural inductive strategy was proposed for extending *Theorem 3.10* beyond the specific $\{0, 1\}^3$ case: assume the $(n-1)$ -dimensional case has already been settled ($\text{Aut} \cong B_{n-1}$ uniquely), and require that any n -dimensional candidate group G , restricted to a coordinate-fixed $(n-1)$ -dimensional section of $\{0, 1\}^n$ (fixing one coordinate to a single value), induce the complete group B_{n-1} on that section — not merely some proper subgroup of it. If this restriction condition could be established, n -dimensional uniqueness would follow from $(n-1)$ -dimensional uniqueness by induction, without needing a fresh exhaustive search or counting argument at every n .

This strategy does not work, and the gap is specific: no general fact — not in permutation group theory, and not derivable from Axioms 10–12’ as stated — guarantees that an n -dimensional group’s restriction to a coordinate-fixed section must equal the full automorphism group that section would have if considered as a standalone $(n-1)$ -dimensional structure in its own right, rather than a proper subgroup of it. “The image of a large group restricted to a substructure” and “the complete automorphism group of that substructure considered independently” are two different objects; nothing forces the first to equal the second.

This gap is not hypothetical. A concrete instance of exactly this failure already appears within this paper’s own witness model: Section 12.3 shows that the even-parity subgroup’s setwise stabilizer of a coordinate-fixed 4-point subset induces only the cyclic group C_4 on that subset — a proper subgroup of D_4 , the full automorphism group that same subset has when considered as an independent 2D structure in its own right (Theorem 3.7). This is not a contradiction of anything already established; it is fully consistent with it. C_4 is a perfectly legitimate subgroup of D_4 (Lemma 2.7 already shows C_4 cannot be the *full* automorphism group of an independent 2D structure at $|S|=4$, which is a different and unrelated claim), and a large group inducing a proper subgroup on a substructure it merely restricts to, rather than the substructure’s own full automorphism group, violates no theorem stated anywhere in this paper.

The strategy is therefore rejected at the level of justification, not at the level of conclusion: it may well be true that B_{n-1} always appears on every coordinate-fixed section in this paper’s specific witness models, but no argument given so far establishes this, and the one concrete check available (Section 12.3’s coordinate-fixed subset) shows the corresponding restriction is a proper subgroup, not the full group, in at least one closely analogous setting. Importing “dimensional consistency” or “downward consistency” as an unstated assumption to bridge this gap would be asserting an axiom, not deriving one from what Axioms 10–12’ already establish.

Section 12.5 below replaces this rejected inductive approach with a direct argument that holds identically at every $n \geq 3$ without relying on, or reducing to, any lower-dimensional case.

12.5 A General-n Obstruction, Independent of Induction

Section 12.4’s rejected “sectional forcing” strategy attempted to reach a general-n uniqueness theorem inductively, by relating the n-dimensional case to the already-settled (n–1)-dimensional case. The following result instead applies directly, separately, and identically at every $n \geq 3$, with no dependence on any other value of n: it shows that Axioms 10–11 alone (typed, symmetric Δ ; transitive, non-2-transitive automorphism group of rank n+1) are never sufficient on their own to force the n-dimensional hyperoctahedral group B_n , at any $n \geq 3$, because a specific, explicitly constructed proper subgroup always achieves the identical rank.

The two structures defined next are not ad hoc constructions invented for this paper. Under the identification $\text{Aut}(S, \Delta) \cong B_n$, E_n is exactly the standard Coxeter group of type D_n — the index-2 subgroup of signed permutations with an even number of sign changes, often called the even-signed (or demihyperoctahedral) permutation group in the literature on Weyl groups (Wilson [13, §2.1.3] gives this exact characterization: the kernel of the homomorphism counting sign changes mod 2 is the index-2 normal subgroup D_n of B_n).

P_n is the corresponding subgroup obtained by restricting the coordinate-permutation factor to the alternating group while leaving every sign change unrestricted — denoted $C_n = Z_2 \wr A_n$ in Jackson [14], where it is used as a maximal subgroup of B_n in that paper's analysis of generating pairs and the strong symmetric genus, a question unrelated to this paper's use of it.

Both groups are thus well-studied objects in their own right; what this paper establishes (Lemma 2.9, Theorem 3.12, Theorem 3.16, and — a finding that surprised this project, recorded in full in Appendix M — a fourth, apparently new structure at $n=5$) concerns their specific role as rank- $(n+1)$ proper subgroups of B_n relative to Axioms 10–11, a question distinct from anything established about them in the cited literature.

Definition 0.8 (Even-Parity Subgroup).

For $n \geq 1$, let $B_n = (Z_2)^n \rtimes S_n$ act on $\{0,1\}^n$ by coordinate flips composed with coordinate permutations. Assign to each element the combined parity of its coordinate permutation (the usual sign of a permutation in S_n) and its number of coordinate flips (mod 2). The even-parity subgroup $E_n \leq B_n$ is the index-2 subgroup of elements with combined parity 0.

Lemma 2.9 (Transitivity of the Even-Parity Subgroup).

For every $n \geq 3$, E_n is transitive on $\{0,1\}^n$.

Proof.

Let $x, y \in \{0,1\}^n$ with Hamming distance d , and let $f = x \oplus y$ (so $\text{popcount}(f) = d$).

Case *deven*

The pure coordinate flip

$$g(v) = v \oplus f$$

has combined parity $\text{popcount}(f) \bmod 2 = 0$, so $g \in E_n$, and

$$g(x) = x \oplus f = y$$

Case *dodd*

Fix once and for all a single transposition τ of two coordinate positions — any two; the choice does not depend on x , y , or d (for concreteness, positions 1 and 2).

Step 1 — Define g .

Let

$$k = x \oplus \tau(x) \oplus f$$

and define

$$g(v) := \tau(v) \oplus k$$

Step 2 — Verify $g(x) = y$.

$$g(x) = \tau(x) \oplus k = \tau(x) \oplus x \oplus \tau(x) \oplus f = x \oplus f = y,$$

since $\tau(x) \oplus \tau(x) = 0$.

Step 3 — Verify g has even combined parity.

The map g is written as “permute then flip,” so its combined parity is

$$\text{sign}(\tau) \oplus (\text{popcount}(k) \bmod 2)$$

For any two coordinate vectors a, b ,

$$\text{popcount}(a \oplus b) \equiv \text{popcount}(a) + \text{popcount}(b) \pmod{2}$$

since $|A \triangle B| = |A| + |B| - 2|A \cap B|$. Applying this to $k = x \oplus \tau(x) \oplus f$:

$$\text{popcount}(k) \equiv \text{popcount}(x \oplus \tau(x)) + \text{popcount}(f) \pmod{2}.$$

Now $\text{popcount}(x \oplus \tau(x))$ is either 0 (if the two transposed positions in x are equal) or 2 (if they differ). Either way it is even, so

$$\text{popcount}(k) \equiv d \pmod{2}$$

because d is odd in this case.

Thus the combined parity is

$$\text{sign}(\tau) \oplus (\text{popcount}(k) \bmod 2) = 1 \oplus 1 = 0$$

so $g \in E_n$.

For every $n \geq 3$ and every Hamming distance d , an element of E_n taking x to y exists — the same single fixed transposition τ works for every odd d , with no further case distinction needed. Since this holds for every pair x, y , E_n is transitive on $\{0,1\}^n$.

□

Theorem 3.12 (Rank Preservation by the Even-Parity Subgroup).

For every $n \geq 3$, E_n has rank $n + 1$ on $\{0,1\}^n$ — the same rank as B_n — not merely transitivity. Equivalently: for every two ordered pairs (x, y) and (x', y') with equal Hamming distance

$$d = \text{dist}(x, y) = \text{dist}(x', y')$$

there exists $g \in E_n$ with $g(x) = x'$ and $g(y) = y'$.

Proof.

Write $\delta = x \oplus y$ and $\delta' = x' \oplus y'$, so $\text{popcount}(\delta) = \text{popcount}(\delta') = d$. Let $\text{supp}(\delta)$ and $\text{supp}(\delta')$ be the sets of coordinates where these vectors equal 1, each of size d , with complements of size $n - d$.

Step 1 — Match the two supports with a coordinate permutation.

Since $|\text{supp}(\delta)| = |\text{supp}(\delta')| = d$, there exists a permutation p of the n coordinates mapping $\text{supp}(\delta)$ bijectively onto $\text{supp}(\delta')$. By construction, applying p to δ yields δ' :

$$p(\delta) = \delta'$$

Step 2 — Construct a flip vector sending x to x' .

Define

$$f = x \oplus p^{-1}(x'), g(v) := p(v \oplus f)$$

This g is an element of B_n (flip by f , then apply p).

Step 3 — Verify that $g(x) = x'$ and $g(y) = y'$.

Image of x .

$$g(x) = p(x \oplus f) = p(x \oplus x \oplus p^{-1}(x')) = p(p^{-1}(x')) = x'$$

Image of y . Since p commutes with $\oplus$:

$$p(y \oplus f) = p(y \oplus x \oplus p^{-1}(x')) = p(y \oplus x) \oplus x'$$

But $y \oplus x = \delta$, and Step 1 gives $p(\delta) = \delta' = x' \oplus y'$. Thus

$$p(y \oplus f) = \delta' \oplus x' = (x' \oplus y') \oplus x' = y'$$

So $g \in B_n$ satisfies $g(x) = x'$ and $g(y) = y'$.

Step 4 — Adjust parity so that $g \in E_n$.

The element g may have odd combined parity. Only one of g and a parity-corrected variant g' will lie in E_n . We now show that such a correction is always available.

Case $d \geq 2$.

$\text{supp}(\delta)$ has at least two positions. Let τ be a transposition of two such positions. Both positions hold value 1 in δ , so

$$\tau(\delta) = \delta$$

Set $p' = p \circ \tau$. Then

$$p'(\delta) = p(\tau(\delta)) = p(\delta) = \delta'$$

so Step 1 still holds. Repeating Steps 2–3 with p' yields a valid g' whose combined parity differs from that of g by exactly 1.

Case $d \leq 1$ and $n \geq 3$.

Here $n - d \geq n - 1 \geq 2$, so the complement of $\text{supp}(\delta)$ has at least two positions. Choose τ as a transposition of two such positions. Both positions hold value 0 in δ , so again

$$\tau(\delta) = \delta$$

and the same correction argument applies.

Case $d = 0$.

Then $x = y$ and $x' = y'$. The map need only satisfy $g(x) = x'$, which is already covered by Lemma 2.9. This case is already included in the $d \leq 1$ analysis.

Every $n \geq 3$ and every d falls into one of these cases, so a parity-correcting move is always available, and one of g or g' lies in E_n . Thus for any two ordered pairs at the same Hamming distance, an element of E_n maps one pair to the other, and E_n has exactly $n + 1$ orbits on $\{0,1\}^n \times \{0,1\}^n$, matching the rank of B_n .

□

This argument is general for every $n \geq 3$ — it does not rely on or extrapolate from the $n=3,4,5$ computations reported earlier in this paper. Those computations ($n=3,4,5,6$, with an explicit implementation of exactly this construction checked against every same-distance ordered pair of pairs — 1216, 17664, 257024, and 3,780,608 cases respectively, zero exceptions) were used to find and verify the construction before it was written up as the general proof above, and now serve as an independent confirmation that the proof's case analysis is complete, rather than as the basis for the claim itself.

Corollary 2.9.1 (Rank and Transitivity Alone Never Suffice).

For every $n \geq 3$, there exists a proper subgroup of B_n (namely E_n) that is transitive on $\{0,1\}^n$ and has rank $n+1$, the same as B_n (Lemma 2.9, Theorem 3.12). Consequently, Axioms 10–11 alone cannot force $\text{Aut}(S,\Delta) \cong B_n$ at any $n \geq 3$; a local-saturation condition in the spirit of Axiom 12' is necessary at every such n , not only at $n=3$.

This is a genuine advance on the question left open in Section 12.4: it does not yet establish sufficiency — a full n -dimensional analogue of Axiom 12', together with a proof excluding every rank- $(n+1)$ proper subgroup of B_n (not only E_n) for every n , remains to be done — but the necessity half of the general- n picture is now fully settled, not merely verified at small n , and it replaces an inductive strategy that was checked and rejected with a direct, non-inductive result holding identically at every n .

12.5.1 Sufficiency, and a Second Index-2 Subgroup

Two further results close most of the remaining gap. First, the no-internal-splitting fact verified directly for $\{0,1\}^3$ in Section 12.2.1's discussion of Theorem 3.9 holds at every n , not just $n=3$, and is worth stating as its own general- n lemma so later sections (Section 13 builds on this one and) can cite it directly rather than re-verifying it at each n .

Lemma 2.12 (No Internal Type-Splitting in $\Gamma_1(x)$, for Every n).

For every $n \geq 2$ and every $x \in \{0,1\}^n$, every pair of distinct points in $\Gamma_1(x)$ is at Hamming distance exactly 2 from each other.

Proof.

Each point in $\Gamma_1(x)$ differs from x in exactly one coordinate position. Two distinct such points differ from x in two different positions (else they would be equal), so they differ from each other in exactly those two positions — the one where the first differs from x but the second does not, and the one where the second differs from x but the first does not — giving Hamming distance exactly 2 between them. This holds independently of n , and was additionally verified by direct computation for $n=3, \dots, 7$.

□

Second, E_n is not the only index-2 subgroup of B_n worth identifying by name. E_n and the order-24 subgroup found in Section 12.3 but never identified there arise from the two independent S_n -invariant ways of building a homomorphism $B_n \rightarrow \mathbb{Z}_2$: combine the coordinate permutation's

own sign with the total flip-count parity (giving E_n), or use the permutation's sign alone, ignoring flips entirely.

Definition 0.9 (Permutation-Even Subgroup).

For $n \geq 1$, let

$$B_n = (\mathbb{Z}_2)^n \rtimes S_n$$

act on $\{0,1\}^n$ as before. The permutation-even subgroup

$$P_n \leq B_n$$

is the subgroup consisting of those elements whose coordinate-permutation component (ignoring the flip component) has even parity. Equivalently,

$$P_n = (\mathbb{Z}_2)^n \rtimes A_n$$

At $n=3$, P_3 is exactly the second order-24 subgroup found in Section 12.3: its element-order profile (1, 2×7 , 3×8 , 6×8) matches the profile reported there exactly, confirming the identification. This resolves the question, left open at the time of that search, of what that subgroup actually is. A third combination — using only the flip-count parity, ignoring the permutation's sign — was checked directly and found not transitive at $n=3$ and $n=4$ (verified computationally), so it does not produce a further candidate.

Theorem 3.16 (Rank Preservation by the Permutation-Even Subgroup).

For every $n \geq 3$, P_n has rank $n+1$ on $\{0,1\}^n$ — the same rank as B_n .

Proof.

Step 1 — Reuse Theorem 3.12's construction of p , f , and g .

Let x, y, x', y' have Hamming distance

$$d = \text{dist}(x, y) = \text{dist}(x', y')$$

Write $\delta = x \oplus y$, $\delta' = x' \oplus y'$, with $\text{supp}(\delta)$ and $\text{supp}(\delta')$ of size d . Choose any permutation p mapping $\text{supp}(\delta)$ bijectively onto $\text{supp}(\delta')$, extending arbitrarily on the complements; then $p(\delta) = \delta'$ exactly as in Theorem 3.12 Step 1.

Define

$$f = x \oplus p^{-1}(x'), g(v) = p(v \oplus f).$$

Exactly as in Theorem 3.12 Steps 2–3,

$$g(x) = x', g(y) = y'$$

Step 2 — P_n 's membership condition is simpler than E_n 's.

Unlike E_n , the subgroup P_n imposes **no restriction** on the flip vector f ; all flips are allowed. Membership in P_n depends **only** on the parity of the coordinate permutation. Thus, if p is already even, then $g \in P_n$ immediately.

Step 3 — If p is odd, correct it using the same transposition device as in Theorem 3.12.

If p is odd, choose a transposition τ that preserves δ :

- if $d \geq 2$, take τ swapping two positions in $\text{supp}(\delta)$;
- if $d \leq 1$ and $n \geq 3$, take τ swapping two positions in the complement of $\text{supp}(\delta)$ (which has size ≥ 2).

In either case, both swapped positions carry the same δ -value, so

$$\tau(\delta) = \delta$$

Set

$$p' = p \circ \tau$$

Then $p'(\delta) = \delta'$ still holds, while the parity of p' is now even.

Recompute

$$f' = x \oplus (p')^{-1}(x'), g'(v) = p'(v \oplus f')$$

The same argument as in Step 1 gives

$$g'(x) = x', g'(y) = y'$$

Since P_n imposes no condition on f' , the element g' lies in P_n with no further verification.

Hence for every same-distance pair of ordered pairs, some $g \in P_n$ connects them: P_n has rank $n+1$.

□

Neither E_n nor P_n satisfies Axiom 12' (the $n=3$ case is exactly Lemma 2.8 above; the same stabilizer-order argument applies at every n , since both groups' point stabilizers have order $|B_n|/2 \div |S|$ — half of B_n 's — and so cannot induce the full $\text{Sym}(n)$ on $\Gamma_1(x)$). Corollary 2.9.1's necessity conclusion therefore holds with E_n and P_n as two independent witnesses, not one.

Theorem 3.15 (General- n Sufficiency: Axiom 12' Forces B_n).

Let $n \geq 2$ and let (S, Δ) satisfy Axioms 10–11 with $S = \{0,1\}^n$ under Hamming distance. If $\text{Aut}(S, \Delta)$ additionally satisfies Axiom 12', then

Proof.

Write $G = \text{Aut}(S, \Delta)$ and fix $x \in S$. Let

$$\varphi: G_x \rightarrow \text{Sym}(\Gamma_1(x))$$

be the restriction homomorphism, sending each element of the point stabilizer G_x to its induced action on $\Gamma_1(x)$.

Step 1 — Axiom 12' forces $\text{im}(\varphi) = \text{Sym}(\Gamma_1(x))$.

By Lemma 2.12, $\Gamma_1(x)$ has no internal type-splitting. By Theorem 3.9, Axiom 12' together with this fact forces **every** permutation of $\Gamma_1(x)$ to be realized as $\varphi(g)$ for some $g \in G_x$: since all pairs in $\Gamma_1(x)$ have the same Δ -type (Lemma 2.12), every permutation of $\Gamma_1(x)$ trivially

preserves internal types and therefore satisfies the hypothesis of Axiom 12', which then requires that it extend to some global automorphism fixing x . Thus

$$\text{im}(\varphi) = \text{Sym}(\Gamma_1(x))$$

not merely some proper subgroup.

Step 2 — This forces $|G_x| \geq n!$.

By the first isomorphism theorem,

$$|G_x| = |\text{im}(\varphi)| \cdot |\ker(\varphi)| = n! \cdot |\ker(\varphi)| \geq n!$$

since $|\ker(\varphi)| \geq 1$ in all cases.

Caution — why this must come from image(φ) specifically, not from $|G_x|$ reaching $n!$ by any route.

The inequality $|G_x| \geq n!$ follows from image(φ) being the full Sym(n) specifically — not from any weaker condition under which $|G_x|$ merely reaches the number $n!$ through some other combination of a smaller image and a larger kernel. For instance, image(φ) = A_n (order $n!/2$) with $|\ker(\varphi)| = 2$ would give $|G_x| = n!/2 \times 2 = n!$ numerically — the same bare count — but such a G would not satisfy Axiom 12' at all, since the odd permutations of $\Gamma_1(x)$ would then not be realized by any element of G_x , violating the extension requirement for those specific permutations. This distinction is essential and is restated here explicitly because it is the exact point at which a careless reading of this proof could go wrong: Step 1 establishes the image, not merely a cardinality, and it is the image that does the work in what follows.

Step 3 — Orbit-stabilizer gives $|G| \geq |B_n|$.

By Axiom 11, G is transitive on S , and $|S| = 2^n$. The orbit-stabilizer theorem gives

$$|G| = |S| \cdot |G_x| \geq 2^n \cdot n! = |B_n|$$

Step 4 — G is also contained in B_n , forcing equality.

Any Δ -preserving bijection of S is a graph automorphism of the hypercube graph Q_n : graph automorphisms preserve graph distance by definition, and Hamming distance coincides with graph distance in Q_n for all n . Two points at Hamming distance d are connected by a path of exactly d edges (flipping the differing coordinates one at a time), and no shorter path exists because each edge changes Hamming weight by exactly ± 1 .

The automorphism group of Q_n is B_n for every n (Harary 2000), so

$$G \leq B_n$$

Combined with $|G| \geq |B_n|$ from Step 3, this yields $|G| = |B_n|$. Since G is a subgroup of B_n with equal order,

$$G = B_n$$

□

This argument is entirely internal to dimension n : it never restricts to, or compares against, any $(n-1)$ -dimensional substructure, and so does not revive the "sectional forcing" strategy rejected in Sections 14.4–14.5 — it is a direct counting argument operating in one fixed dimension throughout. It makes no reference to E_n , P_n , or any other specific candidate: it shows directly that ANY group satisfying Axiom 12' must already be all of B_n , by counting alone, regardless of what such a group might otherwise look like. This is a genuinely different and substantially stronger route than the exhaustive-search strategy used at $n=3$ in Section 12.3, and it resolves the sufficiency direction for every $n \geq 2$ at once — confirming Theorem 3.10's conclusion at $n=3$ by an independent method, and removing any need for an exhaustive search at higher n .

The exhaustive search at $n=3$ (Section 12.3) and the E_n/P_n construction above (Definitions 0.8–0.9, Theorems 3.12 and 3.16, Corollary 2.9.1) remain valuable and are not made redundant by Theorem 3.15: they establish necessity — that Axiom 12' is not a vacuous or superfluous addition to Axioms 10–11, since concrete proper subgroups (E_n , P_n) exist that satisfy Axioms 10–11 but fail Axiom 12'. Theorem 3.15 establishes sufficiency — that Axiom 12', once satisfied, leaves no room for anything other than B_n . Both directions are needed for a complete account, and they were established by genuinely different methods.

12.6 Toward a 3D Analogue of Theorem 3.8

Theorem 3.8 (2D) excludes a third independent type using a Schur ring of dimension 3 generated by the orbital matrices of D_4 . The natural 3D analogue uses the corresponding 4-dimensional algebra generated by the orbital matrices of B_3 (the Bose–Mesner algebra of the Hamming scheme on $\{0,1\}^3$). The following computation, carried out directly rather than asserted from a citation, establishes the dimension claim this analogue depends on.

Lemma 2.10 (Verified Dimension of the Distance Algebra on $\{0,1\}^3$).

Let A_0, A_1, A_2, A_3 be the 8×8 matrices with

$$(A_k)_{ij} = \begin{cases} 1 & \text{if the } i\text{-th and } j\text{-th points of } \{0,1\}^3 \text{ are at Hamming distance } k, \\ 0 & \text{otherwise.} \end{cases}$$

Then:

1. **Linear independence** — *The four matrices A_0, A_1, A_2, A_3 are linearly independent as vectors in the space of 8×8 matrices.*
2. **Partition of the all-ones matrix** — *Their sum $A_0 + A_1 + A_2 + A_3$ is the all-ones matrix.*
3. **Closure under multiplication** — *The algebra they span is closed under matrix multiplication.*

Proof.

All three claims are verified by direct computation on the explicit 8×8 matrices.

1. **Linear independence** — Flattening $A_0, \dots, A_3$ into four 64-dimensional vectors gives rank exactly 4. This is computed directly from the matrices themselves, not inferred from the rank of $\text{Aut}(S, \Delta)$ in Theorem 3.10 (though the two values coincide, as expected).
2. **Sum to the all-ones matrix** — Every pair of points in $\{0,1\}^3$ lies at exactly one Hamming distance in $\{0, 1, 2, 3\}$, so the four matrices partition the all-ones matrix. Verified directly.
3. **Closure under multiplication** — As a witness computation,

$$A_1 A_1 = 3A_0 + 2A_2,$$

obtained by explicit matrix multiplication and exact linear-algebraic decomposition (not approximated). This confirms that the product of two basis matrices lies exactly in their span, establishing closure of the algebra they generate.

□

Theorem 3.11 (No Fourth Independent Type at $\{0,1\}^3$).

Let (S, Δ) satisfy Axioms 10–11 at $n = 3$ with $S = \{0,1\}^3$ under Hamming distance, so that

$$\text{Aut}(S, \Delta) \cong B_3 \text{ (Theorem 3.10).}$$

Then Σ cannot be extended by a fourth type σ_4 , distinguishable from $\sigma_1, \sigma_2, \sigma_3$, that remains compatible with $\text{Aut}(S, \Delta)$.

Proof.

Step 1 — The Hamming-distance algebra at $n=3$ has dimension exactly 4.

By Theorem 3.10,

$$\text{Aut}(S, \Delta) \cong B_3$$

with rank 4 (Section 14.2–Section 14.3). The orbits of B_3 on $S \times S$ correspond exactly to the four Hamming-distance classes, represented by the matrices $A_0, \dots, A_3$. By Lemma 2.10, these four matrices are linearly independent and span an algebra of dimension exactly 4, closed under multiplication.

Step 2 — A fourth type would need a fifth independent matrix, and there is no room for one.

A candidate fourth type σ_4 , distinguishable from $\sigma_1, \sigma_2, \sigma_3$ and compatible with $\text{Aut}(S, \Delta)$, would correspond to a further adjacency matrix M_4 compatible with the same automorphism group — hence constant on each B_3 -orbit on $S \times S$ — and any such matrix is, by definition, some linear combination of $A_0, \dots, A_3$ (since these four matrices, being constant on each orbit and there being exactly four orbits, already span the entire space of matrices constant on B_3 -orbits). Hence M_4 cannot be independent of $A_0, \dots, A_3$: this is a direct consequence of the dimension count in Lemma 2.10, not an appeal to any result established elsewhere in this paper.

□

Theorem 3.11 is established at the specific $\{0,1\}^3$ witness model with its supporting dimension count (Lemma 2.10) independently verified, in exactly the sense Theorem 3.8 was established for D_4 at 2D. The general- n version of this theorem, and the question of which standard reference (Bannai & Ito; Brouwer, Cohen & Neumaier; Godsil) most directly supports it in citable form, remain open and are listed below.

- **Point-set-level characterization (status and sampling evidence)**

- **Point-set-level characterization:**

Axioms 10–12' determine that, once one *assumes* $S = \{0,1\}^3$, the automorphism group is forced to be B_3 (Theorem 3.10). A different and substantially harder question is whether **any** 8-point set S , *not* assumed in advance to be $\{0,1\}^3$, that satisfies Axioms 10–11 with rank exactly 4, must necessarily be isomorphic to $(\{0,1\}^3, \text{Hamming distance})$. This has **not** been proven.

- Sampling check (not a classification).

A full search over all 50 transitive groups of degree 8 would require a permutation-group classification database not available for this work. Instead, four classically known transitive degree-8 actions, each **not** contained in any subgroup of B_3 , were tested directly:

1. **PSL(2,7)** acting on the projective line $\text{PG}(1,7)$ – order 168, rank 2, 2-transitive.
2. **AGL(3,2)** acting on the affine space over $\text{GF}(2)^3$ – order 1344, rank 2, 2-transitive.
3. **Dihedral group** D_8 acting on an 8-cycle – order 16, rank 5, non-2-transitive.
4. **AGL(1,8)** acting on $\text{GF}(8)$ – order 56, rank 2, 2-transitive.

None of these four actions has the required combination (**rank exactly 4, non-2-transitive**) needed to serve as a genuine alternative model.

- Fifth attempted candidate.

A fifth action — S_4 acting on the 8 cosets of a 3-cycle subgroup — was attempted, but the implementation produced a mathematically impossible output (rank 16, exceeding the universal bound $\text{rank} \leq n = 8$ for any transitive action). This confirms a coding error in the coset-action construction rather than a real mathematical phenomenon. It is therefore recorded as **abandoned**, not as a result, consistent with the discipline of treating impossible outputs as bugs rather than surprises (cf. Section 15.7).

- Status.

This sampling narrows the plausible space of alternatives and provides modest supporting evidence, but it **does not** establish the point-set-level characterization and **should not** be cited as doing so.

- **General- n uniqueness: status and remaining gap**

- General- n uniqueness is now complete in both directions (Section 14.5.1).

- Necessity holds for every $n \geq 3$, with two independent witnesses E_n and P_n (Lemma 2.9, Theorem 3.12, Definition 0.9, Theorem 3.16, Corollary 2.9.1).

- Sufficiency holds for every $n \geq 2$ directly: *Theorem 3.15* shows that Axiom 12' forces $G = B_n$ by a counting argument that requires
 1. no enumeration of candidates,
 2. no exhaustive search,
 3. no restriction to small n .
- Still open:

whether E_n and P_n exhaust all rank- $(n+1)$ proper subgroups satisfying Axioms 10–11 without Axiom 12' at general n . This is a classification problem, distinct from uniqueness under Axiom 12'. Theorem 3.15 already settles uniqueness under Axiom 12' independently of any such classification.
- Status at small n .

At $n = 3, 4$, the classification is exhaustively confirmed (Section 14.3 for $n = 3$).

For $n \geq 5$, only the existence of E_n and P_n is established; exhaustiveness is not.

- **Vocabulary discipline (effective immediately, retroactively applied)**

- **Disallowed vocabulary.**

Effective immediately — and retroactively applied to all preceding text — no description of this witness model, this section's arguments, or any future generalization may use “**cube**,” “**vertex/vertices**,” “**face**,” “**slice**,” “**rotation**,” “**orientation**,” “**edge**,” “**diagonal**.” These terms describe how the structure *could be drawn or pictured*, not what the structure *is*. Any reasoning that depends on such imagery (e.g., “restrict to a face,” “rotate about a diagonal”) is reasoning about the **picture**, not about (S, Δ) .

- **Reason for the restriction.**

Section 12.4's rejected inductive strategy originally illustrated exactly this failure: it relied on an external geometric analogy. Upon review, the identical fact is already established **internally**, using compliant vocabulary, by Section 12.3's coordinate-fixed-subset argument. Section 12.4 now cites that internal argument directly, rather than reaching outside the paper's own apparatus for an illustration.

- **Permitted vocabulary for the witness model.**

This section describes its witness model **only** using:

$\{0, 1\}^n$, Hamming distance, coordinate positions, coordinate-fixed subsets.

This restriction must be maintained **without exception** when the section is extended.

- **Acceptable terminology from established fields.**

Terms naming standard, precisely defined mathematical objects — such as “**hypercube graph Q_n** ,” “**hyperoctahedral group**,” “**Hamming scheme**” — are citations of established nomenclature, not appeals to a picture, and therefore

remain acceptable. The distinction is whether the term performs **argumentative work via the image it evokes**; if it does, it is prohibited.

- **3D analogue and general- n dimension counts**

- **3D analogue of Theorem 3.8.** |

- The 3D analogue is **Theorem 3.11** (Section 12.6). Its supporting dimension count — **Lemma 2.10** — is **fully self-contained** and **independently verified by direct computation**, not asserted from any external citation.

- **General- n dimension count.**

- The general- n version is **Lemma 2.13** (Section 13), likewise **self-contained**.

- **External sources not required.**

- Following the decision recorded in Section 13, no specific page/chapter locator in any external source (Brouwer–Cohen–Neumaier; Bannai–Ito; Godsil) is being pursued for this fact, because **the proofs do not depend on one**. A precise locator could be added later as optional polish, but would not change either proof.

- The proposed MacWilliams-identity bridge turned out to be unnecessary: Theorem 3.11's actual proof uses only the direct linear-algebra argument from Lemma 2.10 (the same method as Theorem 3.8's Schur-ring argument at 2D), with no appeal to MacWilliams identities or any other coding-theory tool. This item is recorded as closed by construction, not by verification of the originally proposed tool.
- No directional language; "3D" characterizes a relation to 0D/1D/2D, not a claim about priority (apply the same bidirectionality discipline as Section 10.4's remark, once a genuine 2D/3D threshold theorem is drafted).

What carries forward from this section is not the $\{0,1\}^3$ witness model itself but the route that proved necessary to characterize it: a local-saturation condition (Axiom 12') derived from a more primitive principle, then shown sufficient by direct counting rather than by exhaustive search (Theorem 3.15). Section 13 instantiates exactly this same route at $n=4$, with no fresh search required at that layer — the strongest evidence available within this paper that the obstruction identified here is a genuine feature of every $n \geq 3$, not an artifact of the smallest case.

- Summary

This part extends the typed-differentiation axioms from the binary case (2D) to a concrete three-dimensional witness model, $\{0,1\}^3$ under Hamming distance, and uses it to develop the general- n machinery the rest of the paper depends on. An exhaustive computational search finds exactly three transitive rank-4 subgroups of B_3 satisfying the relevant axioms — B_3 itself, and two further order-24 subgroups later identified as E_n and P_n — closing the question at this specific cardinality before any general argument is attempted. A direct inductive shortcut from 2D to 3D is tested and explicitly rejected, with the reason for its failure traced to an internal fact this paper had already established (Section 12.3), rather than to any external analogy. The part's central result, Theorem 3.15, then supersedes this case-by-case approach entirely: a single direct counting argument shows that a local-saturation axiom (Axiom 12') forces B_n at every $n \geq 2$

simultaneously, not dimension by dimension. The part closes by extending 2D's "no third type" result (Theorem 3.8) to this setting and flagging the open question — resolved, with a surprise, in Appendix M — of whether E_n and P_n are the only proper alternatives once Axiom 12' is dropped.

PART V — THE FOUR-DIMENSIONAL STRUCTURE (4D)

Section 13. Internal anatomy of 4D, with computational verification of the general- n results across $0D-4D$.

13. The Internal Structure of 4D

Section 12 extended the typed-differentiation axioms to general n and established — via Axiom 12' and the general- n sufficiency theorem (*Theorem 3.15*) — that

$$\text{Aut}(S, \Delta) = B_n \text{ for every } n \geq 2,$$

not merely for the specific case $n = 3$ used as the witness model in developing the argument.

This section instantiates the same machinery at $n = 4$. The witness model is

$$S = \{0,1\}^4$$

under Hamming distance (16 points), and *Theorem 3.15* already applies to this case directly. No separate exhaustive search is required at this layer, since the sufficiency argument is uniform in n and does not depend on small- n enumeration.

13.1 What Remains To Be Decided Before Axioms Can Be Drafted

The witness model for 4D required a deliberate decision before any axiom could be drafted, not a default continuation of the $n=2$, $n=3$ pattern: would it continue the $\{0,1\}^n$ pattern at $n=4$, under language that avoids inviting a geometric or physical reading, or would a different combinatorial structure be more appropriate? Two decisions follow:

- **Witness model:** The model is

$$\{0,1\}^4 \text{ under Hamming distance}$$

continuing the same family used at $\{0,1\}^3$. No special name is introduced for it — **not** “tesseract,” and not any term suggesting a four-dimensional geometric solid. It is described only as $\{0,1\}^4$ with $\Delta = \text{Hamming distance}$, exactly as $\{0,1\}^3$ was described in Section 14.

- **Vocabulary:** the discipline established in Section 12 — no “cube,” “vertex/vertices,” “face,” “slice,” “rotation,” “orientation,” “edge,” or “diagonal” — applies here without exception.

Avoidance of geometric terminology and its relation to the tesseract

This avoidance is **not** a claim that the tesseract itself is illegitimate. Schläfli’s 1852 classification of the six regular convex 4-polytopes — the 5-cell, 8-cell (tesseract), 16-cell, 24-cell, 120-cell, and 600-cell — is rigorous, settled mathematics, and nothing in this work disputes it.

What is avoided is selecting the **tesseract**, among the six, as *the* representative object for “four-dimensionality” merely by continuing the visual analogy

point $\rightarrow$ segment $\rightarrow$ square $\rightarrow$ [next],

an analogy that picks out the tesseract because it most resembles the structure one step down, **not** because any independent argument forces it over the other five.

The structure $\{0,1\}^4$ under Hamming distance does indeed share its point set and distance-1 adjacency with the tesseract’s combinatorial skeleton (16 points, 32 distance-1 pairs) — the same fact already cited via *Harary 2000* for the hypercube graph Q_n , here at $n = 4$. This overlap is **by construction**, since Q_n is defined as $\{0,1\}^n$ with distance-1 adjacency.

But **nothing** in the derivation of *Theorem 3.13* or *Theorem 3.15* depends on this overlap. The argument — Axioms 10–12’ force

$$G \cong B_n$$

by a direct counting method — applies identically to **any** appropriately typed n -point structure satisfying the same axioms, regardless of whether it happens to share a skeleton with a regular 4-polytope.

Avoiding “minimal” or “natural” language for $\{0, 1\}^4$

Nor should $\{0,1\}^4$ be described as “**the minimal**” or “**the most natural**” B_n -set. In fact, B_4 acts faithfully and transitively on a **strictly smaller** set: the eight signed coordinate directions

$$\{+e_i, -e_i : i = 1, \dots, 4\}$$

one of the standard representations of the hyperoctahedral group in the literature. This is confirmed directly: there are **384** distinct permutations of these eight points, matching $|B_4|$ exactly (see *hyperoctahedral group*).

What singles out $\{0,1\}^4$ is **not** minimality or naturalness, but the **typed structure required by Axiom 10**. On the 8-point signed-axis set, B_4 has **rank 3**, giving only **two** nontrivial types — the rank-3 structure characteristic of the 2D case. Axiom 10 at $n = 4$ requires a **rank-5** structure with **four** nontrivial types. The signed-axis action cannot realize this; $\{0,1\}^4$ can.

Thus $\{0,1\}^4$ is the B_4 -set on which the required typed- Δ structure is realized — not the unique, smallest, or “most natural” B_4 -set in any unqualified sense. The term “**natural**” is avoided entirely, since it expresses an aesthetic preference rather than a derived property.

13.2 General-n Axioms (Retroactively Formalizing What 2D and 3D Assumed)

Axiom 8 and Axiom 9, as stated in the main paper, are written specifically for exactly two types — they are 2D-specific. Section 12 formally states the general-n version once, as Axiom 10 and Axiom 11 below, rather than separately at every dimension. Axiom 12' (Section 12) was already stated in properly general form.

Axiom 10 and Axiom 11 (the general-n forms) are stated in full in Section 12.3 and used unchanged here.

At $n=2$ these are exactly Axiom 8 and Axiom 9. At $n=3$ these are exactly what Section 12 states formally as well (Section 12.2.1), consistent with the same general-n statement appearing at each instance. At $n=4$ these are stated here for the same purpose, completing the pattern across all three layers so far axiomatized.

13.3 A Systematic Source of "Extra" Candidates (Strengthening the 3D Account)

Section 12 found, by computational search, two proper subgroups of B_3 (each of order 24) that are transitive with rank 4, matching B_3 exactly — the even-parity subgroup E_3 (Definition 0.8) and a second subgroup whose abstract identity was left unresolved there. That second subgroup is now identified: it is the subgroup of B_n generated by all coordinate flips together with only the even permutations of coordinate positions — that is,

$$(\mathbb{Z}_2)^n \rtimes A_n,$$

rather than the full $(\mathbb{Z}_2)^n \rtimes S_n$. Both candidates arise systematically, not coincidentally, from the same source: they are precisely the two rank- $(n+1)$ proper subgroups of B_n compatible with Axioms 10–11 (Lemma 2.9, Theorem 3.12).

Definition 0.9 (the permutation-even subgroup P_n) is stated in full in Section 12.5.1.

Both E_n (Definition 0.8) and P_n (Definition 0.9) are index-2 subgroups of B_n , arising from the two independent S_n -invariant ways of constructing a homomorphism

$$B_n \rightarrow \mathbb{Z}_2.$$

One combines the sign of the coordinate permutation with the parity of the total flip count, producing E_n . The other uses the sign of the permutation alone, ignoring flips entirely, producing P_n .

A third combination — using only the flip-count parity and ignoring the permutation's sign — was checked directly and found **not transitive** at $n = 3$ and $n = 4$ (verified computationally). It therefore does not produce a further rank-preserving candidate and is not considered further.

At $n=3$, P_3 is exactly the second order-24 subgroup found in Section 12's search: its element-order profile $(1, 2 \times 7, 3 \times 8, 6 \times 8)$ matches P_3 's computed profile exactly, confirming the identification. This resolves the open question left open in Section 12 about that subgroup's abstract identity.

13.4 Candidates at $n=4$, and a General- n Sufficiency Theorem

Direct computation confirms that both E_4 and P_4 are transitive on $\{0,1\}^4$ with **rank exactly 5**, matching the rank of B_4 exactly. Both have order

$$|E_4| = |P_4| = 192,$$

each an index-2 subgroup of

$$|B_4| = 2^4 \cdot 4! = 384.$$

Moreover, E_4 and P_4 are confirmed **distinct as sets** by direct computation, not merely presumed distinct from their defining constructions.

Lemma 2.11 (Exclusion of E_4 and P_4 at $n=4$).

Neither E_4 nor P_4 satisfies Axiom 12's consequence of full symmetric local action: for $x = (0,0,0,0)$, $\Gamma_1(x)$ has 4 elements, and the point stabilizer of x within each of E_4 and P_4 has order 12, inducing an action on $\Gamma_1(x)$ of order 12 — not the full $\text{Sym}(\Gamma_1(x))$ of order 24 that B_4 's point stabilizer (order 24) induces.

Proof.

Direct computation gives the following. For $x = (0,0,0,0)$, the first neighborhood is

$$\Gamma_1(x) = \{(1,0,0,0), (0,1,0,0), (0,0,1,0), (0,0,0,1)\}$$

Since B_4 is transitive on the 16 points of $\{0,1\}^4$, the stabilizer of x in B_4 has order

$$|B_4|/16 = 384/16 = 24$$

and it induces **all 24 permutations** of $\Gamma_1(x)$, i.e. the full $\text{Sym}(\Gamma_1(x))$.

For each of E_4 and P_4 , the stabilizer of x has order

$$|E_4|/16 = |P_4|/16 = 192/16 = 12$$

and direct computation shows that each induces an action of **order exactly 12** on $\Gamma_1(x)$. In both cases this is a **proper subgroup** of $\text{Sym}(\Gamma_1(x))$, confirming that neither E_4 nor P_4 satisfies the full symmetric local action required by Axiom 12'.

□

Hence, by *Theorem 3.9*, Axiom 12' forces full symmetric local action whenever $\Gamma_i(x)$ has no internal type-splitting — a condition that holds here by exactly the same Hamming-distance argument used at $n = 3$. Since neither E_4 nor P_4 induces the full

$\text{Sym}(\Gamma_1(x))$ on the 4-point first neighborhood, neither group can be the automorphism group of any structure satisfying Axioms 10–12'. This establishes the $n = 4$ instance of necessity: rank and transitivity alone do not suffice, paralleling *Corollary 2.9.1* at $n = 3$, now confirmed for P_n as well as E_n .

A second, independent question is sufficiency: once Axiom 12' is assumed, does it force

$$G = B_n$$

outright, without enumerating E_n , P_n , or any other candidate? The general argument that follows shows that the answer is yes for every n . The proof proceeds by direct counting, not by exhaustive search, and therefore completes the sufficiency direction in full generality — eliminating the need for the exhaustive $n = 4$ computation this paper originally set out to complete.

Lemma 2.12 (no internal type-splitting, every n) is proved in Section 12.5.1 and is not re-proved here.

Theorem 3.15 (general- n sufficiency) is proved in Section 12.5.1 and is not re-proved here.

This argument is entirely internal to dimension n : it never restricts to, or compares against, any $(n-1)$ -dimensional substructure, and therefore does not revive the *sectional forcing* strategy already rejected in Sections 14.4–14.5 and again in Section 15.7. It is a direct counting argument operating in one fixed dimension throughout.

It also makes no reference to E_n , P_n , or any other specific candidate subgroup. Instead, it shows directly that any group satisfying Axiom 12' must already be the entire B_n , by counting alone, regardless of what such a group might otherwise look like.

This is a genuinely different — and substantially stronger — route than the exhaustive-search strategy used at $n = 3$ (Section 14.3). It resolves the sufficiency direction for every $n \geq 2$ at once, and retroactively confirms the conclusion of *Theorem 3.10* at $n = 3$ by an independent method.

The exhaustive search at $n = 3$ (Section 14.3) and the construction of E_n and P_n (Section 15.3; *Lemma 2.11*; *Corollary 2.9.1*) remain essential and are not made redundant by *Theorem 3.15*. They establish necessity: Axiom 12' is not vacuous or superfluous relative to Axioms 10–11, because concrete proper subgroups — E_n and P_n — exist that satisfy Axioms 10–11 but fail Axiom 12'. This shows that Axiom 12' genuinely rules out real counterexamples.

Theorem 3.15 establishes sufficiency: once Axiom 12' is satisfied, there is no remaining freedom — the automorphism group must already be B_n . Thus necessity and sufficiency are proved by genuinely different methods: • necessity via explicit construction and verification of proper counterexamples, • sufficiency via a dimension-internal counting argument that does not enumerate candidates at all.

Both directions are required for a complete account of the typed-differentiation axioms, and together they give a fully resolved characterization of when

$$\text{Aut}(S, \Delta) = B_n$$

Theorem 3.16 (rank preservation by P_n) is proved in Section 12.5.1 and is not re-proved here; the $n=3$ instance there generalizes directly.

Independent direct computation confirms this pattern at $n = 3, 4, 5$:

$$|P_n| = 24, 192, 1920,$$

respectively, and in each case P_n is transitive on $\{0, 1\}^n$ with rank $n + 1$. These results are fully consistent with the general proof, providing an explicit computational check at the first three nontrivial dimensions.

13.5 Uniqueness at $n=4$

Theorem 3.15 (general- n , Section 15.4) applies directly at $n = 4$, with no exhaustive search and no dependence on classifying E_4 or P_4 as the only candidates. It gives the following immediately: for any (S, Δ) satisfying Axioms 10–12' with

$$S = \{0, 1\}^4 \text{ under Hamming distance}$$

the automorphism group is determined uniquely as

$$\text{Aut}(S, \Delta) \cong B_4$$

Theorem 3.13 (Forced Hyperoctahedral Symmetry at $n=4$).

Any (S, Δ) satisfying Axioms 10–12' with $S = \{0, 1\}^4$ under Hamming distance has $\text{Aut}(S, \Delta) \cong B_4$ uniquely.

Proof.

Immediate from Theorem 3.15 at $n=4$.

□

A Note on What "Characterization" Does and Does Not Mean Here

Theorem 3.13/3.15 should be read as a characterization of the automorphism group, not (yet) as a characterization of the underlying point set. What is established is the following: given that

$$S = \{0, 1\}^n \text{ under Hamming distance,}$$

the axioms do not merely *permit* B_n as one consistent possibility among others — they force it. Once Axioms 10–12' hold, no other group can be $\text{Aut}(S, \Delta)$. In this precise sense, the axioms characterize B_n on this witness model, not merely allow it.

A stronger and genuinely different claim would be that Axioms 10–11 themselves force

$$|S| = 2^n$$

and that *any* (S, Δ) satisfying them — starting from an arbitrary set S , not assumed in advance to be built from binary strings — must be isomorphic to $(\{0, 1\}^n, \text{Hamming})$. This has not been established here. Axiom 11 fixes the rank of $\text{Aut}(S, \Delta)$ on $S \times S$ to be $n + 1$ and requires transitivity, but rank and transitivity alone do not obviously determine $|S|$. It is not immediate that no structure of some other size could satisfy the same orbit-counting requirements.

Establishing this stronger, point-set-level characterization — not just the automorphism-group-level one already proven — is a separate, open question, and is listed explicitly in Section 13.9 (*Open Problem: Point-Set Characterization*).

13.6 Toward No Fifth Type

The Hamming scheme $H(4, 2)$ has Bose–Mesner algebra dimension exactly 5 (rank = $n + 1 = 5$), with the five distance matrices

$$A_0, \dots, A_4$$

forming its basis. The layer sizes — equivalently, the dimensions of the associated primitive idempotents — are given by the binomial coefficients $\binom{4}{k}$ for $k = 0, \dots, 4$:

$$1, 4, 6, 4, 1$$

All of these facts were **independently re-verified by direct computation** for this paper. The verification included:

- constructing the full 16×16 distance matrices $A_0, \dots, A_4$,
- constructing the primitive idempotents via the Hadamard/Walsh basis (*Walsh–Hadamard transform*),
- confirming that the five idempotents are linearly independent,
- confirming that they sum to the identity matrix, and
- confirming that they are mutually orthogonal projectors.

Thus the Bose–Mesner algebra of $H(4, 2)$ was **re-established from first principles**, not merely cited from external sources.

A separate computation tested whether a small set of matrices (the four single-coordinate flip matrices together with one stratification projector, the indicator of the Hamming-weight-4 layer) admits an exact left inverse recovering a single-point indicator within their linear span. The originally reported residual was 1.0 for one point (index 0); re-tested across all 16 points,

the result is not uniform — 15 points give residual 1.0, the remaining one gives exactly 0.0. The reason is elementary: with $n=4$ there is exactly one point of Hamming weight 4 (the all-ones point), and the stratification projector used is, by construction, already that point's own indicator matrix, so testing its "recovery" trivially succeeds with zero residual.

The original choice of test point (index 0) was not representative — it was simply one of the 15 points where this coincidence does not occur — and a proposed structural explanation for the finding (the self-duality of the Hamming graph forcing a uniform unit residual) was checked and rejected; see Section 13.7 for that argument in full.

This numerical result, properly scoped, does not participate in the proof of any theorem in this section (Theorem 3.14 rests entirely on the dimension-counting argument given directly in its own proof above (the $n=4$ instance of the method used in Lemma 2.10 at $n=3$), independent of this computation) and should not be elaborated into a structural narrative beyond what has actually been checked. It is recorded here as an open numerical observation: whether the specific points for which recovery fails (15 of 16, in this particular basis choice) have any structural significance, or are simply an artifact of which stratification layer happens to be a singleton at this specific n , is unresolved and listed as an open item below.

Theorem 3.14 (No Fifth Independent Type at $\{0,1\}^4$).

$\text{Aut}(S,\Delta) \cong B_4$ (Theorem 3.13). Σ cannot be extended by a fifth type σ_5 , distinguishable from $\sigma_1, \dots, \sigma_4$, compatible with $\text{Aut}(S,\Delta)$.

Proof.

The orbits of B_4 on $S \times S$ correspond exactly to the five Hamming-distance classes, with adjacency matrices

$$A_0, \dots, A_4.$$

These five matrices are linearly independent and span an algebra of dimension exactly 5, closed under matrix multiplication — verified directly by the same method used in Lemma 2.10 at $n = 3$.

A candidate fifth type σ_5 would correspond to an additional adjacency matrix M_5 compatible with the same automorphism group. Compatibility with B_4 means that M_5 must be **constant on each B_4 -orbit** on $S \times S$. But the space of all matrices constant on these orbits is **exactly** the span of

$$A_0, \dots, A_4,$$

since these already form a basis for that space.

Therefore any such M_5 is a linear combination of $A_0, \dots, A_4$, and hence cannot be linearly independent of them. Thus no fifth type exists.

□

This argument is structurally identical to Theorem 3.11's at $n=3$, and to Theorem 3.8's at $n=2$, differing only in the dimension count (5 instead of 4 or 3). It now

depends on Theorem 3.13 unconditionally, since Theorem 3.13 itself no longer carries any open dependency (Section 13.5).

Lemma 2.13 (General-n Dimension of the Hamming Distance Algebra).

For every $n \geq 1$, let $A_0, \dots, A_n$ be the $2^n \times 2^n$ matrices with $(A_k)_{xy} = 1$ if $x, y \in \{0, 1\}^n$ are at Hamming distance k , and 0 otherwise. Then $A_0, \dots, A_n$ are linearly independent, their span $\mathcal{A}$ is closed under matrix multiplication, and $\dim(\mathcal{A}) = n+1$.

Proof.

Linear independence: Suppose

$$\sum_{k=0}^n c_k A_k = 0$$

For any fixed pair (x, y) at Hamming distance d , the (x, y) -entry of the sum is exactly c_d , since among the matrices A_k only A_d is nonzero at that entry. Since pairs at every distance $d \in \{0, \dots, n\}$ occur in $\{0, 1\}^n$, we obtain $c_d = 0$ for every d . Thus all coefficients vanish, and the matrices $A_0, \dots, A_n$ are linearly independent.

Multiplicative closure. The group B_n is transitive on $\{0, 1\}^n$ and preserves Hamming distance. Hence each relation

$$R_k = \{(x, y) : \text{dist}(x, y) = k\}$$

is exactly a B_n -orbit on $S \times S$, and A_k is its orbital matrix. Products of orbital matrices in any transitive group action are themselves constant on orbits — a standard fact about centralizer/Bose–Mesner algebras (*Bose–Mesner background*). Concretely, $A_i A_j$ counts the number of intermediate points z such that

$$\text{dist}(x, z) = i, \text{dist}(z, y) = j,$$

and this count depends only on the orbit of (x, y) , i.e. only on $\text{dist}(x, y)$. Therefore

$$A_i A_j = \sum_{k=0}^n p_{ij}^k A_k$$

for some coefficients p_{ij}^k , the intersection numbers of the Hamming scheme. Thus the algebra is closed under multiplication.

Dimension. By linear independence,

$$\dim(\mathcal{A}) \geq n + 1$$

Conversely, B_n has exactly $n + 1$ orbits on $S \times S$, corresponding to the $n + 1$ possible Hamming distances. Any matrix constant on these orbits is, by definition, a linear combination of the orbital matrices

$$A_0, \dots, A_n$$

Hence the space of all B_n -orbit-constant matrices has dimension exactly $n + 1$, and the algebra $\mathcal{A}$ spanned by the A_k is precisely this space. Therefore

$$\dim(\mathcal{A}) = n + 1$$

□

Theorem 3.17 (No (n+2)-th Independent Type, General n).

Let $n \geq 2$ and let (S, Δ) satisfy Axioms 10–12' with

$$S = \{0,1\}^n$$

under Hamming distance, so that

$$\text{Aut}(S, \Delta) \cong B_n$$

by Theorem 3.15. Then the type-set Σ cannot be extended by a further type

$$\sigma_{n+1},$$

Proof.

Any additional type σ_{n+1} would correspond to an adjacency matrix

$$M$$

that is compatible with

$$\text{Aut}(S, \Delta) = B_n.$$

Compatibility means that M must be **constant on each B_n -orbit** on $S \times S$. By Lemma 2.13, the space of all such orbit-constant matrices is exactly the adjacency algebra

$$\mathcal{A} = \text{span}\{A_0, \dots, A_n\},$$

where A_k is the adjacency matrix of the Hamming-distance- k relation. This algebra has **dimension $n + 1$** .

Therefore $M \in \mathcal{A}$, so

$$M = \sum_{k=0}^n c_k A_k$$

for some coefficients c_k . Hence M cannot be linearly independent of $A_0, \dots, A_n$, and thus cannot define a new type σ_{n+1} distinguishable from the existing $\sigma_1, \dots, \sigma_n$.

□

This is the general- n statement that Theorem 3.8 ($n=2$), Theorem 3.11 ($n=3$), and Theorem 3.14 ($n=4$) are each specific instances of; it resolves the "general- n version of Theorem 3.14" item from the open-items list directly, by the same dimension-counting method already used at each specific n , generalized rather than re-derived from scratch.

Citation status: the dimension fact in Lemma 2.13 — the Hamming scheme $H(n,2)$ has $n+1$ classes, with Bose–Mesner algebra dimension equal to the class number — is standard within

algebraic combinatorics. The proof given above does not depend on any external citation: it is self-contained, derived directly from the orbit structure of B_n and elementary linear algebra.

A directly-verified secondary source stating this exact fact has now been found and checked against its actual content (not merely its title or abstract): Chan et al. [6], Section 4, opens by stating that the Bose–Mesner algebra of $H(n,2)$ is spanned by $\{A_0, A_1, \dots, A_n\}$ — exactly the $n+1$ -dimensional spanning set Lemma 2.13 establishes independently.

No specific page-level locator in Bannai & Ito (1984), Brouwer, Cohen & Neumaier (1989), or Godsil has been independently verified against their primary text, and none is claimed here; this remains open, and is recorded as such rather than risking an unconfirmed locator. If one of those primary texts is consulted later and a precise locator is confirmed, it can be added at that point without altering anything else in this section.

13.7 Five Specific Arguments Considered and Rejected

Five shortcuts were proposed for this section and checked; none is used above, and all five are recorded here so they are not reattempted.

- Proposal: “Axiom 4 (Non-Orientation) already excludes E_n and P_n without needing Axiom 12’, because achieving even parity sometimes requires coupling a flip with a coordinate transposition, and this coupling constitutes illegal cross-axis chirality.” This proposal is **incorrect**. The distance Δ (Hamming distance) is fully symmetric **regardless** of which candidate automorphism group is under consideration:

$$\Delta(x, y) = \Delta(y, x)$$

holds identically and never depends on whether B_n , E_n , or P_n is being tested as $\text{Aut}(S, \Delta)$. What Axiom 8(iii)/Axiom 4 (Non-Orientation) actually forbids is an **asymmetry in Δ itself** — for example, assigning different types to (x, y) and (y, x) . The groups E_n and P_n do **not** alter Δ in any way; they only change which permutations are admitted as automorphisms **preserving** that already symmetric Δ .

The phrase “requires a coupled operation rather than a single one” describes a property of a group’s **internal parity structure**, not a property of orientation or chirality in Δ . Conflating these two levels — the symmetry of the distance relation versus the internal composition behaviour of the automorphism group — does **not** yield a valid exclusion argument for E_n or P_n under Axiom 4.

- Proposal: a “Dimensional Consistency principle,” requiring that any n -dimensional candidate group induce the complete $(n-1)$ -dimensional group on every coordinate-fixed $(n-1)$ -point subset, used to exclude candidates without exhaustive search. This is the same “sectional forcing” strategy already proposed, checked, and rejected in Section 12 (Sections 14.3–14.5) for the identical reason: no general fact guarantees that a group’s restriction to a coordinate-fixed subset must equal the full automorphism group that subset would have on its own, rather than a proper subgroup of it (a proper subgroup occurring on a subset is no more illegitimate than C_4 occurring as a subgroup of D_4). Restating this strategy under a new name at a higher dimension does not supply the missing justification.

- Objection: It was argued (Section 15.4) that the proof of *Theorem 3.15* allegedly conflates

$$|G_x| = n!$$

with the induced action of G_x on $\Gamma_1(x)$ being the full $\text{Sym}(n)$. The proposed counterexample was a group whose induced action has image A_n (order $n!/2$) but whose kernel has order 2, giving $|G_x| = n!$ “by another route,” supposedly violating the proof while still failing Axiom 12'. This objection was checked directly and does **not** hold. Such a group does **not** satisfy Axiom 12' in the first place. Axiom 12', together with *Lemma 2.12*, requires that **every** permutation of $\Gamma_1(x)$ — in particular, all **odd** permutations — be realized by some element of G_x . A group whose induced action is only A_n realizes **none** of the odd permutations, regardless of the size of its kernel. It is therefore correctly **outside the hypothesis** of Theorem 3.15, not a counterexample to it. To prevent this exact misreading, the proof of Theorem 3.15 has been rewritten to state the distinction **explicitly**: the relevant quantity is the **image** of the restriction homomorphism

$$\varphi: G_x \rightarrow \text{Sym}(\Gamma_1(x)),$$

not the bare cardinality of G_x . Only when $\text{im}(\varphi) = \text{Sym}(n)$ — not merely $|G_x| = n!$ — does Axiom 12' hold.

- Proposal: an “algebraic mechanism” account of the residual-1.0 finding in Section 13.6, attributing it to the self-duality of the Hamming graph forcing a fixed 1-dimensional nullspace, and asserting this gives “a universal unit residual” at every point as a structural necessity. This was checked directly by re-running the same computation at all 16 points (not just the single point originally tested) and is contradicted by the data: the residual is 1.0 at 15 points but exactly 0.0 at the remaining one, where it turns out the stratification projector used happens to already equal that point’s own indicator matrix by construction — an elementary, basis-specific coincidence, not a uniform algebraic necessity. An elaborate structural narrative should not be built on a numerical finding before checking whether the finding is even uniform across the symmetric points it should apply to if the proposed mechanism were real.
- Proposal: A specific “exotic” counterexample at $n = 5$, claimed to be a subgroup of the affine group

$$\text{AGL}(1,32)$$

of order 160, said to act transitively on the 32 points with rank 6 (matching $n + 1$), with the required independent local transitivity on each $\Gamma_i(x)$, and said to lie outside both E_5 and P_5 .

This proposal fails at the **most elementary level**, long before any rank or transitivity computation is relevant. The order of the affine group is

$$|\text{AGL}(1,32)| = 32 \times 31 = 992.$$

Factorizing,

$$992 = 2^5 \times 31.$$

Since $160 = 2^5 \times 5$, and $5 \nmid 31$, **no subgroup of order 160 can exist** in $\text{AGL}(1,32)$ by Lagrange’s theorem. Thus the claimed example is impossible outright. A named, specific-sounding group with concrete numbers attached is not evidence that the example has been checked. The **first** step in evaluating any such claim must be the most basic divisibility test; if the order of the proposed subgroup does not divide the order of the ambient group, the construction cannot exist. Here that check already eliminates the proposal, and no further structural analysis is needed.

13.8 Computational Verification Suite

Every claim in this section, and the corresponding claims in the oD–2D foundations and Section 12, has been independently re-implemented and tested in a standalone Python package, wgs-verify (not just narrated in self-attack discussion). The package is included with this paper as supplementary material. Full results — every check, its specific numeric output, organized by section — are given in Appendix I; this section states only the scope, in summary form, and what the package deliberately does not attempt.

Twenty-two checks are included: 1 covering oD (Appendix C’s collapse principle, illustrated on a finite toy model — this does not construct oD itself, which has no internal representation by Axioms 1 and 5), 5 covering 1D–2D (forced D₄, Schur-ring dimension 3, Axiom 8(iii)’s chirality necessity), 8 covering Section 12 (the exhaustive Section 12.3 search, E₃/P₃’s profiles, Lemma 2.8/2.10, Theorem 3.9/3.10), and 8 covering this section (E₄/P₄, Lemma 2.11/2.13, the general-*n* sweeps for Theorem 3.15/3.16, the signed-axis counterexample, the AGL(1,32) refutation, and the Reynolds-averaging identity). All 22 pass deterministically; see Appendix I for the full table.

What this package deliberately does NOT do, stated plainly so it is not mistaken for more than it is: it does not validate the axioms against external real-world datasets (genomic sequences, neural connectomes, social networks). Any finite set of equal-length sequences has at most $n+1$ Hamming-distance types as an automatic consequence of how Hamming distance is defined — this holds regardless of whether the data satisfies any of this paper’s group-theoretic axioms, so such a comparison would not test anything specific to this paper’s claims, and is omitted for exactly that reason rather than for lack of effort. It also does not resolve the open classification question (Section 13.9 below) of whether E_n and P_n exhaust all rank- $(n+1)$ candidates for $n \geq 5$; it verifies that E_n and P_n themselves behave as claimed, and that one specific proposed counterexample is impossible, without claiming to have searched the space of all conceivable alternative constructions.

The distinction this section draws — between showing the axioms are not vacuous or self-serving (which this package does, exhaustively, for the finite cases checked) and showing the axioms are "reasonable" in some broader sense (which is not a computationally decidable question, and is not attempted here) — is deliberate. The package answers the first question as completely as the finite cases allow; it does not claim to answer the second.

Separately, and explicitly NOT presented as evidence for the axioms’ truth or reasonableness, a second package, wgs-apps, applies the same verified machinery (B_n , Bose–Mesner algebras, Reynolds averaging) to genuinely falsifiable problems outside the paper — symmetry-based denoising, diagnosing the real automorphism group of a real error-correcting code, and symmetry-deviation anomaly scoring. This package is also included with this paper as supplementary material, separately from wgs-verify. Full methods, results, and a record of related proposals that were tested and rejected, are given in Appendix J. These applications demonstrate the machinery is usable, not that the paper’s structural claims are confirmed by them — the two are different questions and are kept separate throughout.

13.9 Open Items Before This Section Can Be Closed

Two items from this list are now resolved by Theorem 3.15 and Theorem 3.16 above and are recorded here for traceability, not as remaining work.

- Completing an exhaustive search at $n=4$ is not necessary. Theorem 3.15 (Section 13.4) gives sufficiency — Axiom 12' forces $G=B_n$ — directly, for every n , by a counting argument with no dependence on enumerating candidates.
- P_n 's general- n rank preservation is proved as Theorem 3.16 (Section 13.4), mirroring Theorem 3.12's proof for E_n .
- Point-set-level characterization (Section 13.5): determine whether Axioms 10–11 force $|S| = 2^n$ and force any (S, Δ) satisfying them to be isomorphic to $(\{0,1\}^n, \text{Hamming distance})$, starting from an arbitrary set S not assumed in advance to be built from binary strings. What is proven (Theorem 3.13/3.15) is that, given $S = \{0,1\}^n$, the automorphism group is forced to B_n ; this is a different and weaker claim than forcing the point set itself, and the gap between them has not been closed.
- At $n=5$, the classification claim is resolved negatively (full details and verification method in Appendix M): the claim that any $G \leq B_n$, transitive on $\{0,1\}^n$ with rank $n+1$ and satisfying Axiom 11, must be B_n , E_n , or P_n , is false at $n=5$. An exhaustive search of all 953 conjugacy classes of subgroups of B_5 found a fourth structure, of order 640 and index 6, confirmed by direct conjugacy testing to be isomorphic to none of B_5 , E_5 , or P_5 . The classification still holds without exception at $n=3$ and $n=4$; whether a further structure exists at every $n \geq 5$ is a new open question. Theorem 3.15 is unaffected: the order-640 structure satisfies Axioms 10–11 but, like E_n and P_n , does not satisfy Axiom 12', so it does not contradict the separate, stronger claim that Axiom 12' forces $G = B_n$.
- A specific proposed counterexample at $n=5$ (claimed: a subgroup of $\text{AGL}(1,32)$ of order 160, said to act transitively with rank 6 and the required local transitivity) was checked and is false: $|\text{AGL}(1,32)| = 32 \times 31 = 992 = 2^5 \times 31$, which has no factor of 5, so no subgroup of order $160 = 2^5 \times 5$ can exist by Lagrange's theorem alone — this is recorded in Section 13.7 so it is not reintroduced. This false proposal is unrelated to the genuine order-640 structure now confirmed above; the two should not be conflated, and the order-640 structure was found by exhaustive search of subgroups of B_5 itself, not by testing any specific named candidate group.
- Clarify the structural meaning (if any) of the residual-1.0 phenomenon in Section 13.6, now sharpened by direct testing: the residual is 1.0 at 15 of the 16 points and exactly 0.0 at the remaining one (the point whose indicator coincides with the stratification projector used). Determine whether the 15-point/1-point split has any structural significance tied to the specific basis chosen, or is purely an artifact of testing a basis that happens to contain one exact single-point indicator already; either way, this remains outside the proof of any theorem in this section and should not be cited as evidence for irreversibility within the formal centralizer algebra.
- The general- n version of Theorem 3.14 is Theorem 3.17 (Section 13.6), with its supporting dimension count given as Lemma 2.13, generalizing the $n=3$ (Lemma 2.10) and $n=4$ (inline in Theorem 3.14's proof) cases by the same method rather than introducing anything new.
- The specific Brouwer–Cohen–Neumaier locator for Lemma 2.13 will not be pursued further or used. The book is a real source, and corroborating evidence for a "Chapter 9" location was found, but the exact section/page was never independently confirmed against the primary text. Since Lemma 2.13's proof is fully self-contained and does not

depend on this citation, Lemma 2.13 now states the fact as standard without naming a specific external locator (Section 13.6). A precise locator could be added later without changing the proof; this is optional polish, not a gap.

- No directional language; "4D" characterizes a relation to 0D/1D/2D/3D, not a claim about priority (apply the same bidirectionality discipline as Section 10.4's remark, once a genuine 3D/4D threshold theorem is drafted).
- Relation to existing classification literature: the classification question above (which groups $G \leq B_n$, transitive with rank $n+1$ and satisfying Axiom 11, can arise) sits within an active area of permutation group theory, and a closer look identifies a substantially better-targeted reference than the two noted in an earlier pass. $B_n = S_2 \text{ wr } S_n$ in product action on $\{0,1\}^n$ is, in the terminology of Praeger & Schneider's monograph on permutation groups and Cartesian decompositions [2], precisely the full stabilizer of a homogeneous Cartesian decomposition of $\{0,1\}^n$ into n binary coordinates; that monograph develops, via O'Nan–Scott-style theory for primitive, quasiprimitive, and innately transitive groups, exactly the machinery for classifying subgroups that preserve or partially preserve such a decomposition, and explicitly covers automorphism groups of Hamming graphs and their generalizations. This is a markedly closer match to this paper's question than Liebeck & Saxl's minimal-degree classification [7] or Kivva's rank-4 result [8], both retained below as still relevant but more tangential context.
- This match has not been confirmed against the monograph's primary text — it has not been independently obtained and read, and no specific theorem or page within it is cited here, consistent with this paper's standard for unverified locators (cf. the Brouwer–Cohen–Neumaier locator left unresolved in Section 13.6). A genuine obstacle to a quick resolution is visible even without consulting the primary text: the order-640 structure found at $n=5$ (Appendix M) is built from the Frobenius group $\text{AGL}(1,5)$, order 20, acting on the 5 coordinates — a construction available specifically because 5 is prime, not an instance of the wreath-product-in-product-action pattern the Cartesian-decomposition machinery most directly classifies (that pattern covers subgroups that themselves decompose as wreath products over a smaller alphabet; the order-640 group does not obviously do so). Whether the monograph's broader O'Nan–Scott-style classification nonetheless covers this case is exactly the open question, not yet settled here.
- Tested, negatively, for the natural $n=7$ candidate (full derivation in Appendix N): the construction underlying Appendix M's order-640 structure, $(\mathbb{Z}_2)^n \rtimes \text{AGL}(1,n)$, does not generalize to $n=7$ — it gives rank 10, not the required 8 — and a derived criterion explains why: this construction can satisfy Axiom 11's rank- $(n+1)$ condition only when $n \leq 5$, with $n=5$ sitting exactly at that boundary. This rules out one specific, natural family of candidates and narrows, without resolving, the general classification question.

- Summary

Part V confirms that nothing further needs to be added once the general- n machinery established in Part IV is in hand: Theorem 3.15 forces B_4 at $\{0,1\}^4$ directly, without repeating the exhaustive search Part IV needed at $n=3$. Most of this part is devoted to ruling out a fifth independent difference type by the same Bose–Mesner dimension-counting method (Theorem 3.14), and to a detailed record of five distinct shortcuts to this exclusion that were tried and rejected, each for a specific, stated reason (Section 13.7). Every numerical claim in this part is independently re-verified by a standalone computational package (Appendix I). The part closes by listing what remains open: most significantly, whether the classification of rank- $(n+1)$

subgroups under Axioms 10–11 alone continues to hold beyond $n=4$ — a question Appendix M answers negatively at $n=5$, and Appendix N narrows further at $n=7$.

14. Discussion

This section steps back from the proofs of Sections 2–13 to address several standing concerns now that oD through 4D have each been established in turn: the self-referential character of describing a structural base state, how oD's scope relates to the rest of this paper and to the longer research programme beyond it, and two methodological observations — one about the pattern common to every layer's forcing argument, one about the specific role exhaustive computation played in finding what no axiom in this paper anticipated — that only become visible once the whole development is in view.

14.1 Why oD Must Be Self-Referential

External references are impossible for oD in the ordinary sense, since there is nothing prior to oD to refer to. The paper resolves the resulting self-reference concern not by denying it but by stratifying it. The meta-language used to describe oD is explicitly separated from oD's internal content (Appendix A). The apparent infinite regress this might seem to invite is shown to terminate after a single step (Appendix B).

14.2 oD and Structural Language

Within the structural language developed here, oD occupies the base position: the unique entity exhibiting no distinction. Theorem 3.3 establishes that the minimal distinction relating oD and 1D is unique; the analogous pattern from 1D onward — how each layer's automorphism group is forced once the next axiom in the sequence is imposed — is established directly in this paper, layer by layer through Sections 11–13 and then for every n simultaneously from Section 12 onward (Theorem 3.15). What remains open is not the pattern within oD–4D itself, but its extension to general cardinality at 2D and its point-set-level characterization at each n (Section 16.1).

14.3 oD and WEMP

oD is the only state with zero structural energy (Theorem 3.1). Its relation to the WEMP energy functional is a matter of external compatibility, not internal proof; see Appendix D for the precise scope of this relation.

14.4 oD and the Iterated Distinction Hierarchy

Theorem 3.4 establishes an iterated distinction relation linking oD to higher-dimensional layers. Each stage's distinguishing capacity is grounded in the structure already established at the prior stage.

This paper characterizes that structure through 4D directly: Sections 11–13 establish 1D and 2D individually, and Sections 14–15 establish 3D and 4D via the general- n axioms, with Theorem 3.15 forcing the pattern at every $n \geq 2$ simultaneously. Dimensions 5 and beyond fall under the exploratory tier described in Section 1.7.

14.5 Methodological Note on the Two-Tier Programme

We flag explicitly the asymmetry in rigor across the planned sequence. Heuristic treatments — proposing that a given dimensional layer behaves analogously to a known structure in physics, mathematics, or information theory — are valuable for identifying promising directions, but they do not constitute proof and should not be cited as if they were.

Any analogy drawn beyond the foundational tier of this programme (Section 1.7) is to be read as a conjecture inviting rigorous treatment, not as an established result. This note applies to all future papers in the sequence that explicitly identify themselves as belonging to the exploratory tier, whatever range of dimensions that tier is eventually found to cover.

14.6 On the Joint Specification of Adjacent Layers

We flag a structural feature of the results already established, rather than introducing a new claim: the axiom systems characterizing adjacent layers in this sequence are not independently stipulable. Theorem 3.6 already states this precisely as a logical biconditional, and §9.4 notes explicitly that the relation it expresses is symmetric in its logical content. This section names the general pattern that instance illustrates.

Axiom 6, taken together with Axiom 7, is exactly what forces $\text{Aut}(S, \Delta)$ to be the full symmetric group on S (Proposition 1.5), and exactly what Theorem 3.5 uses to exclude any coordinatizing partition. Theorem 3.6 identifies the relaxation of that same axiom — not a separately chosen condition, but the precise negation of the restriction doing the forcing — as the condition under which a coordinatizing partition first becomes possible. These are not two facts about two separate objects that happen to coincide; they are one logical relation, stated from either side.

It follows that a layer's axioms cannot be fixed in isolation from a specification of what its forced structure excludes. To state Axioms 6–7 fully is already to state, implicitly, what must fail for them to hold — and Theorem 3.6 makes that implicit content explicit by identifying it exactly. The same holds one level up: Axioms 8–9 are stated as the relaxation Theorem 3.6 requires, not

as an independent specification arrived at by some other route, and §10.4 (Theorem 3.8) shows in turn what those axioms, once adopted, forbid adding.

This is a claim about logical dependency among specifications. It is not a claim about which layer is prior, more fundamental, or temporally first. A specification of forced structure and a specification of what relaxes it are two readings of a single biconditional — in the same sense that "P implies Q" and "not-Q implies not-P" are two readings of one fact, not a temporal sequence in which one is established before the other. (Section 1.7 and the note following Theorem 3.3 in §1.7 already establish this distinction; nothing in this section revises it.)

14.7 Across 0D–4D: A Common Methodological Pattern

Stepping back from the layer-by-layer development of Sections 2–13, one pattern recurs at every threshold: each time a layer's automorphism group is fully forced (1D's full symmetric group, 2D's D_4 , then B_n at every $n \geq 2$ simultaneously), the tool needed to rule out one further independent difference type escalates with the rank already forced — 2-transitivity and primitivity at rank 2 (Lemma 2.6), a rank-3 Schur ring at 2D (Theorem 3.8), and a rank- $(n+1)$ Bose–Mesner algebra from 3D onward (Lemma 2.10, Lemma 2.13, Theorems 3.11/3.14/3.17). No single tool is reused unchanged across layers; each is the one a structure of that specific rank requires, not a default carried over from the layer before. This is recorded here as an observation about the proofs already given, not as a new claim requiring separate justification.

14.8 The $n=5$ Exception and the Limits of Axiomatic Anticipation

Theorem 3.15's counting argument forces B_n under Axiom 12' at every $n \geq 2$ without exception, but the weaker classification question — whether E_n and P_n exhaust the alternatives once Axiom 12' is dropped — was not settled by any axiom in this paper; it was settled by exhaustive computation, and it was settled negatively, at $n=5$ (Appendix M). No step in the axiomatic development up to that point predicted this exception or excluded it. This is worth stating plainly: the computational verification reported throughout this paper (Appendix I, twenty-two checks) is not only a check on claims already derived by other means; at $n=5$ it is the sole means by which a genuine structural fact — a fourth structure existing where exactly two were expected — was established at all. The subsequent test at $n=7$ (Appendix N), by contrast, was guided by a hypothesis derived in advance (the rank condition $n \leq 5$) and confirmed it computationally; the two appendices illustrate different relationships between proof and computation within the same paper, not a single uniform method applied twice.

15. Implications

Having established axioms and internal properties for each layer from oD through 4D, this section considers what they imply beyond the immediate axiomatic development — for mathematics generally, for the specific literature this paper's general-n results connect to, for other fields, and for the methodological stance this paper takes throughout.

15.1 For Mathematics

oD admits an analogy to an initial object in the category of structural states [10]: a structural state admitting (up to meta-linguistic equivalence) no nontrivial internal structure, standing in a unique relation to every distinction-bearing structure. We note that "initial object" is itself a directional term within category theory (it carries a unique morphism to every other object); the analogy is offered only to situate oD within an established mathematical vocabulary, not to assert that this directionality holds of oD itself. No categorical apparatus is assumed to act internally on oD.

15.2 For Physics

oD serves as the structural counterpart against which any physical description is defined: a configuration in which no degree of freedom, no observable, and no symmetry action exist. Physical theories require at least one distinguishable degree of freedom; the minimal distinction relating oD and 1D (Theorem 3.3) characterizes the boundary between their absence and their presence.

In this sense, oD is not a physical state but the logical precondition for the existence of physical states, and the minimal distinction identified in Theorem 3.3 marks the boundary at which physical formalism has something to act on. The precise dimension at which physical formalism becomes fully applicable is not asserted in this paper; see Appendix F.7.

15.3 For Information Theory

oD is the zero-information state (Axiom 5): the unique state exhibiting no distinguishability, related to the minimal non-zero unit of distinguishability — one degree of freedom — characterized in Appendix E.

15.4 For Structural Language

oD occupies the base position in the iterated distinction structure developed within this structural language, with each higher layer characterized by iterated distinction rather than by any presupposed geometric background.

15.5 For Finite Group Theory and Algebraic Combinatorics

The order-640 structure found at $n=5$ (Appendix M) is, as far as a search of the readily available literature can establish, a new addition to the record of rank- $(n+1)$ subgroups of B_n satisfying Axiom 11's local condition (Section 1.4 places it relative to Liebeck & Saxl's minimal-degree classification [7] and Kivva's rank-4 result [8]; Section 13.9 identifies Praeger & Schneider's Cartesian-decomposition machinery [2] as a closer match still unconfirmed against the primary text). Independently of where it eventually finds its place within that literature, the construction technique — a Frobenius group $AGL(1,p)$ acting on p coordinates, available specifically when p is prime — together with the derived criterion in Appendix N restricting this exact technique to $n \leq 5$, is offered as a concrete addition to the toolkit for testing whether a given rank- $(n+1)$ classification is genuinely exhaustive at a specific n , independent of this paper's own axiomatic motivation for asking the question.

15.6 For Computational Group Theory

Section 14.8 already notes the methodological point this implies more broadly: an exhaustive search of a specific group's subgroup lattice (here, all 953 conjugacy classes of subgroups of B_5) is reported as a discovery procedure in its own right, not merely a confirmation step performed after a result is otherwise in hand. The standalone verification package accompanying this paper (Appendix I; Data and Code Availability) reproduces this search independently, in the ordinary sense that computational group theory requires — not as a record of having been run once by the author, but as a check any reader can re-run and recompute for themselves.

- Summary

oD is definable without circularity, axiomatizable in a small mutually consistent set of axioms (Axioms 0–5), and uniquely determined (Theorem 3.2). The minimal distinction relating oD to a one-dimensional structure is likewise unique (Theorem 3.3), established by two independent arguments — one meta-linguistic, one object-linguistic — that converge on the same conclusion. This same axiomatic discipline, applied layer by layer through Sections 9–13, is what forces B_n at every $n \geq 2$ and surfaces the one exception this paper did not anticipate at $n=5$; Section 16 closes with what remains open at every layer.

16. Conclusion

We have shown that the zero-dimensional state is definable without circularity. It is axiomatizable in a small and mutually consistent set of axioms. It is self-contained once the

meta-language/object-language separation is made explicit. It is uniquely determined, in the sense made precise by Theorem 3.2. We have further shown that the minimal distinction relating 0D and 1D is unique (Theorem 3.3), with an explicit construction given in Appendix E.

Beyond this, we have characterized the internal anatomy of 1D directly, via a state space and a binary differentiation relation (Axioms 6–7), and shown that the automorphism group of any such structure is forced to be maximal (Proposition 1.5). From this, by an argument grounded in standard permutation group theory rather than in the meta-linguistic apparatus of Section 7 — via a cardinality-independent primitivity lemma (Lemma 2.6) — we obtained a second, independent confirmation, compatible with that given in Theorem 3.3(ii), that no one-dimensional structure admits a nontrivial coordinatizing partition (Theorem 3.5). We then identified the unique minimal relaxation — abandoning the restriction to a single undifferentiated type of difference — under which this limitation no longer applies (Theorem 3.6), and exhibited an explicit structure witnessing that this relaxed condition is genuinely satisfiable and not vacuous (Appendix H).

Building on this witness, we gave a self-contained axiomatization of 2D’s own internal anatomy (Axioms 8–9). This required a symmetry clause excluding a chirality counterexample — one that would otherwise smuggle in orientation through type assignment. We showed by explicit construction that two further candidate automorphism groups, C_4 and V_4 , cannot be realized as the full automorphism group of any structure satisfying these axioms at the smallest non-trivial finite cardinality (Lemma 2.7).

This forces the automorphism group uniquely to the dihedral group of order 8, with rank exactly 3 (Theorem 3.7) — a 2D analogue of Proposition 1.5. Via the associated Schur ring, it also rules out any third independent difference type (Theorem 3.8) — a 2D analogue of Theorem 3.5.

Section 11 connects this back to the general- n framework developed independently in Sections 14–15: Axioms 8–9 are exactly the $n=2$ instance of Axioms 10–11, and the general- n sufficiency theorem proved there (Theorem 3.15) already re-derives $\text{Aut}(S, \Delta) \cong D_4 (\cong B_2)$ at $|S|=4$ by an entirely independent route. The reason 2D never needed a saturation axiom like Axiom 12’ is now precise: at the sizes $|\Gamma_1(x)|=2$ and $|\Gamma_2(x)|=1$ forced at 2D, Lagrange’s theorem alone makes transitivity and full symmetric action the same condition, a coincidence that breaks at $n \geq 3$.

The general, arbitrary-cardinality case of 2D remains open. So does the threshold from 2D to 3D. The full axiomatization of 3D and 4D themselves, by contrast, is no longer subsequent work: Sections 14–15 carry it out, at the smallest non-trivial finite cardinality for each, together with a general- n sufficiency theorem (Theorem 3.15) covering every $n \geq 2$ at once.

At 3D, extending the typed-differentiation axioms to general n (Axioms 10–11) was not by itself sufficient to force the hyperoctahedral group B_3 : an exhaustive search at the smallest cardinality (Section 12.3) uncovered two further candidates of order 24, requiring a local-saturation axiom (Axiom 12’, derived from a more primitive principle as Theorem 3.9) to exclude them (Lemma 2.8, Theorem 3.10). The same obstruction recurs at every $n \geq 3$, not merely at $n=3$ (Corollary 2.9.1), via the even-parity subgroup E_n , which we showed shares B_n ’s exact rank without being B_n itself. With Axiom 12’ in place, a direct counting argument — not an exhaustive search — establishes sufficiency at every $n \geq 2$ simultaneously (Theorem 3.15), and rules out a fourth

independent type at $|S|=8$ via the same dimension-counting method used at 2D (Lemma 2.10, Theorem 3.11).

At 4D, the same general- n axioms, now instantiated at $|S|=16$, were shown from the outset to require no further saturation axiom beyond what 3D already established: Theorem 3.15 applies unchanged, forcing B_4 directly (Theorem 3.13), and a fifth independent type is excluded by the same Bose–Mesner dimension-counting argument, generalized to arbitrary n (Lemma 2.13, Theorem 3.14, Theorem 3.17). Considerable care was taken throughout Sections 14–15 to exclude geometric vocabulary — no description of $\{0,1\}^3$ or $\{0,1\}^4$ as a cube or tesseract — since the witness model is a purely combinatorial object and reasoning that leans on a geometric picture of it is reasoning about the picture, not about the structure (S, Δ) itself. Every numerical claim in Sections 14–15 is independently re-verified by a standalone computational package, reported in full in Appendix I.

One finding sits outside this otherwise uniform picture. Theorem 3.15 settles sufficiency under Axiom 12' at every n , but the weaker classification question — whether E_n and P_n exhaust the rank- $(n+1)$ subgroups of B_n satisfying only Axioms 10–11 — turned out to be false at $n=5$: an exhaustive search of B_5 's entire subgroup lattice produced a fourth structure, of order 640, distinct from B_5 , E_5 , and P_5 alike (Appendix M). The most natural way of extending this structure's construction to the next prime, $n=7$, was tested directly and shown to fail, for a reason now fully derived rather than merely observed: the construction satisfies the required rank condition if and only if $n \leq 5$ (Appendix N). Whether some different construction produces a further exception at $n=7$ or beyond is the one classification question this paper leaves genuinely open, rather than merely unattempted.

The self-referential character of a theory of a structural base state is not a defect to be hidden. It is a feature to be made precise; Appendices A and B are offered in that spirit. Likewise, two independent arguments converge on a compatible conclusion about 1D — one meta-linguistic, one object-linguistic. This convergence is offered not as redundancy but as a consistency check, appropriate to a foundational layer that later work will build upon.

16.1 Open Questions

Three questions are stated as open at the points in this paper where they arise; they are collected here for reference, not restated with new content.

- The general, arbitrary-cardinality case of 2D, and the threshold from 2D to 3D, are not characterized in this paper (Section 1.7, Abstract).
- Whether E_n and P_n exhaust every rank- $(n+1)$ proper subgroup of B_n satisfying Axioms 10–11 (without Axiom 12') holds without exception at $n=3$ and $n=4$, but is confirmed false at $n=5$ (Appendix M): an exhaustive search of B_5 's subgroup lattice found a fourth structure, of order 640, isomorphic to none of B_5 , E_5 , or P_5 . The most natural way of extending that structure's construction to $n=7$ was tested directly and fails, for a precise, derived reason rather than by accident (Appendix N): the construction satisfies Axiom 11's rank condition only when $n \leq 5$. Whether some other construction produces a further exception at $n=7$ or beyond remains open. Theorem 3.15 makes the question lower-stakes regardless of how it resolves — no candidate can satisfy Axiom 12' without

being all of B_n — but the classification question for Axioms 10–11 alone, without Axiom 12', is open at every $n \geq 6$.

- No specific page-level locator in Bannai & Ito (1984), Brouwer, Cohen & Neumaier (1989), or Godsil has been independently verified against their primary text for the Bose–Mesner algebra dimension fact used in Lemma 2.13; a directly-checked secondary source [6] has been cited instead (Section 12, Citation status).

16.2 Future Work

Four directions, beyond the open questions of Section 16.1, are recorded here as concrete extensions this paper does not attempt.

First, a unified specification connecting the axiom systems of 0D through 4D — making explicit, in one place, the logical relations already established piecewise throughout this paper (the biconditional between 1D's forced maximality and 2D's relaxation in Section 9.4; the $n=2$ identification between Axioms 8–9 and Axioms 10–11 worked out in Section 11) — would extend Section 11's treatment of 2D to all five layers at once.

Second, the rank- $(n+1)$ classification question left open at $n \geq 6$ admits at least one further avenue beyond the $AGL(1,n)$ construction ruled out in Appendix N: the other 2-transitive group of degree 7, $PSL(3,2)$ acting on the Fano plane, was identified as a candidate but not tested directly (Appendix N). A complete answer at $n=6$ and $n=7$ would require either testing this and other sporadic 2-transitive actions directly, or settling whether Praeger and Schneider's general machinery for groups preserving a Cartesian decomposition [2] resolves this specific classification question — a connection identified in Section 13.9 but not yet confirmed against the primary text.

Third, the point-set-level characterization left open throughout Sections 14–15 — whether Axioms 10–11 force $|S| = 2^n$ in the first place, rather than only fixing the automorphism group once $S = \{0,1\}^n$ is assumed — remains a genuinely separate, harder question. The sampling evidence gathered in Section 13.5 (four further degree-8 transitive actions tested directly and excluded) is suggestive but falls short of a proof.

Fourth, the general, arbitrary-cardinality case of 2D (Section 16.1) would require extending the primitivity argument of Lemma 2.6, used at 1D to rule out a coordinatizing partition there, to a structure with more than two independent types of marked difference — a genuinely different and harder symmetry question than the binary, $|S|=4$ case treated in Section 10.

Appendix A — Full Axiom System

Axioms 0–5 are restated here in full formal notation for reference (notation to be finalized in the formal write-up). See Section 3 for the discursive statement of each axiom.

Appendix A.0 — Meta-Language Disclaimer

Tools Used to Describe oD Do Not Belong to oD

The formal apparatus used throughout this paper — set, object, element, singleton — belongs to the meta-language in which the zero-dimensional state is described, not to the internal content of that state. This distinction is standard in foundational mathematics: the empty set, for instance, is described using set-theoretic vocabulary without that vocabulary becoming part of the empty set's content.

Definition 0.1 is a meta-linguistic designation, a way of pointing at an object from outside; it does not assert that oD internally contains the structure used to point at it. Axiom 2 (Absence of Boundary) is a claim about the internal constitution of oD, not a constraint on the language used to discuss it. The two levels — what oD contains, and what we use to talk about what it contains — are categorically distinct and must not be conflated.

Appendix B — Proofs of Lemmas and Theorems

Full formal proofs are collected here; see Sections 6–7 for proof sketches given in context.

Appendix B.0 — Structural Minimality and the Absence of Internal Morphisms

oD admits no internal morphisms: there is no structure-preserving map from oD to itself other than the trivial identity, since there is no internal structure to preserve or permute (Axioms 1–5). The meta-language, by contrast, can speak of equivalence between structures satisfying Axioms 0–5 (Lemma 2.5) without this equivalence constituting an internal morphism.

"Isomorphism" in Theorem 3.2 refers to equivalence in the meta-language, not to an internal morphism of oD.

Appendix B.1 — Defense Against the Infinite Regress Objection

A natural objection to part (ii) of Theorem 3.3 runs as follows. Distinguishing two relations from each other requires a further relation. Does distinguishing that relation from the first two not require yet another? This would generate an infinite regress that undermines the entire argument.

This objection misidentifies where the regress would have to occur. The argument in (ii) does not claim that distinguishing relations in general is impossible without an infinite hierarchy. It

claims something narrower, entirely local to the situation at hand: a single atomic application of a distinction operator to oD cannot output two relations as a joint, simultaneous product. The reason is specific: doing so would require the meta-language to already possess a way of telling those two relations apart, prior to and independent of the very operation meant to produce them.

This is not a claim about the general impossibility of multiplicity; it is a claim about the order in which distinguishing-power is licensed. Once $1D$ exists (i.e., once a single distinction has been marked), the resources to distinguish further relations from each other and from the first become available at the next stage, where they are themselves licensed by the structure already present. There is no regress because each stage's distinguishing power is grounded in the distinction-content already established at the prior stage — this is precisely the content of Theorem 3.4, which describes exactly this iterative grounding. The regress threat dissolves once one observes that distinguishing-power is itself characterized stage by stage, not presupposed all at once prior to any structure existing.

This pattern has a recognized precedent in foundational mathematics and logic. Type theory stratifies levels to avoid self-reference paradoxes [15, 16]. Category theory draws a related distinction, between morphisms (which relate objects) and 2-morphisms (which relate morphisms). Both rely on the same principle: the apparatus needed to distinguish entities at one level is introduced only at the next level, never presupposed at the level it is meant to operate on.

Theorem 3.3 invokes this same stratification locally, for exactly one step. It does not posit an infinite tower. The claim is bounded and terminates immediately, since only one application of the distinction operator is under consideration.

This local, single-step stratification should not be confused with the self-referential structures addressed by Gödel's incompleteness theorems [17]. Those theorems concern the provability of arithmetic statements within a fixed formal system, not the meta-language/object-language separation at issue here. The two phenomena share the word "self-reference" but are otherwise unrelated; no inference from one to the other is drawn or required in this paper.

Appendix C — Structural Algebra of oD

This appendix completes Definition 0.2 by deriving, rather than merely asserting, that oD 's algebra is necessarily minimal. Lemma 2.4 and Corollary 4.4 establish this for one specific operation (folding); what follows generalizes that result to every operation of the relevant kind, using the same collapse method already used by Lemma 2.5 and Theorem 3.3(ii).

Definition C.1 (Internally-Grounded Operation).

An operation σ taking oD as an argument is internally-grounded if its output is required to be derived from structural features oD is hypothesized to supply — a boundary, a tension, an orientation, a substructure — rather than stipulated by the meta-language independently of oD 's own content.

This restriction is necessary and is not a technicality to be glossed over: an externally stipulated constant map — "always output 1D, regardless of argument" — trivially takes oD as input and returns something other than oD, but its output does not derive from anything oD supplies; it is simply declared by the meta-language. Such a map is excluded from this appendix's scope by Definition C.1, and is in any case never proposed anywhere in this paper as a genuine operation belonging to oD's own algebra. Folding (Lemma 2.4) is internally-grounded in the relevant sense: it is defined as an attempt to extract a reduced internal representation of whatever structure is given to it, not as an externally stipulated relabeling.

Proposition C.2 (Closure of oD Under Every Internally-Grounded Operation).

Let σ be any internally-grounded operation (Definition C.1), of any arity, taking oD in one or more argument positions. Then $\sigma(oD, \dots) = oD$.

Proof.

Suppose toward contradiction that $\sigma(oD) = O$ for some $O \neq oD$. Since σ is internally-grounded, O must be derived from some structural feature oD supplies. By Axiom 1 (Indivisibility) and Corollary 4.1, oD contains no substructure for σ to extract from. By Axiom 2 (Absence of Boundary) and Axiom 3 (Zero Tension), oD supplies neither a boundary nor a tension. By Axiom 4 (Non-Orientation), oD supplies no orientation. By Axiom 5 (Minimal Information), oD supplies no further structural resource beyond these. σ therefore has nothing internal to derive O from.

Since O is asserted distinct from oD, this difference must itself be grounded in something — by hypothesis, in some feature σ derived from oD's own content. No such feature exists, by the exhaustion just given. By the same collapse argument used in Lemma 2.5 and Theorem 3.3(ii): a difference with no internal grounding has no distinguishing predicate, and anything lacking a distinguishing predicate is meta-linguistically identified with what it cannot be distinguished from. O therefore collapses into identification with oD, contradicting $O \neq oD$.

Hence $\sigma(oD) = oD$ for every internally-grounded σ , of any arity.

□

Corollary C.3 (Folding Is the Special Case).

Lemma 2.4 and Corollary 4.4 (oD is the fixed point of folding) are the single-operation, unary instance of Proposition C.2, with σ taken to be the folding operator.

Corollary C.4 (oD Is the Null Element of the Structural Algebra).

For any algebra of internally-grounded operations on structural states — whatever such an algebra is eventually developed for higher-dimensional layers, a question left to subsequent work (Section 1.7, Section 14.2) — its restriction to oD has exactly one element, oD itself, and every operation in that algebra, applied to oD, returns oD.

This is the precise content of Definition 0.2: oD is the null element not by further stipulation, but because Proposition C.2 leaves no other possibility. The scope of this claim should not be overstated: this appendix establishes only that oD's own algebra, restricted to itself, is trivial. It does not address operations combining oD with some non-oD argument (e.g. a binary operation

taking oD and a 1D structure) — such operations belong to the still-undeveloped general algebra of higher layers, and nothing here constrains what they might return.

Appendix D — External Framework Compatibility

D.o — WEMP Compatibility, Not Internal Proof

The energy functional $E(oD) = 0$ belongs to the Well-formed Energy Mapping Protocol (WEMP), an external framework not assumed by Axioms 0–5 or required by Theorems 3.1–3.4. Theorem 3.1 (Zero Structural Energy) is derived purely from Axioms 2–3 via Lemmas 2.1–2.2, internal to this paper’s formalism.

This appendix’s role is restricted to compatibility verification: it shows that if one separately adopts the WEMP energy functional, the zero-dimensional state as axiomatized here is consistent with — and maps onto — the zero point of that functional. This compatibility check has no bearing on, and contributes nothing to, the internal proofs of Section 7.

A specific objection this guards against, stated explicitly so it is not mistaken for an oversight: citing WEMP’s formula for $E(oD) = 0$ might seem to concede that WEMP’s apparatus has been smuggled into oD’s internal content. That apparatus includes derivatives, operators, an integration manifold, and the degrees of freedom such a functional presupposes. On this objection, oD would then be a disguised higher-dimensional manifold, not the structureless state Axioms 0–5 describe.

This does not follow. The compatibility check in this appendix relates two independently-defined objects: oD as axiomatized internally (Axioms 0–5), and the zero point of an external functional defined entirely on its own terms. That functional requires no reference to oD for its own definition.

Mapping one object onto the other does not transfer the apparatus of either into the other. A function having a value of zero at some point in its domain does not require that point to itself contain the function’s machinery — a thermometer reading zero degrees does not import the thermometer’s internal mechanism into the temperature it measures.

WEMP’s degrees of freedom belong to WEMP’s domain, not to oD. The mapping recorded here is external, in the same sense and for the same reason that this paper’s relation to the broader WEMP framework is external throughout.

Appendix E — Formal Construction of the Minimal Distinction Between oD and 1D

(Distinction-operator construction; meta-linguistic; framework-independent.)

E.1 Purpose of This Appendix

This appendix provides a fully formal, meta-linguistic construction of the minimal distinction relating the zero-dimensional state to a one-dimensional structure. The construction is external: it is performed in the meta-language and does not introduce internal structure into oD itself. It does not use the word "offset" or any term implying displacement, shift, movement, or direction; it does not assume any directional or metric structure; and it does not rely on any framework beyond the axioms already established. Its purpose is to show that the smallest possible act of differentiation is necessarily characterized by a one-dimensional structure, and to instantiate explicitly the abstract argument of Theorem 3.3. The construction characterizes a relation; it asserts no priority or process between the two structures it relates (Section 1.7).

E.2 What a Distinction Is (Meta-Linguistic Only)

Definition E.1 (Meta-Distinction).

A distinction is a meta-linguistic act that marks a difference between two referential positions, without assuming any internal structure of the objects being distinguished. We adopt the minimal notion of a distinction in the sense of Spencer-Brown's Laws of Form [18], stripped of all algebraic content beyond the bare act of marking.

Important clarifications:

- A distinction is not a structure inside oD.
- A distinction is marked by the meta-language, not by oD.
- A distinction marks two positions — the marked position and the unmarked position — but these are not elements and not points; they are pure referential placeholders, not members of any set.
- No ordering, metric, topology, or orientation is assumed.

This is the minimal notion of "difference" compatible with Axioms 0–5.

E.3 The Minimal Distinction Construction

Step 1 — oD

Let oD be the unique object satisfying Axioms 0–5: no parts, no boundary, no orientation, no internal relations, no degrees of freedom.

Step 2 — A single meta-distinction

We mark a distinction in the meta-language, written $\langle \square | \blacksquare \rangle$ in the notation of Laws of Form. This notation does not imply set membership, ordering, spatial separation, or direction. It simply indicates that two referential positions are distinguishable.

Step 3 — The marked and unmarked positions

The marked position ■ and the unmarked position □ together constitute the distinction: the unmarked position is the referential trace of oD, and the marked position is what is distinguished from it. Neither position is treated as prior to or productive of the other.

Definition E.2 (Minimal Distinction-Pair).

A minimal distinction-pair is a second referential position (the marked position) distinguishable from the position referencing oD (the unmarked position), without any further structure.

Crucially: this is not a displacement, not a shift, and not a transformation of oD. It is purely a meta-linguistic differentiation, expressed as the relation $oD \leftrightarrow \langle \square | \blacksquare \rangle$, where $\leftrightarrow$ denotes correspondence rather than a one-way process.

A clarification on Axiom 4 is in order. The labeling of one position as "marked" and the other as "unmarked" is not an ordered pair. It does not constitute a relation internal to oD or to the distinguished structure. It is instead a meta-linguistic labeling convention, recording which referential position is currently under discussion. This asymmetry is epistemic — a feature of how the meta-language refers — rather than structural or geometric — a feature of what is referred to. Because the labeling is external to oD and encodes no internal relation between the two positions, it does not introduce the orientation that Axiom 4 forbids.

E.4 Why Two Positions Necessitate Exactly One Degree of Freedom — and Define One-Dimensionality

Theorem E.3 (Minimal Representational Capacity of a Distinction).

Let the meta-language distinguish two positions □ and ■ as in E.3. We claim the minimal representational capacity required to host this distinction, and no more, consists of exactly one degree of freedom — the capacity to register which of the two positions is in view.

Necessity.

By Axioms 1–5, oD possesses zero degrees of freedom and therefore cannot register any distinction whatsoever; some non-zero representational capacity is required.

Sufficiency.

One degree of freedom — a single binary register of "which position" — suffices to encode the distinction $\langle \square | \blacksquare \rangle$ completely; no further structure (metric, order, topology, orientation) is invoked or needed.

Exactness.

By Lemma 2.3' (Atomicity of the Distinction Operator), a single atomic distinction cannot mark more than one degree of freedom: any additional degree of freedom would itself require a second-order distinction to tell the resulting relations apart, a capacity unavailable prior to the first distinction's occurrence (cf. Appendix B.1).

We define this minimal, necessary, and exact capacity — one degree of freedom, free of metric, order, topology, or orientation — as the one-dimensional structure characterizing the minimal distinction involving oD (Definition E.4 below). One-

dimensionality is thus identified with this minimal capacity, not assumed in advance and subsequently shown to fit.

□

E.5 Formal Definition of the 1D Structure

Definition E.4 (1D Structure (Minimal Distinction Case)).

The one-dimensional structure is the minimal structure that can represent the meta-linguistic distinction relating it to oD, as introduced in Definition E.1.

Properties:

- It contains exactly one degree of freedom: the capacity to differentiate the two positions □ and ■.
- It possesses no metric, no orientation, no topology, and no ordering — these are not latent within the construction but would require further, independent acts of distinction to introduce.
- It is the minimal non-trivial structure in the dimensional hierarchy.

E.6 Summary

This appendix establishes a chain of four claims. A distinction is a meta-linguistic act, not an internal structure. Marking a distinction yields two referential positions. Two positions necessitate, and are exactly met by, a one-degree-of-freedom representational capacity. This capacity is what the paper defines as the minimal distinction relating oD and 1D.

No "offset," no "shift," no "movement," and no "direction" is involved at any point in this construction. This construction instantiates, at the level of explicit meta-linguistic apparatus, the abstract argument given in the proof of Theorem 3.3; the distinction operator δ of Definition 0.5 is realized concretely here as the marked/unmarked distinction $\langle \square | \blacksquare \rangle$. The construction characterizes a relation between oD and 1D, not a process, priority, or direction between them; it is axiom-compatible, framework-independent, and dimension-minimal.

Appendix F — Why Physical Formalism Cannot Apply at oD

(Foundational, pre-physical layer.)

F.1 Purpose of This Appendix

The arguments in this appendix operate entirely in the meta-language. Logical vocabulary — conditionals, presupposition, categorical distinctions — is used to explain why certain structures cannot belong to oD internally; this does not imply that logical structure is absent from the

meta-language itself, nor does it contradict the meta-language/object-language separation of Appendix A.o. It is the same distinction by which the empty set is described using the full apparatus of set-theoretic language without that apparatus becoming part of the empty set's content. This appendix applies the meta-language to state which structures cannot be internal to oD; it does not claim that no structure can be used to discuss oD at all.

This appendix clarifies why the mathematical structures standard in physical theories — Hilbert spaces, operator algebras, symmetry groups, observables, expectation values — do not appear anywhere in this paper's treatment of the zero-dimensional state. The reason is not methodological restraint or postponement, but a categorical mismatch: these structures belong to a different ontological tier than the one in which oD is defined.

The zero-dimensional state is a pre-physical object. Physical formalisms presuppose the existence of structures that oD, by definition, precedes.

F.2 Ontological Tiers and the Scope of Physical Mathematics

Physical theories operate on what may be called a layer of appearance: the layer of measurement and observable structure. Even the most abstract physical frameworks — quantum field theory, statistical mechanics, gauge theory — presuppose a space of states, a notion of observable quantities, a symmetry group acting on those states, and a rule for extracting measurable predictions. These are not optional conveniences; they are the minimal prerequisites for any physical description.

By contrast, the zero-dimensional state belongs to a layer of structural origin prior to the emergence of observables, states, or measurable quantities. This origin layer is not a physical layer; it is the layer from which the possibility of physical description eventually emerges.

Physical mathematics cannot apply at oD because the objects it describes do not yet exist.

F.3 Why Hilbert-Space Methods Cannot Apply

A Hilbert space presupposes linear structure, an inner product, completeness, a space of states, operators acting on those states, expectation values, and a measurement theory [19, 20]. None of these notions are meaningful at oD. By Axioms 0–5, oD has no internal degrees of freedom, no linear structure, no notion of superposition, no metric or inner product, no operator domain, and no observable quantities.

Any Hilbert-space representation of oD would be a representation of the meta-language, not of oD itself. This would violate the meta-language discipline established in Appendix A.o.

F.4 Why Operator Algebras Cannot Apply

Operator algebras (CCR, CAR, C*-algebras, von Neumann algebras) presuppose a domain of action, a non-trivial algebraic structure, commutation or anticommutation relations, and a

notion of excitation or transformation [21]. oD has no domain, no algebraic structure, no relations to preserve or break, and no excitations.

There is no meaningful sense in which an operator can "act" on oD. Any such action would be an operation in the meta-language, not a transformation of oD.

F.5 Why Symmetry Groups Cannot Apply

Symmetry groups require a space on which the group acts, a notion of invariance, and a structure that can be preserved or broken [22]. oD has no internal structure, no degrees of freedom, no coordinates, no relations, and no action space.

The only symmetry group applicable to oD is the trivial group, and even this is a statement about the meta-language, not about oD itself. Any non-trivial symmetry would presuppose structure that oD does not possess.

F.6 Why Observables and Expectation Values Cannot Apply

An observable requires a state space, a measurable quantity, a spectrum of possible outcomes, and a rule for extracting expectation values [23]. oD has no measurable quantities, no outcomes, no probability structure, and no measurement theory.

The expression $\langle \text{oD} | \text{O} | \text{oD} \rangle$ is not merely undefined; it is category-incorrect. It presupposes the existence of a Hilbert space and an observable algebra — structures that belong to the layer of appearance, not to the layer in which oD is defined.

This appendix discusses the expression $\langle \text{oD} | \text{O} | \text{oD} \rangle$ only as an object of meta-linguistic analysis — to show why it fails to refer — never as a notation asserting something true of oD itself, consistent with the discipline of Appendix A.o.

F.7 The Correct Interpretation: oD as Pre-Physical

oD should not be understood as a physical vacuum, a quantum state, a symmetry-invariant configuration, or a limit of a physical theory. It is the structural origin from which the very possibility of physical description eventually emerges.

Physical formalism becomes a candidate tool only once the prerequisites discussed in F.2–F.6 — a degree of freedom, a state space, an observable capable of a non-zero value — are no longer categorically absent. This paper does not assert at which dimension that threshold is crossed; it asserts only that it has not been crossed at oD. The question of where physical formalism first becomes applicable is left to the later papers in the sequence described in Section 1.7.

F.8 Summary

Physical formalisms cannot apply at oD because they presuppose structures — states, observables, symmetries — that oD does not contain, and because applying them to oD would

be a category error rather than a simplification. Any physical notation used at oD would describe the meta-language, not oD itself. The emergence of physical structure begins only after the first distinction (1D), and the present paper takes no position on how soon thereafter such formalism becomes genuinely applicable.

oD is not a physical object.

Appendix G — Worked Finite Examples for Section 9

(Illustrative only; the general statements of Proposition 1.5, Theorem 3.5, and Theorem 3.6 do not depend on any particular cardinality.)

G.1 Smallest Instance: $|S| = 2$

Let $S = \{a, b\}$. By Axioms 6–7, $\Delta(a, b)$ holds and $\Delta(a, a)$, $\Delta(b, b)$ do not. The only bijections of S are the identity and the transposition swapping a and b ; both trivially preserve Δ , so $\text{Aut}(S, \Delta)$ is the full symmetric group on two elements, consistent with Proposition 1.5. With only one non-trivial pair available, there is no room to test independence directly; this case establishes only that Proposition 1.5 holds at the smallest possible cardinality.

G.2 $|S| = 3$

Let $S = \{s_1, s_2, s_3\}$, with $\Delta(s_i, s_j)$ holding for all $i \neq j$. Every permutation of S preserves Δ , so $\text{Aut}(S, \Delta)$ is the full symmetric group on three elements, of order six. Given any candidate distinguishing operator δ_1 compatible with this group — for instance, a 3-cycle — its non-trivial powers exhaust the cyclic subgroup it generates; a second operator independent of δ_1 , in the sense of Definition 0.5's discipline, would have to mark a division of S not respected by the full symmetric group's action. Direct inspection of the six elements of this group shows no such division survives: any subset other than S itself or the empty set is moved into a position only partially overlapping its original position by some element of the group, so no candidate partition is invariant. This is the $|S| = 3$ instance of Theorem 3.5.

G.3 $|S| = 4$ and the Threshold Theorem

Let $S = \{s_1, s_2, s_3, s_4\}$. This is the smallest cardinality at which a structure could in principle have room for a nontrivial coordinatizing partition (e.g., by labeling states with pairs from $\{0,1\} \times \{0,1\}$). Under Axioms 6–7, $\Delta(s_i, s_j)$ holds for every $i \neq j$, with no further qualification; $\text{Aut}(S, \Delta)$ is forced to the full symmetric group on four elements, by the same argument as Proposition 1.5, and Theorem 3.5 excludes any coordinatizing partition exactly as before.

Two routes might seem to avoid this forced collapse while staying close to the original axioms; both fail, illustrating the proof of Theorem 3.6:

- Allowing $\Delta(s_i, s_j) = 0$ for some pair of distinct states (relaxing Axiom 7 instead of Axiom 6): by Lemma 2.5, two states with no distinguishing predicate between them are meta-

linguistically identified, so this pair of states collapses into one, reducing the structure rather than enriching it.

- Retaining Axiom 7 but assigning different magnitudes or sub-types to different pairs without declaring this typing as a feature of Δ itself (e.g., an undeclared asymmetry among which pairs are "more different" than others): this implicitly assigns Δ more than the two values Axiom 6 permits, which is precisely the relaxation of Axiom 6 that Theorem 3.6 identifies as the necessary condition — not a way of avoiding it.

Both routes confirm Theorem 3.6: there is no way to admit a coordinatizing partition while retaining both axioms unchanged. The only route is the one the theorem identifies — relaxing Axiom 6 to admit more than one type of marked difference — which is exactly the interface condition given in Definition 0.6 for 2D.

Appendix H — A 2D Witness Structure Admitting a Nontrivial Coordinatizing Partition

The full construction is given in a separate supplementary document, "Supplementary Appendices H\2013K," submitted alongside this paper: Definitions H.1\2013H.4, Lemmas H.5\2013H.7, and Proposition H.8, in full with proofs.

Condensed summary: the witness structure is $S\backslash\mathbb{N}\backslash\mathbb{N} = \{0,1\}\backslash\mathbb{N}$ under a relaxed differentiation relation $\backslash\mathbb{N}\backslash\mathbb{N}\backslash\mathbb{N}$ with three values (Hamming distances 0, 1, 2). Its automorphism group is shown, by explicit construction and a matching upper bound, to have order exactly 8 and to be isomorphic to D_8 , the dihedral group (Lemma H.6). The coordinate-fiber partition $B\backslash\mathbb{N}$ (equivalently $B\backslash\mathbb{N}$) is shown to be invariant under this full group (Lemma H.7), giving a coordinatizing partition compatible with $\text{Aut}(S\backslash\mathbb{N}\backslash\mathbb{N}, \backslash\mathbb{N}\backslash\mathbb{N}\backslash\mathbb{N})$ (Proposition H.8) \2014 confirming Theorem 3.6\2019s relaxed condition is non-vacuous: at least one structure satisfying it exists (Corollary H.9). This establishes existence only, not that every structure satisfying the relaxed condition admits such a partition.

This pointer exists so the two files do not silently diverge. Do not re-add Appendix H content directly into this file.

Appendix I — Full Computational Verification Results

The complete, unabridged output is given in a separate supplementary document, "Supplementary Appendices H\2013K," submitted alongside this paper.

Condensed summary: the verification package (wgs-verify) referenced in Section 13.8 comprises twenty-two checks in total — one covering 0D (Appendix C's collapse principle, on a finite toy model), five covering 1D–2D, eight covering Section 12 (3D), and eight covering Section 13 (4D).

All twenty-two passed on every run; the supplementary document also includes Appendix I.1, six minimal standalone scripts (14–15 lines each, no dependency beyond itertools and numpy) verifying the same key claims by hand-checkable means, for readers who would rather not install and run the full package. The package itself (source code, not just its results) is included as supplementary material alongside this paper.

Appendix J — Genuine Applications of the Verified Machinery

The complete record is given in a separate supplementary document, "Supplementary Appendices H\u2013K," submitted alongside this paper.

Condensed summary: kept explicitly separate from Appendix I, this appendix records whether the verified machinery (B_n , Bose–Mesner algebras, Reynolds averaging) solves genuine problems outside this paper’s own claims. Results there are not evidence for or against this paper’s structural claims and are not to be read as such; the appendix records three accepted applications (tested and working) and four rejected proposals, each with its specific refutation, on the principle that a proposed application that cannot fail does not belong in the record as an accepted result. The accompanying package (wgs-apps), kept and distributed separately from wgs-verify, is included as supplementary material alongside this paper.

Appendix K — Minimal Standalone Scripts (Section 12)

The complete scripts are given in a separate supplementary document, "Supplementary Appendices H\u2013K," submitted alongside this paper.

Condensed summary: four minimal standalone scripts (11–16 lines each, using only itertools and basic arithmetic) verifying Section 12’s key claims directly: the E_3/P_3 identity confirmation resolving the previously-unidentified second order-24 subgroup, Lemma 2.8’s exclusion argument, Theorem 3.15’s premises at $n=3$, and a point-set-level sampling check (the $AGL(3,2)$ case) referenced in Section 12’s open items. Each script is independently runnable without the full verification package described in Appendix I.

Appendix L — Non-Uniqueness of Axioms 8–9 Beyond $|S|=4$

This appendix records, in full, the computational verification behind §11.4’s conjecture. It includes the methodological missteps actually made during this verification, not only the final

clean result, on the principle already followed for Appendix H: an error caught and corrected is more informative, and more checkable by a reader, than a silently-repaired final answer.

L.1 $|S|=6$: A Complete Brute-Force Enumeration

At $|S|=6$ there are 15 unordered point-pairs, hence $2^{15} = 32768$ candidate colorings of $S \times S$ into two types. All 32768 were checked by direct computation. 22130 satisfy Axiom 8's nonempty- Γ condition (i)–(ii) alone. A first check of Axiom 9 was attempted that tested only orbit-internal color consistency, without separately requiring transitivity and non-2-transitivity on S itself; this check is vacuously satisfied whenever the automorphism group is small, since a group with trivial or near-trivial action has only singleton or near-singleton orbits on pairs, which are trivially color-consistent. That flawed check reported all 22130 structures as satisfying Axiom 9, which is incorrect. The corrected check additionally requires $\text{Aut}(S, \Delta)$ to be transitive on S , not 2-transitive, and for the point stabilizer G_x to act transitively on $\Gamma_1(x)$ and $\Gamma_2(x)$ separately, for every x . Under the corrected check, exactly 50 of the 32768 colorings satisfy Axioms 8–9.

Grouping these 50 colorings into isomorphism classes (orbits under the action of $\text{Sym}(6)$ on colorings, which correctly identifies colorings differing only by relabeling points or by swapping which type is called σ_1 versus σ_2) yields exactly two classes: one of size 30 with $|\text{Aut}(S, \Delta)| = 48$, and one of size 20 with $|\text{Aut}(S, \Delta)| = 72$. These were identified directly: the order-48 structure is the automorphism group of the octahedron graph (equivalently, the hyperoctahedral group B_3 acting on the 6 signed coordinate directions $\{\pm e_1, \pm e_2, \pm e_3\}$, rather than on the 8-point Hamming cube used in Section 12); the order-72 structure is the automorphism group of the complete bipartite graph $K_{3,3}$. Both identifications were confirmed by constructing each graph explicitly and verifying its automorphism group's order and element-order profile match the discovered structures exactly: profile $(1^1, 2^{19}, 3^8, 4^{12}, 6^8)$ for the order-48 case and $(1^1, 2^{21}, 3^8, 4^{18}, 6^{24})$ for the order-72 case, in both directions.

L.2 $|S|=8$: Why Brute Force No Longer Applies, and What Replaces It

At $|S|=8$ there are 28 unordered point-pairs, hence $2^{28} \approx 2.68 \times 10^8$ candidate colorings — too many to check exhaustively by the $|S|=6$ method in reasonable time. Two specific candidates, constructed directly by analogy with the $|S|=6$ result, were checked first and confirmed: B_4 acting on the 8 signed coordinate directions (order 384, verified by checking all 40320 permutations of $\text{Sym}(8)$ against the resulting coloring directly, not assuming the candidate group equals the full automorphism group in advance) and the automorphism group of $K_{4,4}$ (order 1152, verified the same way). This established that at least two non-isomorphic structures exist at $|S|=8$, matching the $|S|=6$ pattern, but did not establish that exactly two exist — ruling out a third, unconsidered family requires checking against every transitive group of degree 8, not only the two specific candidates suggested by the $|S|=6$ pattern.

An intermediate attempt used randomized search — generating subgroups of $\text{Sym}(8)$ from randomly sampled small generating sets, and checking each for transitivity and rank 3 — as a partial substitute for full enumeration. This method can find additional solutions if they exist, but cannot rule out their existence: a solution requiring an unlikely or larger generating set could be missed regardless of how long the search runs. This method is recorded as a step actually taken, but its results are not used to support any claim of completeness; the conjecture in §11.4 rests entirely on the exhaustive method described next.

L.3 $|S|=8$: Exhaustive Verification via the Complete List of Degree-8 Transitive Groups

The mathematically complete alternative to brute-force coloring enumeration is to enumerate transitive permutation groups directly, since any group G that is the full automorphism group of some Axioms-8–9-satisfying structure must itself be transitive on points and have rank exactly 3 on ordered pairs (one diagonal orbit plus exactly two non-diagonal orbits, corresponding to σ_1 and σ_2). The complete list of transitive permutation groups of degree 8, up to conjugacy in $\text{Sym}(8)$, is a standard, independently-maintained reference object in computational group theory (the TransGrp library); it contains exactly 50 groups. This list was obtained directly via GAP’s `AllTransitiveGroups(NrMovedPoints, 8)` and used as the complete search space, replacing the infeasible 2^{28} -coloring enumeration with a feasible 50-group check.

Filtering the 50 groups for rank exactly 3 on ordered pairs (computed via `Orbits(g, Cartesian([1..8],[1..8]), OnTuples)` for each g) yields 13 candidates, ranging in order from 48 to 1152. All 13 were then checked for Axiom 8(iii)’s symmetry condition — each non-diagonal orbit must be closed under reversing the pair, i.e. (i,j) in the orbit if and only if (j,i) is — and all 13 satisfy it; this condition eliminated none of the candidates at $|S|=8$, unlike what might have been expected by analogy with the relatively restrictive Axiom 8(iii) discussion in Section 10.1.

The remaining step is essential and easy to omit: a candidate group G appearing in this list with rank 3 does not by itself establish that G is the full automorphism group of the resulting structure (S,Δ) . G might be a proper subgroup of a larger group that also preserves the same coloring, in which case G is not the answer to “what is $\text{Aut}(S,\Delta)$ ” — the larger group is. Each of the 13 candidates was therefore checked directly: construct the coloring its rank-3 orbit structure induces, then compute the true automorphism group of that coloring by testing all 40320 permutations of $\text{Sym}(8)$ against it (the same direct method used in §L.2 for the two original candidates, not a method that assumes the candidate group is already correct). Of the 13, only two have a true automorphism group equal to themselves: the orders 384 and 1152. The other 11 are proper subgroups of one or the other of these two — they preserve the same colorings, but are not those colorings’ full automorphism groups, and so do not represent additional, independent solutions.

The two surviving solutions were identified, not merely numerically matched: B_4 (constructed explicitly via generators for the four axis-flips and three axis-

transpositions) was confirmed conjugate, in $\text{Sym}(8)$, to the order-384 group on the list (GAP’s structure description for this group, “[2⁴]S(4)”, independently names the same signed-axis structure); the automorphism group of $K_{4,4}$ (constructed explicitly via generators for the within-side permutations and one side-swap) was confirmed conjugate to the order-1152 group (GAP’s description “[S(4)²]2” likewise names this structure independently). Conjugacy in $\text{Sym}(8)$ is the precise condition for “the same permutation group, up to relabeling the 8 points” — a strictly stronger and more reliable confirmation than matching group order or matching the heuristic StructureDescription string alone, both of which are necessary but not sufficient conditions that earlier steps of this verification relied on before this final check.

This establishes, with the same completeness and rigor as the $|S|=6$ case (full enumeration of the relevant search space, not sampling), that exactly two non-isomorphic structures satisfy Axioms 8–9 at $|S|=8$.

L.4 What This Does and Does Not Establish

Confirmed, by exhaustive methods, at two independent cardinalities ($|S|=6$ and $|S|=8$): Axioms 8–9 do not force a unique automorphism group beyond the smallest case, and the two structures that do arise at each cardinality checked are the signed-axis representation of B_m and the automorphism group of $K_{m,m}$. Not established: that this pattern holds for every $m \geq 2$ (only $m=2,3,4$ have been checked at all, and $m=2$ is degenerate since the two families coincide there); that these are the only two families possible at any m not yet checked; or any general argument explaining why exactly these two families arise rather than some other pair, or three, or one. §11.4’s conjecture should be read with this scope, and no more.

Appendix M — A Counterexample to E_n/P_n Exhaustiveness at $n=5$

This appendix records the discovery of a structure satisfying every condition of the open classification question recorded in §13.9 — transitive, rank exactly $n+1$, stabilizer-transitive on each $\Gamma_i(x)$, a subgroup of B_n — that is isomorphic to neither B_n , E_n , nor P_n , at $n=5$. As with Appendix L, the methodological missteps actually made are recorded alongside the final result.

M.1 Scope of the Question, and a First Search That Was Not Correctly Scoped

§13.9’s open classification question is stated for $G \leq B_n$ specifically — subgroups of the hyperoctahedral group itself, not arbitrary transitive groups of degree 2^n . At $n=5$, $S = \{0,1\}^5$ has 32 points, and B_5 has order 3840. A first search instead queried

the complete list of transitive permutation groups of degree 32 (a standard library, analogous to the degree-8 list used in Appendix L, but containing 2{,}801{,}324 groups at degree 32 rather than 50 at degree 8) restricted to a narrow order window centered on 1920 (E_5 and P_5 's order). This found exactly two structures with rank 6 in that window, and both were confirmed (by direct conjugacy testing) to be E_5 and P_5 themselves — a correct but narrowly-scoped result: it ruled out a third rank-6 structure only among degree-32 transitive groups of order within roughly 25 of 1920, not among all subgroups of B_5 , and not at any other order, including B_5 's own order 3840. This was recognized as insufficiently scoped to the actual question before being relied upon for any conclusion.

M.2 The Correctly-Scoped Search: All Subgroups of B_5

The question as stated concerns subgroups of B_5 specifically, so the correct search space is the subgroup lattice of B_5 itself — a group of order 3840, directly enumerable, rather than the unrelated and far larger space of all degree-32 transitive groups. B_5 was constructed explicitly (generators: five coordinate-flip permutations and four adjacent-coordinate-swap permutations on the 32 points of $\{0,1\}^5$, verified to generate a group of order 3840 exactly matching $2^5 \times 5!$). Its subgroup lattice contains 36{,}430 subgroups in 953 conjugacy classes. Each of the 953 conjugacy class representatives was checked directly for transitivity on the 32 points and for rank exactly 6 on ordered pairs; this filtering was independently re-run via a second, differently-written function operating on the same 953 representatives, with identical results both times (953 representatives checked, 4 satisfying both conditions), ruling out an error specific to either implementation.

Exactly four conjugacy classes satisfy both conditions, of orders 640, 1920, 1920, and 3840. The order-3840 class is B_5 itself (confirmed by direct conjugacy testing against the explicit B_5 construction). Of the two order-1920 classes, one is conjugate to E_5 and the other to P_5 (E_5 and P_5 were each constructed independently — not assumed — via generating sets verified by incremental closure to produce groups of exactly order 1920 matching each definition in §0.8–§0.9, then confirmed by direct conjugacy testing, not by order or name alone, against the two order-1920 candidates). The order-640 class is conjugate to none of B_5 , E_5 , or P_5 .

M.3 Direct Verification of the Order-640 Structure

The order-640 structure was not accepted on the strength of rank and order alone. Its point stabilizer was computed directly and confirmed to have order 20 (consistent with the orbit-stabilizer theorem: $640/32 = 20$). For the fixed point 1, the five non-diagonal $\Gamma_i(1)$ sets (of sizes 5, 10, 10, 5, 1, summing with the fixed point itself to 31, correctly accounting for all 32 points) were each tested directly for transitivity under the stabilizer; all five passed. This directly confirms Axiom 11's full stated condition for this structure, rather than relying only on the general argument (already verified independently in Appendix L, §L.1) that single-orbit transitivity on points automatically implies stabilizer-transitivity on each corresponding $\Gamma_i(x)$.

GAP's StructureDescription for this group is " $C_2 \times ((C_2 \times C_2 \times C_2 \times C_2) : (C_5 : C_4))$ ", of order $2 \times 2^4 \times 20 = 640$. The factor $C_5:C_4$ (order 20) is a standard, named object — the Frobenius group F_5 , also known as $AGL(1,5)$, the holomorph of C_5 , or $SmallGroup(20,3)$ — confirming this is not an arbitrary or malformed group, but is built from a recognizable component. The group has index 6 in B_5 and is not normal in B_5 (E_n and P_n , by contrast, are always normal in B_n , being index-2 subgroups; this order-640 structure is of a structurally different kind from either).

A literature search did not locate any prior discussion of this specific structure — an index-6, order-640 subgroup of B_5 with this rank-6 action — under any name, in published work on hyperoctahedral group subgroup structure, association schemes, or Frobenius group embeddings. This is reported as the result of a search of readily-available academic literature, not as a claim that no such discussion exists anywhere; an unfalsifiable negative of that kind cannot be established by search and is not asserted here.

M.4 What This Establishes, and What It Does Not

Established: at $n=5$, the classification claim of §13.9 (every $G \leq B_n$, transitive, rank $n+1$, satisfying Axiom 11, is B_n , E_n , or P_n) is false. A fourth, non-conjugate structure exists, confirmed by exhaustive search of B_5 's full subgroup lattice (953 conjugacy classes, not a sample), independently re-run, and confirmed to satisfy Axiom 11's full stated condition by direct computation rather than by the general argument alone.

Not established: whether a comparable fourth (or further) structure exists at every $n \geq 5$, at infinitely many n , or only at $n=5$; what general construction, if any, produces such structures at other n ; or any structural reason why this specific order-640 structure exists where it does. These are new questions, raised by this finding, for future work — not resolved by it.

Not affected: Theorem 3.15. The order-640 structure satisfies Axioms 10–11 but not Axiom 12' — this follows directly from Theorem 3.15's own proof (any group satisfying Axiom 12' is forced to equal B_n by direct counting, with no exception for any $n \geq 2$), and was not separately re-verified here since re-deriving it would only repeat Theorem 3.15's existing argument. The distinction between this paper's two central claims about B_n — that Axiom 12' forces uniqueness (Theorem 3.15, unaffected), and that the weaker Axioms 10–11 alone admit only $B_n/E_n/P_n$ as rank- $(n+1)$ possibilities (§13.9's classification question, now false at $n=5$) — is exactly the distinction this paper has maintained throughout between sufficiency and a separate, narrower classification question (§12.5.1).

M.5 — Relation to Kantor's Classification of k -Homogeneous, Non- k -Transitive Groups ($k=2,3,4$)

This subsection records a literature parallel flagged during self-review. It does not alter, extend, or depend on any theorem stated above, and is included for disclosure rather than as new mathematical content.

Kantor determined all k -homogeneous, non- k -transitive permutation groups for $k=2,3,4$ [25, 26] — a classification completed by direct group-theoretic argument, case by case, with no single formula covering all three values of k , and established before the Classification of Finite Simple Groups. This paper's own small- n results — the forced D_4 at $n=2$ (Theorem 3.7), the two extra transitive rank-4 subgroups of B_3 found by exhaustive search at $n=3$ (Section 12.3), and the confirmation that E_4 and P_4 are exactly the rank-5 candidates at $n=4$ (Section 13.4) — were likewise obtained case by case, one dimension at a time, before Theorem 3.15's general- n counting argument closed the sufficiency question uniformly.

The parallel is acknowledged here explicitly for a reason specific to this paper's stated aims (Section 1; Section 14): the classification of which permutation groups can arise under a given homogeneity or transitivity condition is not, on its own, the contribution this paper is making, and is not claimed as such. Kantor's result already settles a closely related — though not identical — question about k -homogeneous actions in general, using tools (character theory, the structure of 2-transitive groups) this paper does not use and does not need. What this paper adds is independent of that classification literature: an axiomatic route (Axioms 8–12') by which the same small-case, then-uniform, then-breaks-again-at- $n=5$ pattern is derived from typed-differentiation conditions alone, rather than obtained from the ambient permutation-group classification. The forced appearance of B_n , E_n , P_n , and the order-640 exception is, in this paper, a consequence of Sections 2–15's axioms; it is not presented as a new addition to the k -homogeneity classification literature, and no claim is made that Kantor's methods or results were used, needed, or improved upon here.

A precise formal identification between Kantor's k -homogeneous/ k -transitive dichotomy on an arbitrary set and this paper's rank- $(n+1)$ -subgroup-of- B_n question on $\{0,1\}^n$ specifically has not been attempted and is not asserted; the two conditions are stated over different actions (k -subsets of an arbitrary degree- m set, versus ordered pairs under the Hamming-cube action), and nothing here should be read as claiming they coincide. The parallel drawn is at the level of pattern — small cases resolved individually, uniformity emerging past a threshold, and that threshold's own failure (Appendix M itself, at $n=5$) — not at the level of a shared theorem.

Appendix N — A Tested Negative Result: The $n=5$ Construction Does Not Generalize to $n=7$

This appendix records a direct computational test of whether the order-640 structure found at $n=5$ (Appendix M) generalizes to the next prime, $n=7$, via the most natural analogous construction. The result is negative — but unlike a typical unresolved open item, the reason is fully derived, not merely observed: a general criterion is established explaining exactly when this specific construction can succeed, and $n=5$ turns out to be the boundary case, not an arbitrary instance.

N.1 The Candidate Construction

The order-640 structure at $n=5$ (Appendix M) has GAP structure description “ $C_2 \times ((C_2 \times C_2 \times C_2 \times C_2) : (C_5 : C_4))$ ”. The factor $C_5:C_4$, of order 20, is the Frobenius group $AGL(1,5)$ — the

affine group of the line over F_5 , acting on the 5 coordinates of $\{0,1\}^5$ via maps $x \mapsto ax + b \pmod{5}$, for $a \in \{1,2,3,4\}$ and $b \in \{0,1,2,3,4\}$. Combined with the full flip group $(Z_2)^5$, this gives exactly $(Z_2)^5 \rtimes \text{AGL}(1,5)$, of order $32 \times 20 = 640$, matching the discovered structure precisely. Since $\text{AGL}(1,n)$ is defined for every prime n , the most direct generalization to test is whether $(Z_2)^n \rtimes \text{AGL}(1,n)$ yields an analogous rank- $(n+1)$ structure outside E_n and P_n at the next prime, $n=7$.

N.2 Validating the Method Against the Known $n=5$ Result

Before trusting any conclusion at $n=7$, the construction method itself was validated by reproducing the known $n=5$ result independently, in a standalone Python implementation (mirroring the conventions of the wgs-verify package's core module, without reusing any of its code directly): points are the 2^n elements of $\{0,1\}^n$; the group $(Z_2)^n \rtimes K$, for K a set of coordinate-permutations, consists of every (permutation, flip-mask) pair, acting by flipping coordinates and then permuting positions. Constructing $\text{AGL}(1,5)$ directly (20 affine maps) and forming $(Z_2)^5 \rtimes \text{AGL}(1,5)$ gives a group of order 640, confirmed transitive on the 32 points, with rank exactly 6 on ordered pairs — exactly the order, transitivity, and rank already established for the mystery structure in Appendix M. This agreement, obtained independently and by a different implementation than the original GAP-based search, confirms both that the order-640 structure is exactly this construction, and that the verification method is sound before applying it at $n=7$.

N.3 The Test at $n=7$

$\text{AGL}(1,7)$, the affine group of the line over F_7 , has order $7 \times 6 = 42$ ($a \in \{1,\dots,6\}$, $b \in \{0,\dots,6\}$). Forming $(Z_2)^7 \rtimes \text{AGL}(1,7)$ gives a group of order $128 \times 42 = 5376$. Direct computation confirms this group is transitive on the 128 points of $\{0,1\}^7$. Its rank on ordered pairs, however, is exactly 10 — not 8, the value Axiom 11 requires to match B_n 's own rank $n+1$ at $n=7$. The construction therefore does not satisfy Axiom 11 at $n=7$, and is not a candidate analogous to the $n=5$ structure.

N.4 Why It Fails: A General Criterion, Not a Coincidence

The discrepancy is not an isolated numerical accident; it has a complete explanation. For $G = (Z_2)^n \rtimes K$ with $K \leq S_n$ acting on the n coordinates, the rank of G on ordered pairs of $\{0,1\}^n$ equals exactly the number of K -orbits on the power set of the n coordinates: a pair (x,y) and any of its images under G have the same set of coordinates on which they differ, up to the action of K on that difference set, and conversely two pairs with K -equivalent difference sets lie in the same G -orbit (since the flip group $(Z_2)^n$ is unrestricted). Hence $\text{rank}(G)$ equals the sum, over $k = 0,\dots,n$, of the number of K -orbits on k -element subsets of $\{1,\dots,n\}$.

$\text{AGL}(1,n)$ is sharply 2-transitive on its n points for every prime n (order $n(n-1)$, matching the sharp 2-transitivity bound exactly: for any two ordered pairs of distinct points, there is exactly one affine map carrying one to the other). Sharp 2-transitivity guarantees a single orbit on 0-, 1-, and 2-element subsets, and — by the elementary fact that K -orbits on k -subsets correspond bijectively to K -orbits on their $(n-k)$ -element complements — also a single orbit on n -, $(n-1)$ -,

and $(n-2)$ -element subsets. The subset sizes not covered by this argument are exactly $k = 3, 4, \dots, n-3$ — a range that is empty precisely when $n-3 < 3$, i.e., $n \leq 5$.

This is exactly the boundary observed. At $n=5$, the uncovered range $\{3, \dots, n-3\} = \{3, \dots, 2\}$ is empty, so 2-transitivity alone already forces all six subset sizes (0 through 5) into single orbits, giving rank exactly 6. At $n=7$, the uncovered range is $\{3, 4\}$, and 2-transitivity gives no guarantee there; 3-transitivity would be required to guarantee it, and $\text{AGL}(1, n)$ — a sharply 2-transitive group of order $n(n-1)$, far smaller than the $n!/(n-3)!$ needed for sharp 3-transitivity — is not 3-transitive for any $n > 2$. Direct computation confirms the uncovered sizes do split at $n=7$: the 35 three-element subsets of $\{0, \dots, 6\}$ fall into exactly two $\text{AGL}(1, 7)$ -orbits, with representatives $\{0, 1, 2\}$ (a length-3 arithmetic progression) and $\{0, 1, 3\}$ (not one); by the complement correspondence, the four-element subsets split into the same two orbits. This gives rank $1+1+1+2+2+1+1+1 = 10$, exactly the value computed directly in Section N.3.

N.5 What This Establishes, and What It Does Not

Established: the specific construction $(\mathbb{Z}_2)^n \rtimes \text{AGL}(1, n)$ yields a structure satisfying Axiom 11's rank- $(n+1)$ condition if and only if $n \leq 5$; this is a complete, derived criterion, not an empirical pattern fitted after the fact. The $n=5$ case (Appendix M) sits exactly at this boundary, which is itself worth recording: the order-640 structure is not an arbitrary coincidence but the largest n for which this particular construction can possibly work.

Not established: that no rank- $(n+1)$ structure outside E_n and P_n exists at $n=7$, or at any $n \geq 6$, by some other construction. The argument here rules out exactly one specific, natural family — coordinate-permutation groups built from $\text{AGL}(1, n)$ — and explains precisely why that family fails beyond $n=5$; it says nothing about coordinate-permutation groups built from other 2-transitive groups of degree n . For $n=7$, the only other 2-transitive group of degree 7 besides S_n and A_n themselves is $\text{PSL}(3, 2)$, of order 168, acting on the seven points of the Fano plane; this group is likewise not 3-transitive — its 3-element subsets split into the seven Fano lines versus the remaining twenty-eight non-collinear triples, two orbits, the same count as $\text{AGL}(1, 7)$ — and was not separately tested here in full, since the same rank-counting argument predicts the same rank-10 outcome without requiring a fresh computation. Nor does this address constructions of an entirely different kind, not built from a coordinate-permutation subgroup acting on a smaller alphabet at all.

What this narrows: any further rank- $(n+1)$ structure at $n \geq 6$, should one exist, requires a coordinate-permutation subgroup K of S_n that achieves transitivity on the “middle” subset sizes $\{3, \dots, n-3\}$ without being all of S_n — a strictly stronger condition than the 2-transitivity sufficient at $n=5$, and one that ordinary 2-transitive groups, sharply 2-transitive groups in particular, are not expected to satisfy. This is recorded as the most concrete obstruction identified so far to extending the $n=5$ finding, narrowing rather than resolving the general classification question left open in §13.9.

Appendix O — Notation Summary

This is a reference guide to nomenclature, gathered in one place for convenience; it is not a substitute for the definitions given at each symbol's first use, which the section pointers below indicate.

- **S** — a state space: a set, with no internal structure presupposed beyond what the axioms at each layer add (Section 2 onward).
- **Δ** — the typed differentiation relation on S, distinguishing pairs of points by type (Axiom 6, generalized at Axioms 10–11).
- **$\Sigma = \{\sigma_1, \sigma_2, \dots\}$** — the set of non-trivial Δ -types (Axiom 6).
- **$\Gamma_i(\mathbf{x})$** — the set of points $y \in S$ with $\text{type}(\Delta(x,y)) = \sigma_i$, for fixed x (Section 10).
- **$\text{Aut}(\mathbf{S}, \Delta)$** — the group of Δ -preserving bijections of S (Definition 0.7).
- **G_x** — the stabilizer of x within $\text{Aut}(\mathbf{S}, \Delta)$: $\{g \in \text{Aut}(\mathbf{S}, \Delta) : g(x) = x\}$ (Section 12.2.1).
- **B_n** — the hyperoctahedral group, $(\mathbb{Z}_2)^n \rtimes S_n$, of order $2^n \cdot n!$ (Section 12.2).
- **E_n** — the even-parity subgroup of B_n , index 2 (Definition 0.8).
- **P_n** — the permutation-even subgroup of B_n , index 2 (Definition 0.9).
- **Q_n** — the hypercube graph: vertex set $\{0,1\}^n$, an edge between every pair at Hamming distance 1 [1]. This names a precisely defined combinatorial graph, cited for its known automorphism group ($\text{Aut}(Q_n) = B_n$); it carries no claim that $\{0,1\}^n$ has spatial extent, position, or geometric shape, and no proof in this paper depends on picturing it as one (Section 1, opening paragraph; Section 1.4).
- **δ** — a distinction operator: a meta-language device marking a difference relative to oD (Definition 0.5).

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Data and Code Availability. The computational verification package (`wgs_verify`), independently re-implementing and checking every numerical and structural claim made in this paper — twenty-two checks in total, documented in full in Appendix I — is openly available [24] at <https://doi.org/10.5281/zenodo.20839560> and is sufficient on its own to reproduce every computational result this paper reports. A separate, independent package (`wgs_applications`), demonstrating applications of the general-purpose linear-algebra tools used in this paper — group-averaging and Bose–Mesner-algebra residual scoring, both standard representation-theoretic techniques — to symmetry-based signal denoising, automorphism-group diagnosis of a real error-correcting code, and anomaly detection, is supplied at the same location for independent interest only; no claim, theorem, or numerical result in this paper depends on it.

Appendix P — Relation to WGS

This paper is written, and can be read in full, as a self-contained result in finite permutation group theory; nothing in Sections 1–15 or Appendices A–O depends on what follows. This appendix records, separately, how the paper's forced-layer results sit inside a broader generative framework the author develops elsewhere, referred to here as WGS (Wujie Generation Stratum). Framing it this way is a statement about the author's own model, not a claim about the world; nothing in this appendix is offered as evidence for WGS, and no theorem stated above is weakened, strengthened, or reinterpreted by it.

WGS is the appearance-rendering half of a larger ontology (WLM) organizing dimensional layers $0D$ through $27D+$ across three orthogonal strata: D (invisible generative structure), E (invisible rendering mechanism), and $S = \Delta(D, E)$, the stable mismatch that is the only one of the three directly visible at any layer. Within that model, each forced automorphism group proved in this paper — the symmetric group at $1D$, the dihedral group D_4 at $2D$ (Theorem 3.7), and the hyperoctahedral group B_n at every $n \geq 2$ once Axiom 12' is imposed (Theorem 3.15) — is read as the rigidity constraint governing what a given dimensional layer is permitted to render: the E -layer mechanism responsible for producing the specific D/E mismatch S that appears at that layer. On this reading, the $0D \rightarrow 4D$ sequence developed here is the skeleton of WGS's dimensional arc at its lowest five rungs — not a description of what appears at each layer (that is D , outside this paper's scope) and not the mismatch itself (that is S), but the group-theoretic envelope of what is even permitted to appear at all.

Two features of this paper's own results carry additional meaning under that reading, though neither depends on it for mathematical validity. First, that E_n and P_n persist as systematic alternatives to B_n under the weaker Axioms 10–11, excluded only by the added local-saturation Axiom 12', reads as two competing rigidity structures the E -layer could in principle admit, with Axiom 12' marking the specific condition WGS's dimensional arc is taken to actually satisfy at each of these layers. Second, that a fourth, order-640 structure survives at $n=5$ and fails to generalize to $n=7$ (Appendices M–N) reads, under this model, as the E -layer skeleton itself changing character at the fifth rung — consistent with, though not offered as evidence for, this being close to where WGS's own construction is understood to require qualitatively different treatment.

None of this appendix is intended to be evaluated by, or is relevant to, a reader assessing the paper's mathematical content on its own terms. It is included because the author's broader research programme is public, and readers who encounter both are entitled to know how the author himself relates them.

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